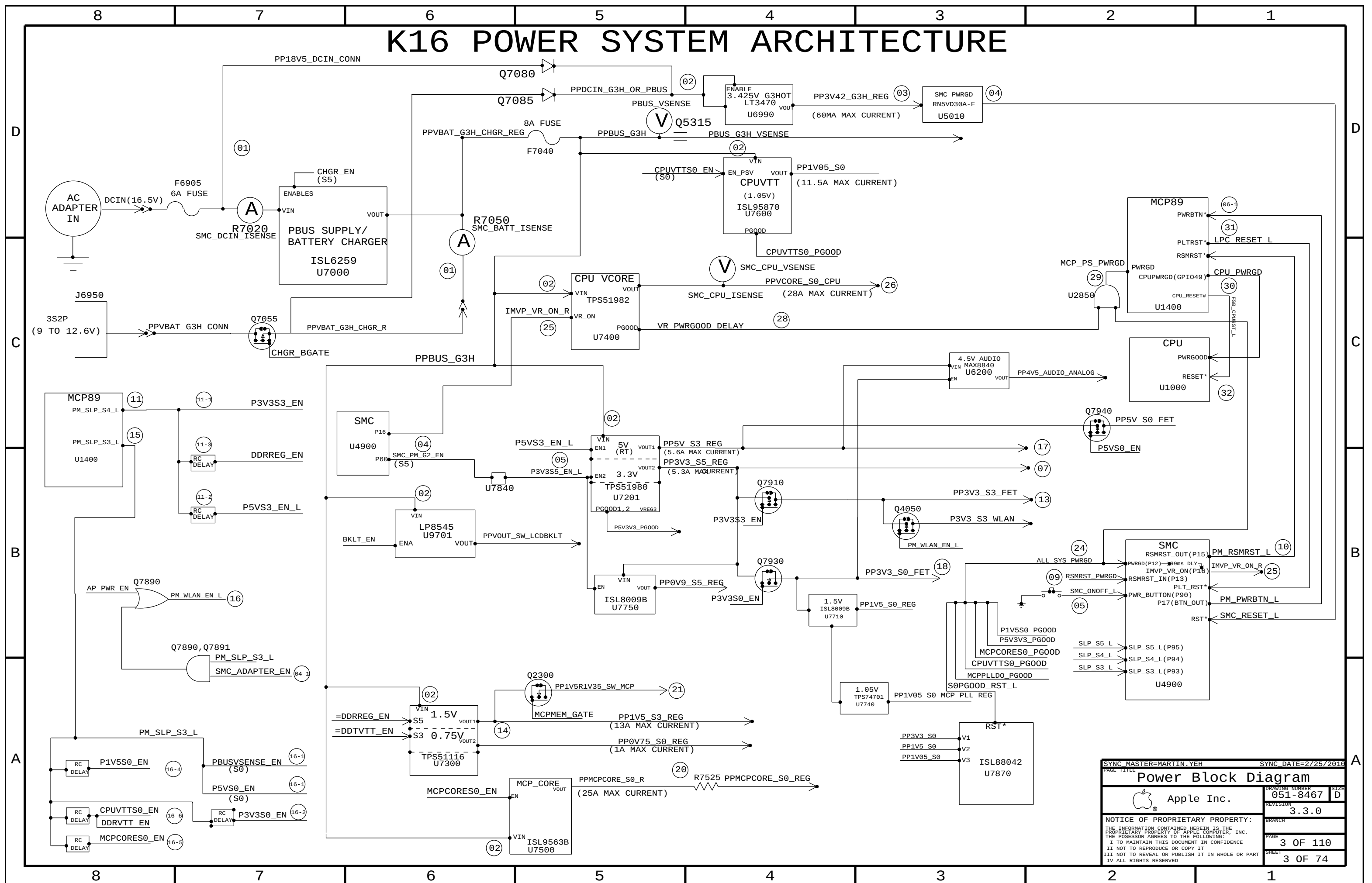






# K16 POWER SYSTEM ARCHITECTURE



|  | VENDOR  | CFG 0 | CFG 1 |
|--|---------|-------|-------|
|  | HYNIX   | 0     | 0     |
|  | SAMSUNG | 0     | 1     |
|  | MICRON  | 1     | 0     |
|  | ELPIDA  | 1     | 1     |

| PART NUMBER | QTY | DESCRIPTION                              | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|------------------------------------------|---------------|----------|------------|
| 337S3820    | 1   | IC, MCP89U-AB1, 24.5MMX24.5MM, 1244FCBGA | U1400         | CRITICAL | MCP89U:A01 |
| 337S3868    | 1   | IC, MCP89U-AB2, 24.5MMX24.5MM, 1244FCBGA | U1400         | CRITICAL | MCP89U:A02 |
| 337S3938    | 1   | IC, MCP89U-AB3, 24.5MMX24.5MM, 1244FCBGA | U1400         | CRITICAL | MCP89U:A03 |

| BOM GROUP        | BOM OPTIONS                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| K16_COMMON       | COMMON, ALTERNATE, PROJ:K16, K16_MISC, MCP89U:A83, K16_DEBUG:ENG, K16_PROGPARTS, SPI:41MHZ, LVDDR3:YES, WLAN_PCTL:HW, IPD_SV:SS_INT, IPD_3V3:SS |
| K16_MISC         | DP_ESD, DP_PWR:SMC, VFRQ:SLPS3, HVDDLDO:FIXED, MCPHVDD:P2V5, MCPPLL_R:REG, S0PGOOD_BJT, ISL6259_SCREENED:YES, DPI2C:SMC                         |
| K16_PROGPARTS    | BOOTROM:UNLOCKED, SMC:PROG                                                                                                                      |
| K16_DEVEL:ENG    | BKLT:ENG, BMON:ENG, XDP_CONN, LPCPLUS, VREFMRGN:YES, EFI_DEBUG, S0PGOOD_ISL, MCPPLL_LDO, S3_S0_LED                                              |
| K16_DEVEL:PVT    | LPCPLUS                                                                                                                                         |
| K16_DEBUG:ENG    | DEVEL_BOM, SMC_DEBUG:YES, XDP                                                                                                                   |
| K16_DEBUG:PVT    | DEVEL_BOM, BKLT:PROD, BMON:PROD, SMC_DEBUG:YES, XDP, VREFMRGN:NO                                                                                |
| K16_DEBUG:PROD   | BKLT:PROD, BMON:PROD, SMC_DEBUG:YES, XDP, VREFMRGN:NO                                                                                           |
| DDR3:HYNIX_2GB   | DRAM_CFG0:L, DRAM_CFG1:L, DRAM_CFG2:L, DRAM_CFG3:L, DRAM_TYPE:HYNIX_2GB                                                                         |
| DDR3:SAMSUNG_2GB | DRAM_CFG0:L, DRAM_CFG2:L, DRAM_CFG3:L, DRAM_TYPE:SAMSUNG_2GB                                                                                    |
| DDR3:MICRON_2GB  | DRAM_CFG0:H, DRAM_CFG1:L, DRAM_CFG2:L, DRAM_CFG3:L, DRAM_TYPE:MICRON_2GB                                                                        |
| DDR3:ELPIDA_2GB  | DRAM_CFG0:H, DRAM_CFG2:L, DRAM_CFG3:L, DRAM_TYPE:ELPIDA_2GB                                                                                     |
| DDR3:ELPIDA_4GB  | DRAM_CFG0:H, DRAM_CFG2:H, DRAM_CFG3:L, DRAM_TYPE:ELPIDA_4GB                                                                                     |
| DDR3:HYNIX_4GB   | DRAM_CFG0:L, DRAM_CFG1:L, DRAM_CFG2:H, DRAM_CFG3:L, DRAM_TYPE:HYNIX_4GB                                                                         |
| DDR3:SAMSUNG_4GB | DRAM_CFG0:L, DRAM_CFG2:H, DRAM_CFG3:L, DRAM_TYPE:SAMSUNG_4GB                                                                                    |
| DDR3:MICRON_4GB  | DRAM_CFG0:H, DRAM_CFG1:L, DRAM_CFG2:H, DRAM_CFG3:L, DRAM_TYPE:MICRON_4GB                                                                        |
| CAPS:SS          | SS_CAP_2_2UF, SS_CAP_10UF, SS_CAP_1UF, SS_CAP_22UF                                                                                              |
| CAPS:MU          | MU_CAP_2_2UF, MU_CAP_10UF, MU_CAP_1UF, MU_CAP_22UF                                                                                              |
| CAPS:TY          | TY_CAP_2_2UF, TY_CAP_10UF, TY_CAP_1UF, TY_CAP_22UF                                                                                              |

|          |   |                                            |       |          |               |
|----------|---|--------------------------------------------|-------|----------|---------------|
| 338S0563 | 1 | IC, SMC, HS8/2117, 9X9MM, TLP, HF          | U4900 | CRITICAL | SMC:BLANK     |
| 335S0610 | 1 | IC, FLASH, SPI, 32MBIT, 3.3V, 86MHZ, 8-SOP | U6100 | CRITICAL | BOOTROM:BLANK |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|-------------|---------------------------|------------|---------|-----------|
|-------------|---------------------------|------------|---------|-----------|

|             |                         |                                     |                                                           |                           |                  |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
|-------------|-------------------------|-------------------------------------|-----------------------------------------------------------|---------------------------|------------------|-------------------------------|---------------|----------|-----------------------------|---------------|----------|--------------|-------------|-----|-----------------------------|---------------|----------|--------------|
|             | 8                       | 7                                   | 6                                                         | 5                         | 4                | 3                             | 2             | 1        |                             |               |          |              |             |     |                             |               |          |              |
| D           | BOM Variants            |                                     |                                                           | Bar Code Labels / EEE #'s |                  |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
|             | BOM NUMBER              | BOM NAME                            | BOM OPTIONS                                               | PART NUMBER               | QTY              | DESCRIPTION                   | REFERENCE DES | CRITICAL | BOM OPTION                  |               |          |              |             |     |                             |               |          |              |
|             | 639-1070                | PCBA,MLB,1.86GHZ HY 2GB,MU CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCWQ,CAPS:MU,DDR3:HYNIX_2GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWM]   | CRITICAL | EEEE:DCWM                   |               |          |              |             |     |                             |               |          |              |
|             | 639-0837                | PCBA,MLB,1.86GHZ, HY 2GB,SS CAP,K16 | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXW,CAPS:SS,DDR3:HYNIX_2GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWN]   | CRITICAL | EEEE:DCWN                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1096                | PCBA,MLB,1.86GHZ HY 2GB,TY CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXN,CAPS:TY,DDR3:HYNIX_2GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWP]   | CRITICAL | EEEE:DCWP                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1101                | PCBA,MLB,1.86GHZ HY 4GB,MU CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXV,CAPS:MU,DDR3:HYNIX_4GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWQ]   | CRITICAL | EEEE:DCWQ                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1098                | PCBA,MLB,1.86GHZ HY 4GB,SS CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXQ,CAPS:SS,DDR3:HYNIX_4GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWR]   | CRITICAL | EEEE:DCWR                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1068                | PCBA,MLB,1.86GHZ HY 4GB,TY CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCWN,CAPS:TY,DDR3:HYNIX_4GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWT]   | CRITICAL | EEEE:DCWT                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1083                | PCBA,MLB,1.86GHZ MI 2GB,MU CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX6,CAPS:MU,DDR3:MICRON_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWV]   | CRITICAL | EEEE:DCWV                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1078                | PCBA,MLB,1.86GHZ MI 2GB,SS CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX3,CAPS:SS,DDR3:MICRON_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWW]   | CRITICAL | EEEE:DCWW                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1090                | PCBA,MLB,1.86GHZ MI 2GB,TY CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXG,CAPS:TY,DDR3:MICRON_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWX]   | CRITICAL | EEEE:DCWX                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1088                | PCBA,MLB,1.86GHZ MI 4GB,MU CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXD,CAPS:MU,DDR3:MICRON_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCWY]   | CRITICAL | EEEE:DCWY                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1067                | PCBA,MLB,1.86GHZ MI 4GB,SS CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCWM,CAPS:SS,DDR3:MICRON_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX0]   | CRITICAL | EEEE:DCX0                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1077                | PCBA,MLB,1.86GHZ MI 4GB,TY CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX0,CAPS:TY,DDR3:MICRON_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX1]   | CRITICAL | EEEE:DCX1                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1080                | PCBA,MLB,1.86GHZ SA 2GB,MU CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX3,CAPS:MU,DDR3:SAMSUNG_2GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX2]   | CRITICAL | EEEE:DCX2                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1095                | PCBA,MLB,1.86GHZ SA 2GB,SS CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXM,CAPS:SS,DDR3:SAMSUNG_2GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX3]   | CRITICAL | EEEE:DCX3                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1071                | PCBA,MLB,1.86GHZ SA 2GB,TY CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCWR,CAPS:TY,DDR3:SAMSUNG_2GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX4]   | CRITICAL | EEEE:DCX4                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1097                | PCBA,MLB,1.86GHZ SA 4GB,MU CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXP,CAPS:MU,DDR3:SAMSUNG_4GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX5]   | CRITICAL | EEEE:DCX5                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1084                | PCBA,MLB,1.86GHZ SA 4GB,SS CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCX7,CAPS:SS,DDR3:SAMSUNG_4GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX6]   | CRITICAL | EEEE:DCX6                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1091                | PCBA,MLB,1.86GHZ SA 4GB,TY CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DCXH,CAPS:TY,DDR3:SAMSUNG_4GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX7]   | CRITICAL | EEEE:DCX7                   |               |          |              |             |     |                             |               |          |              |
| C           | 639-1092                | PCBA,MLB,2.13GHZ HY 2GB,MU CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXJ,CAPS:MU,DDR3:HYNIX_2GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX8]   | CRITICAL | EEEE:DCX8                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1082                | PCBA,MLB,2.13GHZ, HY 2GB,SS CAP,K16 | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX5,CAPS:SS,DDR3:HYNIX_2GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCX9]   | CRITICAL | EEEE:DCX9                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1085                | PCBA,MLB,2.13GHZ HY 2GB,TY CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX8,CAPS:TY,DDR3:HYNIX_2GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXC]   | CRITICAL | EEEE:DCXC                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1089                | PCBA,MLB,2.13GHZ HY 4GB,MU CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXF,CAPS:MU,DDR3:HYNIX_4GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXD]   | CRITICAL | EEEE:DCXD                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1075                | PCBA,MLB,2.13GHZ HY 4GB,SS CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWX,CAPS:SS,DDR3:HYNIX_4GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXF]   | CRITICAL | EEEE:DCXF                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1079                | PCBA,MLB,2.13GHZ HY 4GB,TY CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX2,CAPS:TY,DDR3:HYNIX_4GB   | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXG]   | CRITICAL | EEEE:DCXG                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1099                | PCBA,MLB,2.13GHZ MI 2GB,MU CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXR,CAPS:MU,DDR3:MICRON_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXH]   | CRITICAL | EEEE:DCXH                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1087                | PCBA,MLB,2.13GHZ MI 2GB,SS CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXC,CAPS:SS,DDR3:MICRON_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXJ]   | CRITICAL | EEEE:DCXJ                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1069                | PCBA,MLB,2.13GHZ MI 2GB,TY CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWP,CAPS:TY,DDR3:MICRON_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXK]   | CRITICAL | EEEE:DCXK                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1100                | PCBA,MLB,2.13GHZ MI 4GB,MU CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXT,CAPS:MU,DDR3:MICRON_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXL]   | CRITICAL | EEEE:DCXL                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1093                | PCBA,MLB,2.13GHZ MI 4GB,SS CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXK,CAPS:SS,DDR3:MICRON_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXM]   | CRITICAL | EEEE:DCXM                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1076                | PCBA,MLB,2.13GHZ MI 4GB,TY CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWY,CAPS:TY,DDR3:MICRON_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXN]   | CRITICAL | EEEE:DCXN                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1074                | PCBA,MLB,2.13GHZ SA 2GB,MU CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWW,CAPS:MU,DDR3:SAMSUNG_2GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXP]   | CRITICAL | EEEE:DCXP                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1072                | PCBA,MLB,2.13GHZ SA 2GB,SS CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWT,CAPS:SS,DDR3:SAMSUNG_2GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXQ]   | CRITICAL | EEEE:DCXQ                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1086                | PCBA,MLB,2.13GHZ SA 2GB,TY CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX9,CAPS:TY,DDR3:SAMSUNG_2GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXR]   | CRITICAL | EEEE:DCXR                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1073                | PCBA,MLB,2.13GHZ SA 4GB,MU CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCWV,CAPS:MU,DDR3:SAMSUNG_4GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXT]   | CRITICAL | EEEE:DCXT                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1081                | PCBA,MLB,2.13GHZ SA 4GB,SS CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCX4,CAPS:SS,DDR3:SAMSUNG_4GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXV]   | CRITICAL | EEEE:DCXV                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1094                | PCBA,MLB,2.13GHZ SA 4GB,TY CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DCXL,CAPS:TY,DDR3:SAMSUNG_4GB | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DCXW]   | CRITICAL | EEEE:DCXW                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1450                | PCBA,MLB,1.86GHZ EL 2GB,MU CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DG4W,CAPS:MU,DDR3:ELPIDA_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG4W]   | CRITICAL | EEEE:DG4W                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1451                | PCBA,MLB,1.86GHZ EL 2GB,SS CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DG4Y,CAPS:SS,DDR3:ELPIDA_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG4Y]   | CRITICAL | EEEE:DG4Y                   |               |          |              |             |     |                             |               |          |              |
| B           | 639-1455                | PCBA,MLB,1.86GHZ EL 2GB,TY CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DG53,CAPS:TY,DDR3:ELPIDA_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG53]   | CRITICAL | EEEE:DG53                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1453                | PCBA,MLB,2.13GHZ EL 2GB,MU CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DG51,CAPS:MU,DDR3:ELPIDA_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG51]   | CRITICAL | EEEE:DG51                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1454                | PCBA,MLB,2.13GHZ EL 2GB,SS CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DG52,CAPS:SS,DDR3:ELPIDA_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG52]   | CRITICAL | EEEE:DG52                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1452                | PCBA,MLB,2.13GHZ EL 2GB,TY CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DG50,CAPS:TY,DDR3:ELPIDA_2GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG50]   | CRITICAL | EEEE:DG50                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1458                | PCBA,MLB,1.86GHZ EL 4GB,MU CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DG5P,CAPS:MU,DDR3:ELPIDA_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG5P]   | CRITICAL | EEEE:DG5P                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1463                | PCBA,MLB,1.86GHZ EL 4GB,SS CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DG5W,CAPS:SS,DDR3:ELPIDA_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG5W]   | CRITICAL | EEEE:DG5W                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1460                | PCBA,MLB,1.86GHZ EL 4GB,TY CAP,K16  | K16_CMNPTS,CPU:1.86GHZ,EEEE:DG5T,CAPS:TY,DDR3:ELPIDA_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG5T]   | CRITICAL | EEEE:DG5T                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1462                | PCBA,MLB,2.13GHZ EL 4GB,MU CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DG5Q,CAPS:MU,DDR3:ELPIDA_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG5Q]   | CRITICAL | EEEE:DG5Q                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1459                | PCBA,MLB,2.13GHZ EL 4GB,SS CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DG5V,CAPS:SS,DDR3:ELPIDA_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG5V]   | CRITICAL | EEEE:DG5V                   |               |          |              |             |     |                             |               |          |              |
|             | 639-1461                | PCBA,MLB,2.13GHZ EL 4GB,TY CAP,K16  | K16_CMNPTS,CPU:2.13GHZ,EEEE:DG5R,CAPS:TY,DDR3:ELPIDA_4GB  | 825-7557                  | 1                | LBL,P/N LABEL,PCB,28MM X 6 MM | [EEEE_DG5R]   | CRITICAL | EEEE:DG5R                   |               |          |              |             |     |                             |               |          |              |
|             | 607-6915                | CMN PTS,PCBA,MLB,K16                | K16_COMMON                                                |                           |                  |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
|             | 085-1327                | K16 MLB DEVELOPMENT BOM             | K16_DEVEL :ENG                                            |                           |                  |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
|             | SAMSUNG                 |                                     |                                                           | MURATA                    |                  |                               | TAIYO YUDEN   |          |                             |               |          |              |             |     |                             |               |          |              |
|             | PART NUMBER             | QTY                                 | DESCRIPTION                                               | REFERENCE DES             | CRITICAL         | BOM OPTION                    | PART NUMBER   | QTY      | DESCRIPTION                 | REFERENCE DES | CRITICAL | BOM OPTION   | PART NUMBER | QTY | DESCRIPTION                 | REFERENCE DES | CRITICAL | BOM OPTION   |
|             | 138S0632                | 1                                   | CAP, 2.2UF, 6.3V, 20%, 0402                               | C4807                     | CRITICAL         | SS_CAP_2_2UF                  | 138S0633      | 1        | CAP, 2.2UF, 6.3V, 20%, 0402 | C4807         | CRITICAL | MU_CAP_2_2UF | 138S0634    | 1   | CAP, 2.2UF, 6.3V, 20%, 0402 | C4807         | CRITICAL | TY_CAP_2_2UF |
|             | K16-Specific BOM Tables |                                     |                                                           |                           |                  |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
|             | PART NUMBER             | QTY                                 | DESCRIPTION                                               | REFERENCE DES             | CRITICAL         | BOM OPTION                    |               |          |                             |               |          |              |             |     |                             |               |          |              |
|             | 337S3751                | 1                                   | PDC,SLGAB,PRQ,1.86,17W,1066,ED,6M,BGA                     | U1000                     | CRITICAL         | CPU:1.86GHZ                   |               |          |                             |               |          |              |             |     |                             |               |          |              |
|             | 337S3758                | 1                                   | PDC,SLGEQ,PRQ,2.13,17W,1066,ED,6M,BGA                     | U1000                     | CRITICAL         | CPU:2.13GHZ                   |               |          |                             |               |          |              |             |     |                             |               |          |              |
|             | 341T0276                | 1                                   | IC ASSY,SMC EXTERNAL,K16                                  | U4900                     | CRITICAL         | SMC:PROG                      |               |          |                             |               |          |              |             |     |                             |               |          |              |
| 341T0275    | 1                       | IC ASSY,EFI UNLOCKED,K16            | U6100                                                     | CRITICAL                  | BOOTROM:UNLOCKED |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
| 341S2785    | 1                       | IC EFI ROM,PVT,LOCKED,K16           | U6100                                                     | CRITICAL                  | BOOTROM:LOCKED   |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
| Sub-BOMs    |                         |                                     |                                                           |                           |                  |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
| PART NUMBER | QTY                     | DESCRIPTION                         | REFERENCE DES                                             | CRITICAL                  | BOM OPTION       |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
| 085-1327    | 1                       | K16 MLB DEVELOPMENT BOM             | DEVEL                                                     | CRITICAL                  | DEVEL_BOM        |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
| 607-6915    | 1                       | CMN PTS,PCBA,MLB,K16                | CMNPTS                                                    | CRITICAL                  | K16_CMNPTS       |                               |               |          |                             |               |          |              |             |     |                             |               |          |              |
| 8           | 7                       | 6                                   | 5                                                         | 4                         | 3                | 2                             | 1             |          |                             |               |          |              |             |     |                             |               |          |              |

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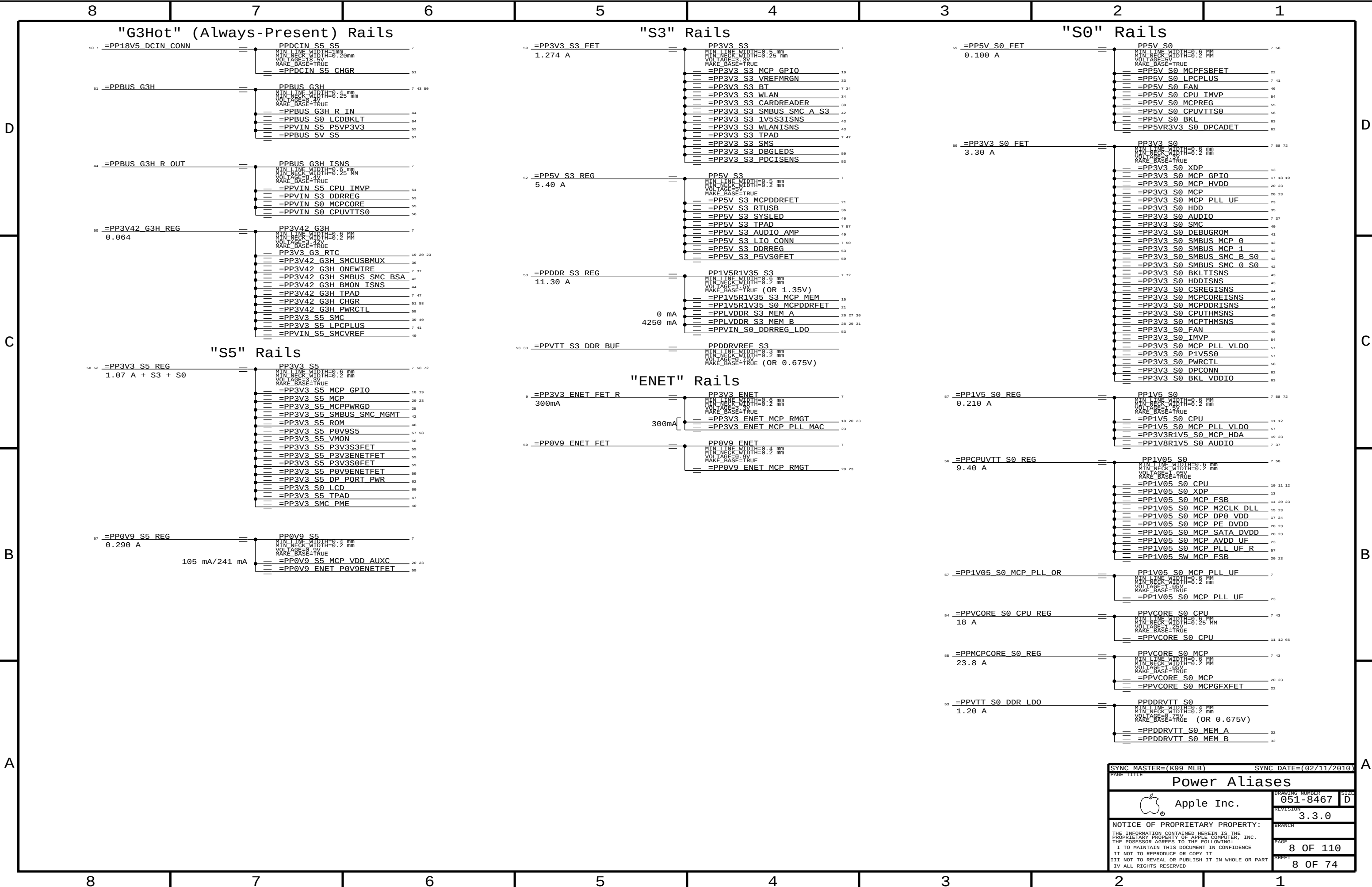
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| D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div>Revision History</div> <div>Proto 0 (ECO #0000876215, v1.0.0, P4 change #210266, 03/16/2010)</div> <div><div>v1.1.0 (P4 change #211399, 03/24/2010)</div><div>MCP:<br/>7742015 - Added RC to DDC pass FETs to avoid glitch (pp. 7, 93).<br/>7788138 - Added feedback divider and BOM tables for more HVDD LD0s (pp. 4, 25).<br/>SMC:<br/>7761747 - Added resistors to connect TCON to SMC or MCP SMBus (pp. 4, 52, 90).<br/>7787883 - Added support for DP HPD wake / S4 state (pp. 4, 7, 8, 19, 49, 50, 78, 94).<br/>SMS:<br/>7765466 - Added S3 pull-up to SMS_INT_L to prevent leakage path (pp. 50, 59).<br/>7769139 - Unstuffed SMS circuit (pg. 4).<br/>General:<br/>7787897 - Property/page fixes to reduce CheckPlus warnings/errors (pp. 7, 8, 12, 17, 74, 93, 108).</div><div>v1.2.0 (P4 change #211839, 03/26/2010)</div><div>USB:<br/>7796626 - Changed port switch from TPS2052B to TPS2069 (pg. 46).<br/>SMC:<br/>7787883 - Added PLACE_NEAR property on R5022 to avoid stub (pg. 50).<br/>SMBus:<br/>7761747 - Added TCON I2C nets to FUNC_TEST list for J9000 (pg. 7).<br/>Power:<br/>7796648 - Changed DP and LCD power from PP3V3_S3 to PP3V3_S5 (pp. 8, 90).<br/>7796658 - Changed backlight driver to E00 version (pg. 97).<br/>BOM:<br/>7796661 - Set up primary &amp; alternate for power supply FET (pp. 4, 72).<br/>7796654 - Consolidated SSM6N15FE to SSM6N37FE (pg. 48).<br/>7796658 - Changed RCs on some SMC analog inputs (pg. 54).<br/>7796683 - Stuffed RC on backlight driver PWM input (pg. 97).<br/>General:<br/>7796631 - Sorted BOM variants for easier verification (pg. 5).<br/>7796631 - Cosmetic clean-up (pg. 76).</div><div>v1.3.0 (P4 change #212050, 03/26/2010)</div><div>SMBus:<br/>7761747 - Added isolation FET and unstuffed series R's on TCON I2C for now (pp. 4, 90, 108).<br/>Power Supply:<br/>7796661 - Removed alternate FET, made some FETs primary to other APN (pp. 4, 72, 73, 76).<br/>7798425 - R/C value changes for 3.42V G3Hot power supply (pg. 69).<br/>7798399 - R/C value changes for 5V/3.3V power supply (pg. 72).<br/>7800179 - R value changes for CPU VCore power supply (pg. 74).<br/>7798445 - R value changes for 0.9V S5 power supply (pg. 77).<br/>7796658 - Changed backlight driver back to non-E00 version (pp. 4, 97).<br/>BOM:<br/>7796658 - Added alternates for two caps per GSM and removed unused alternates (pg. 4).<br/>7798399 - Consolidated 100pF caps (pp. 74, 75).</div><div>v1.4.0 (P4 change #212757, 03/31/2010)</div><div>MCP SPI:<br/>7809733 - Changed strapping to select 62.5MHz SPI bus frequency (pg. 4).<br/>SMBus:<br/>7796631 - Added XDP connection to SMBus aliases page (pp. 13, 52).<br/>7808530 - Changed SMC 'MGMT' SMBus pull-ups from 4.7K to 2K (pg. 52).<br/>7761747 - Documented SMBus addresses for panel (pg. 52).<br/>SD Card:<br/>7800415 - Changed SD Card discharge R to more standard value (pg. 48).<br/>Power Supplies:<br/>7803283 - Changed 5V S3 regulator output from 5.02V to 5.12V nominal (pg. 72).<br/>7809760 - Stuffed C9799 and clarified tables/BOMOPTIONS around these parts (pg. 97).</div><div>Proto 1 (ECO #0000884508, v2.0.0, P4 change #212783, 03/31/2010)</div><div><div>v2.1.0 (P4 change #??????, ??/??/2010)</div><div>BOM:<br/>7796658 - Changed OMITs to OMIT_TABLES (pp. 10-11, 14-20, 26, 31-36, 49, 61).</div></div></div> |   |   |   |   |   |   |   | D |
| C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |   |   |   |   | C |   |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |   |   |   |   | B |   |
| A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |   |   |   |   | A |   |
| <div>NOTE: All page numbers are .csa, not PDF. See page 1 for .csa -&gt; PDF mapping.</div> <div><div><div>SYNC MASTER=N/A</div><div>SYNC DATE=N/A</div><div>Revision History</div><div><div> Apple Inc.</div><div>DRAWING NUMBER<br/>051-8467</div><div>SIZE<br/>D</div></div><div><div>NOTICE OF PROPRIETARY PROPERTY:<br/>THE INFORMATION CONTAINED HEREIN IS THE<br/>PROPRIETARY PROPERTY OF APPLE COMPUTER, INC.<br/>THE POSSESSOR AGREES TO THE FOLLOWING:<br/>I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE<br/>II NOT TO REPRODUCE OR COPY IT<br/>III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART<br/>IV ALL RIGHTS RESERVED</div><div>BRANCH</div><div>PAGE<br/>6 OF 110</div><div>SHEET<br/>6 OF 74</div></div></div></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |   |   |   |   |   |   |
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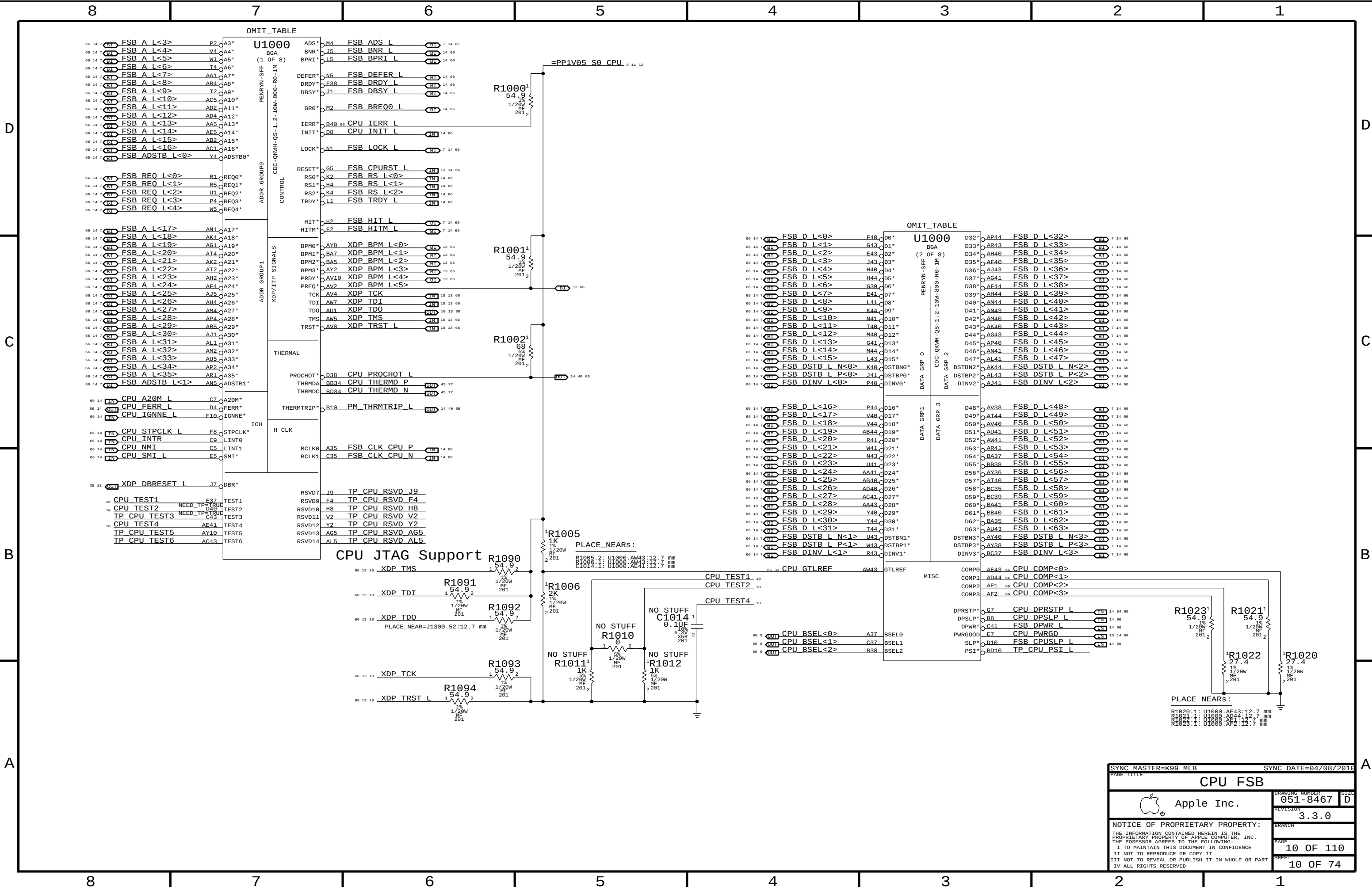


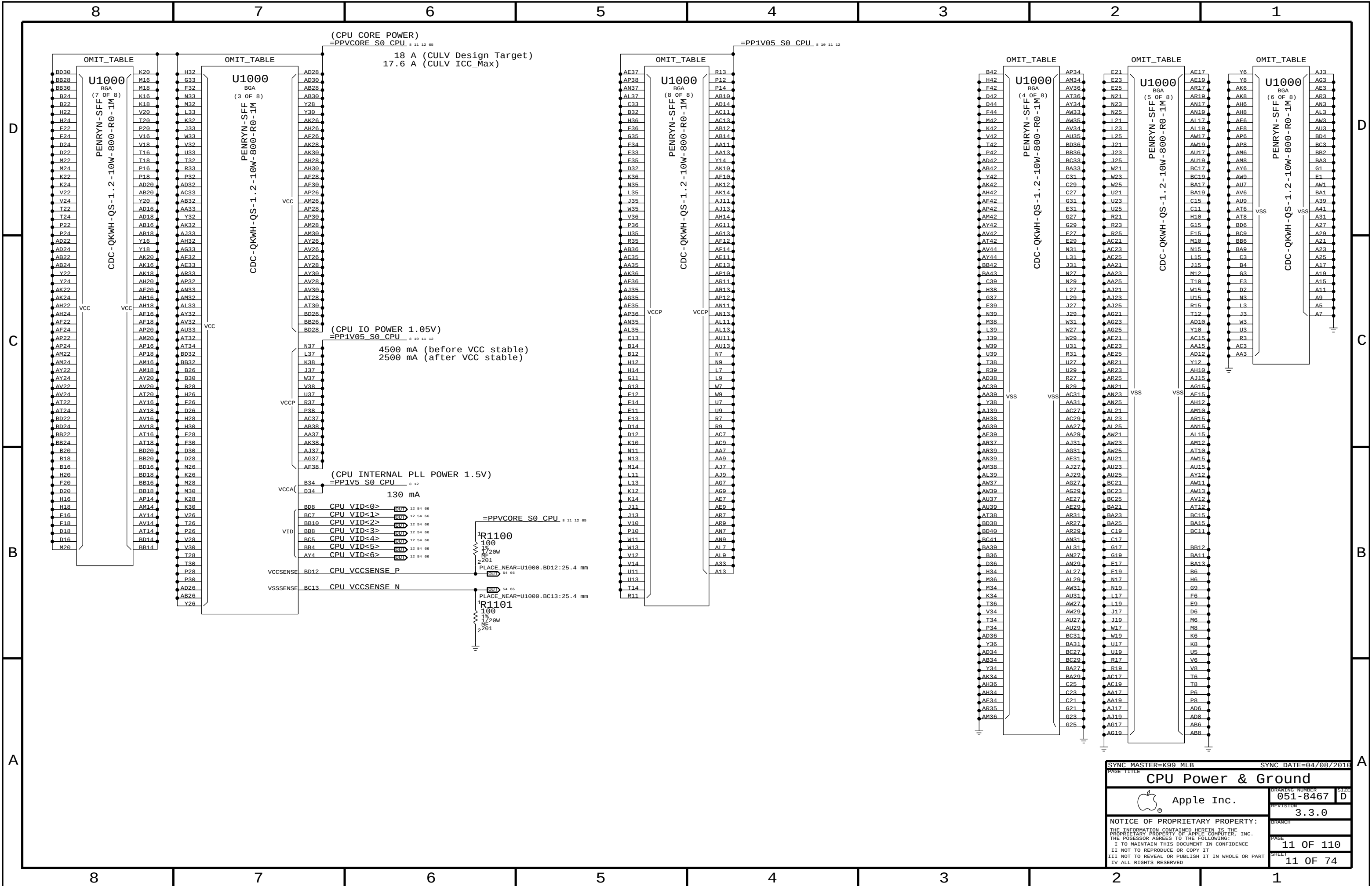
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|--|-----------------------------------------------------|---|---|---|---|----------------------------------------|---|---|
|  | 8                                                   | 7 | 6 | 5 | 4 | 3                                      | 2 | 1 |
|  | Functional Test Points                              |   |   |   |   | NO_TEST Nets                           |   |   |
|  | J4001: AirPort / BT Connector                       |   |   |   |   | FSB Signals (Covered via CPU/MCP JTAG) |   |   |
|  | FUNC_TEST                                           |   |   |   |   | NO_TEST                                |   |   |
|  | TRUE PP3V3 WLAN_F 7 34 40                           |   |   |   |   | TRUE FSB A_L<35..3> 10 14 66           |   |   |
|  | TRUE WIFI_EVENT_L 34 39 40                          |   |   |   |   | TRUE FSB ADS_L 10 14 66                |   |   |
|  | TRUE PCIE AP R2D_N 34 68                            |   |   |   |   | TRUE FSB ADSTB_L<1..0> 10 14 66        |   |   |
|  | TRUE PCIE AP R2D_P 34 68                            |   |   |   |   | TRUE FSB D_L<63..0> 10 14 66           |   |   |
|  | TRUE PCIE CLK100M AP_N 10 34 68                     |   |   |   |   | TRUE FSB DINV_L<3..0> 10 14 66         |   |   |
|  | TRUE PCIE CLK100M AP_P 10 34 68                     |   |   |   |   | TRUE FSB DSTB_L_N<3..0> 10 14 66       |   |   |
|  | TRUE USB_BT_P 10 34 69                              |   |   |   |   | TRUE FSB DSTB_L_P<3..0> 10 14 66       |   |   |
|  | TRUE USB_BT_N 10 34 69                              |   |   |   |   | TRUE FSB HIT_L 10 14 66                |   |   |
|  | TRUE PCIE AP D2R_P 10 34 68                         |   |   |   |   | TRUE FSB HITM_L 10 14 66               |   |   |
|  | TRUE PCIE AP D2R_N 10 34 68                         |   |   |   |   | TRUE FSB LOCK_L 10 14 66               |   |   |
|  | TRUE PCIE_WAKE_L 10 34                              |   |   |   |   | TRUE FSB_REQ_L<4..0> 10 14 66          |   |   |
|  | TRUE AP_RESET_CONN_L 34                             |   |   |   |   |                                        |   |   |
|  | TRUE AP_CLKREQ_0_L 34                               |   |   |   |   |                                        |   |   |
|  | TRUE =PP3V3_S3_BT 8 34                              |   |   |   |   |                                        |   |   |
|  | (Need to add 6 GND TPs)                             |   |   |   |   |                                        |   |   |
|  | J4501: SATA SSD Connector                           |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE PP3V3_S0_HDD_R 7 35                            |   |   |   |   |                                        |   |   |
|  | TRUE SATA_HDD_D2R_C_P 35 68                         |   |   |   |   |                                        |   |   |
|  | TRUE SATA_HDD_D2R_C_N 35 68                         |   |   |   |   |                                        |   |   |
|  | TRUE SATA_HDD_R2D_N 35 68                           |   |   |   |   |                                        |   |   |
|  | TRUE SATA_HDD_R2D_P 35 68                           |   |   |   |   |                                        |   |   |
|  | TRUE SMC_HDD_Q0B_TEMP 35 39                         |   |   |   |   |                                        |   |   |
|  | TRUE SMC_HDD_TEMP_CTL 35 39                         |   |   |   |   |                                        |   |   |
|  | (Need to add 6 GND TPs)                             |   |   |   |   |                                        |   |   |
|  | J4700: LIO Connector                                |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE =PP3V42_G3H_ONEWIRE 8 37                       |   |   |   |   |                                        |   |   |
|  | TRUE =PP3V3_S0_AUDIO 8 37                           |   |   |   |   |                                        |   |   |
|  | TRUE =PP1V8R1V5_S0_AUDIO 8 37                       |   |   |   |   |                                        |   |   |
|  | TRUE SYS_ONEWIRE 37 39                              |   |   |   |   |                                        |   |   |
|  | TRUE SMC_BC_ACOK 9 37 39 48                         |   |   |   |   |                                        |   |   |
|  | TRUE =USB_PWR_EN 36 37 58                           |   |   |   |   |                                        |   |   |
|  | TRUE SMC_LID 7 37 39 48 47                          |   |   |   |   |                                        |   |   |
|  | TRUE =I2C_LIO_SDA 37 42                             |   |   |   |   |                                        |   |   |
|  | TRUE =I2C_LIO_SCL 37 42                             |   |   |   |   |                                        |   |   |
|  | TRUE =I2C_MIKEY_SCL 37 42                           |   |   |   |   |                                        |   |   |
|  | TRUE =I2C_MIKEY_SDA 37 42                           |   |   |   |   |                                        |   |   |
|  | TRUE AUD_IPHS_SWITCH_EN 19 37                       |   |   |   |   |                                        |   |   |
|  | TRUE AUD_IP_PERIPHERAL_DET 17 37                    |   |   |   |   |                                        |   |   |
|  | TRUE AUD_I2C_INT_L 19 37                            |   |   |   |   |                                        |   |   |
|  | TRUE AUD_GPIO_3 37 48                               |   |   |   |   |                                        |   |   |
|  | TRUE SPKRAMP_INR_N 37 49 72                         |   |   |   |   |                                        |   |   |
|  | TRUE SPKRAMP_INR_P 37 49 72                         |   |   |   |   |                                        |   |   |
|  | TRUE USB_EXTD_N 18 37 69                            |   |   |   |   |                                        |   |   |
|  | TRUE USB_EXTD_P 18 37 69                            |   |   |   |   |                                        |   |   |
|  | TRUE USB_CAMERA_N 18 37 69                          |   |   |   |   |                                        |   |   |
|  | TRUE USB_CAMERA_P 18 37 69                          |   |   |   |   |                                        |   |   |
|  | TRUE HDA_SDOUT 19 37 69                             |   |   |   |   |                                        |   |   |
|  | TRUE HDA_BIT_CLK 19 37 69                           |   |   |   |   |                                        |   |   |
|  | TRUE HDA_SDIN0 19 37 69                             |   |   |   |   |                                        |   |   |
|  | TRUE USB_EXTD_OC_L 19 37                            |   |   |   |   |                                        |   |   |
|  | TRUE HDA_RST_L 19 37 69                             |   |   |   |   |                                        |   |   |
|  | TRUE HDA_SYNC 19 37 69                              |   |   |   |   |                                        |   |   |
|  | (Need to add 5 GND TPs)                             |   |   |   |   |                                        |   |   |
|  | J4800: SD Card Connector                            |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE PP3V3_SW_SD_PWR 38                             |   |   |   |   |                                        |   |   |
|  | TRUE SD_CLK 38 70                                   |   |   |   |   |                                        |   |   |
|  | TRUE SD_CMD 38 70                                   |   |   |   |   |                                        |   |   |
|  | TRUE SD_D<7..0> 38 70                               |   |   |   |   |                                        |   |   |
|  | TRUE SD_CD_L 38                                     |   |   |   |   |                                        |   |   |
|  | TRUE SD_WP 38                                       |   |   |   |   |                                        |   |   |
|  | (Need to add 2 GND TPs)                             |   |   |   |   |                                        |   |   |
|  | J5100: LPC+SPI Connector                            |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE =PP3V3_S5_LPCPLUS 8 41                         |   |   |   |   |                                        |   |   |
|  | TRUE =PP5V_S0_LPCPLUS 8 41                          |   |   |   |   |                                        |   |   |
|  | TRUE LPC_AD<3..0> 19 39 41 69                       |   |   |   |   |                                        |   |   |
|  | TRUE SPI_ALT_MOSI 41 69                             |   |   |   |   |                                        |   |   |
|  | TRUE SPI_ALT_MISO 41 69                             |   |   |   |   |                                        |   |   |
|  | TRUE LPC_FRAME_L 19 39 41 69                        |   |   |   |   |                                        |   |   |
|  | TRUE PM_CLKRUN_L 19 39 41                           |   |   |   |   |                                        |   |   |
|  | TRUE SMC_TMS 39 48 41                               |   |   |   |   |                                        |   |   |
|  | TRUE LPCPLUS_RESET_L 25 41                          |   |   |   |   |                                        |   |   |
|  | TRUE SMC_TDO 39 48 41                               |   |   |   |   |                                        |   |   |
|  | TRUE SMC_TRST_L 39 41                               |   |   |   |   |                                        |   |   |
|  | TRUE SMC_MD1 39 41                                  |   |   |   |   |                                        |   |   |
|  | TRUE SMC_TX_L 36 39 48 41                           |   |   |   |   |                                        |   |   |
|  | TRUE LPC_CLK33M_LPCPLUS 25 41 69                    |   |   |   |   |                                        |   |   |
|  | TRUE SPIROM_USE_MLB 19 41 48                        |   |   |   |   |                                        |   |   |
|  | TRUE SPI_ALT_CLK 41 69                              |   |   |   |   |                                        |   |   |
|  | TRUE SPI_ALT_CS_L 41 69                             |   |   |   |   |                                        |   |   |
|  | TRUE LPC_SERIRQ 19 39 41                            |   |   |   |   |                                        |   |   |
|  | TRUE LPC_PWRDWN_L 19 39 41                          |   |   |   |   |                                        |   |   |
|  | TRUE SMC_TDI 39 48 41                               |   |   |   |   |                                        |   |   |
|  | TRUE SMC_TCK 39 48 41                               |   |   |   |   |                                        |   |   |
|  | TRUE SMC_RESET_L 39 48 41 51                        |   |   |   |   |                                        |   |   |
|  | TRUE SMC_NMI 39 41                                  |   |   |   |   |                                        |   |   |
|  | TRUE SMC_RX_L 36 39 48 41                           |   |   |   |   |                                        |   |   |
|  | TRUE LPCPLUS_GPIO 19 41                             |   |   |   |   |                                        |   |   |
|  | (Need to add 6 GND TPs)                             |   |   |   |   |                                        |   |   |
|  | J5600: Fan Connector                                |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE PP5V_S0 7 8 58                                 |   |   |   |   |                                        |   |   |
|  | TRUE FAN_RT_TACH 46                                 |   |   |   |   |                                        |   |   |
|  | TRUE FAN_RT_PWM 46                                  |   |   |   |   |                                        |   |   |
|  | (Need to add 1 GND TP)                              |   |   |   |   |                                        |   |   |
|  | J5700: IPD Flex Connector                           |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE =PP5V_S3_TPAD 8 57                             |   |   |   |   |                                        |   |   |
|  | TRUE =PP3V42_G3H_TPAD 8 47                          |   |   |   |   |                                        |   |   |
|  | TRUE =PP3V3_S3_TPAD 8 47                            |   |   |   |   |                                        |   |   |
|  | TRUE USB_TPAD_CONN_P 47 72                          |   |   |   |   |                                        |   |   |
|  | TRUE USB_TPAD_CONN_N 47 72                          |   |   |   |   |                                        |   |   |
|  | TRUE =I2C_TPAD_SDA 42 47                            |   |   |   |   |                                        |   |   |
|  | TRUE =I2C_TPAD_SCL 42 47                            |   |   |   |   |                                        |   |   |
|  | TRUE SMC_ONOFF_L 39 48 47                           |   |   |   |   |                                        |   |   |
|  | TRUE SMC_LID 7 37 39 48 47                          |   |   |   |   |                                        |   |   |
|  | TRUE SMC_TPAD_RST_L 48 47                           |   |   |   |   |                                        |   |   |
|  | (Need to add 5 GND TPs)                             |   |   |   |   |                                        |   |   |
|  | J6900: DC-In Connector                              |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE =PP18V5_DCIN_CONN 8 58                         |   |   |   |   |                                        |   |   |
|  | TRUE =PP5V_S3_LIO_CONN 8 58                         |   |   |   |   |                                        |   |   |
|  | (Need to add 6 GND TPs)                             |   |   |   |   |                                        |   |   |
|  | J6903: Speaker Connector                            |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE SPKRAMP_R_P_OUT 49 58                          |   |   |   |   |                                        |   |   |
|  | TRUE SPKRAMP_R_N_OUT 49 58                          |   |   |   |   |                                        |   |   |
|  | J6950: Battery Connector                            |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE PPVBAT_G3H_CONN 59 51                          |   |   |   |   |                                        |   |   |
|  | TRUE SMBUS_SMC_BSA_SCL 42 71                        |   |   |   |   |                                        |   |   |
|  | TRUE SMBUS_SMC_BSA_SDA 42 71                        |   |   |   |   |                                        |   |   |
|  | TRUE SYS_DETECT_L 58                                |   |   |   |   |                                        |   |   |
|  | (Need to add 4 GND TPs near J6950 and 1 for shield) |   |   |   |   |                                        |   |   |
|  | J9000: Internal DP Connector                        |   |   |   |   |                                        |   |   |
|  | FUNC_TEST                                           |   |   |   |   |                                        |   |   |
|  | TRUE PPVOUT_SW_LCDBKLT 7 43 68 63                   |   |   |   |   |                                        |   |   |
|  | TRUE PP3V3_SW_LCD 68                                |   |   |   |   |                                        |   |   |
|  | TRUE =I2C_TCON_SDA 42 68                            |   |   |   |   |                                        |   |   |
|  | TRUE LED_RETURN_6 68 63                             |   |   |   |   |                                        |   |   |
|  | TRUE LED_RETURN_5 68 63                             |   |   |   |   |                                        |   |   |
|  | TRUE LED_RETURN_4 68 63                             |   |   |   |   |                                        |   |   |
|  | TRUE LED_RETURN_3 68 63                             |   |   |   |   |                                        |   |   |
|  | TRUE LED_RETURN_2 68 63                             |   |   |   |   |                                        |   |   |
|  | TRUE LED_RETURN_1 68 63                             |   |   |   |   |                                        |   |   |
|  | TRUE DP_INT_HPD_CONN 68                             |   |   |   |   |                                        |   |   |
|  | TRUE DP_INT_AUX_CH_C_N 68 72                        |   |   |   |   |                                        |   |   |
|  | TRUE DP_INT_AUX_CH_C_P 68 72                        |   |   |   |   |                                        |   |   |
|  | TRUE DP_INT_ML_F_P<0> 68 72                         |   |   |   |   |                                        |   |   |
|  | TRUE DP_INT_ML_F_N<0> 68 72                         |   |   |   |   |                                        |   |   |
|  | TRUE DP_INT_ML_F_P<1> 68 72                         |   |   |   |   |                                        |   |   |
|  | TRUE DP_INT_ML_F_N<1> 68 72                         |   |   |   |   |                                        |   |   |
|  | TRUE =I2C_TCON_SCL 42 68                            |   |   |   |   |                                        |   |   |



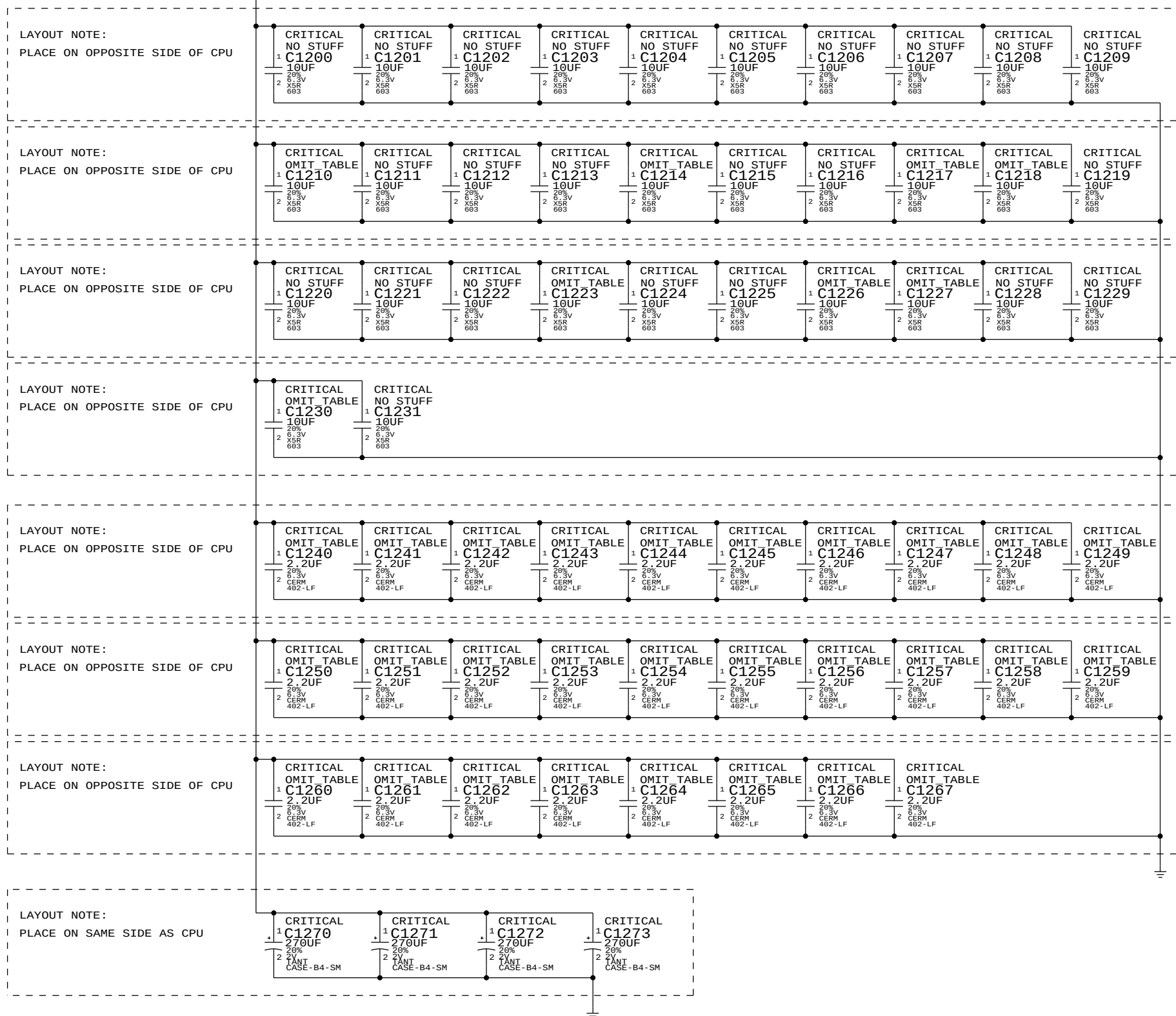




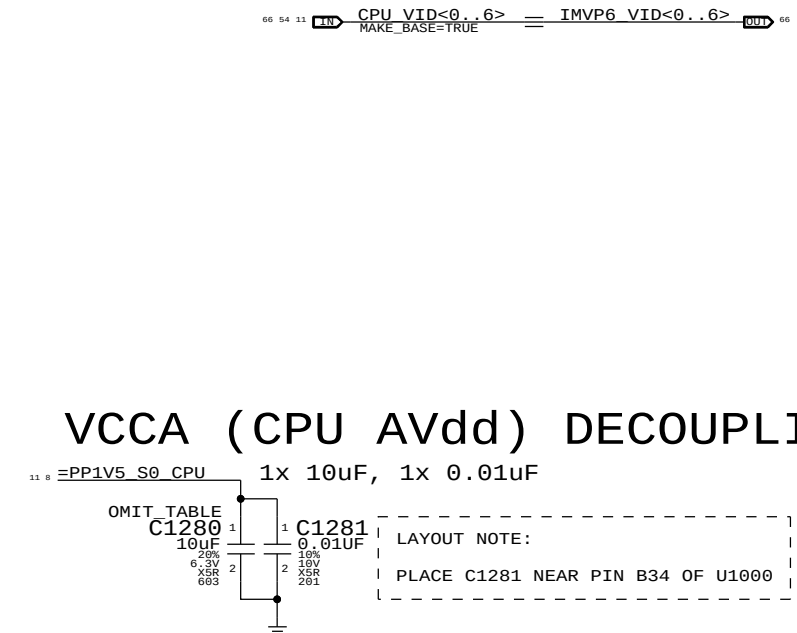




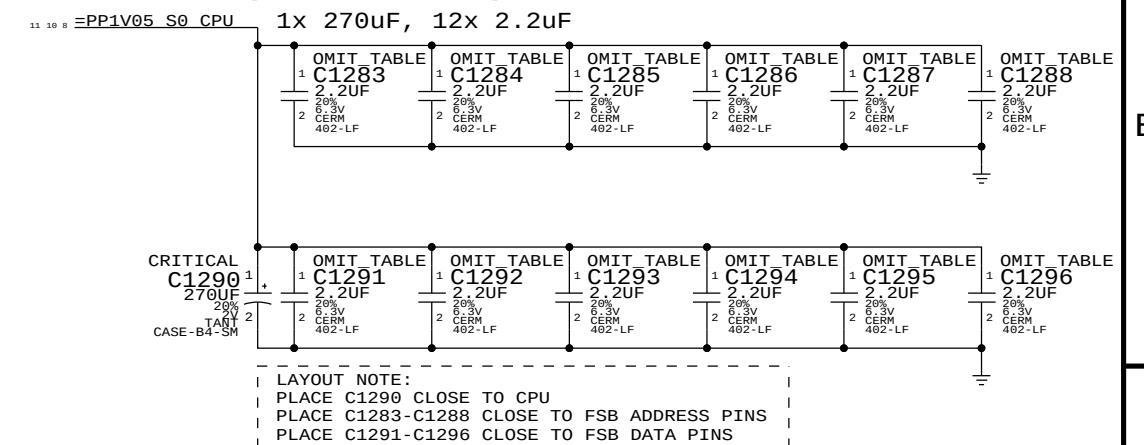
4x 270uF. 32x 10uF 0603, 28x 2.2uF 0402 + 40x 2.2uF 0402



## CPU VCORE VID CONNECTIONS



## VCCP (CPU I/O) DECOUPLING



## D

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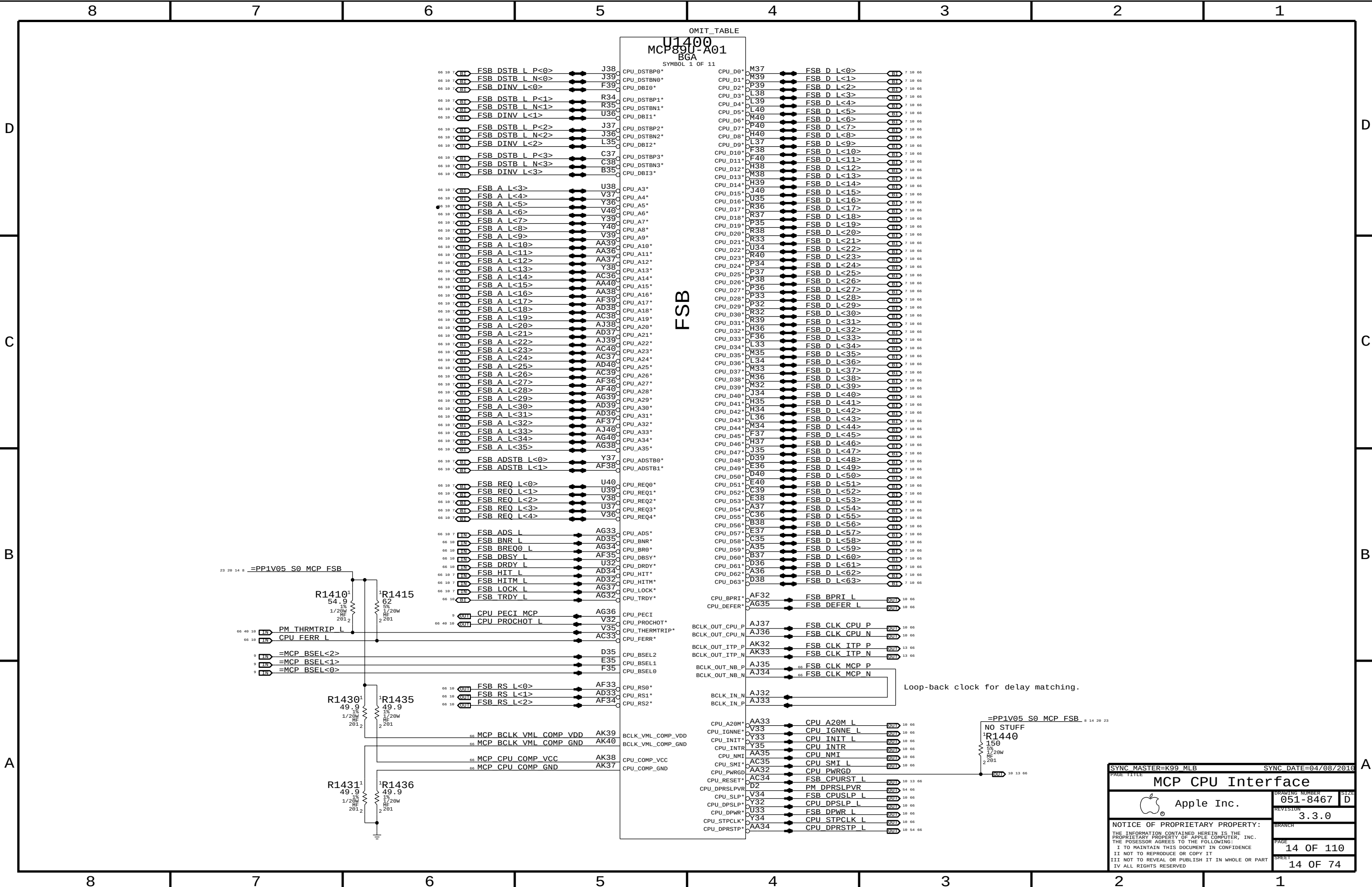
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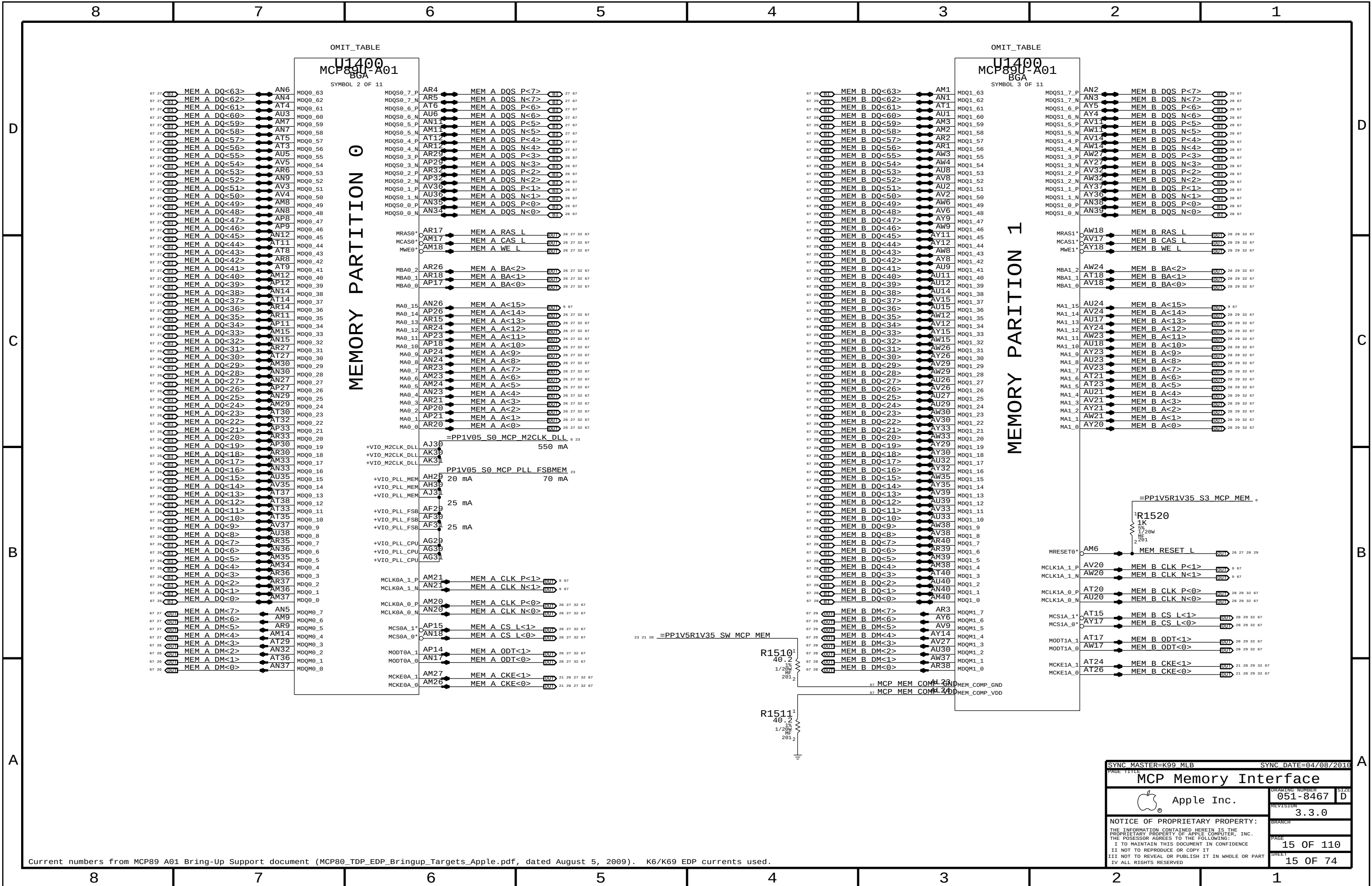
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| PAGE TITLE                                                                                                                  |  | DRAWING NUMBER       |           |
| MCP CPU Interface                                                                                                           |  | 051-8467             | D         |
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MCP Memory Interface

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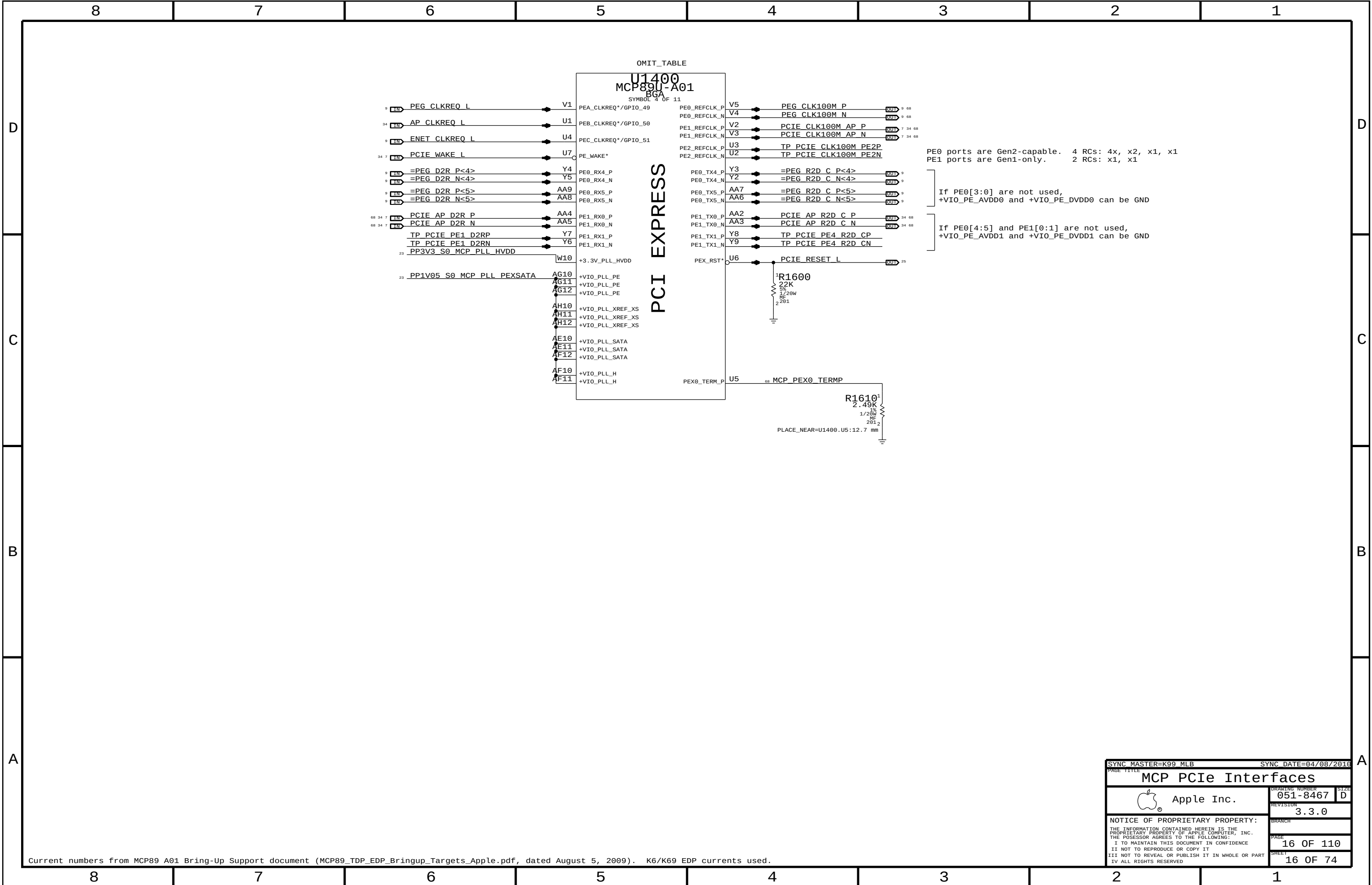
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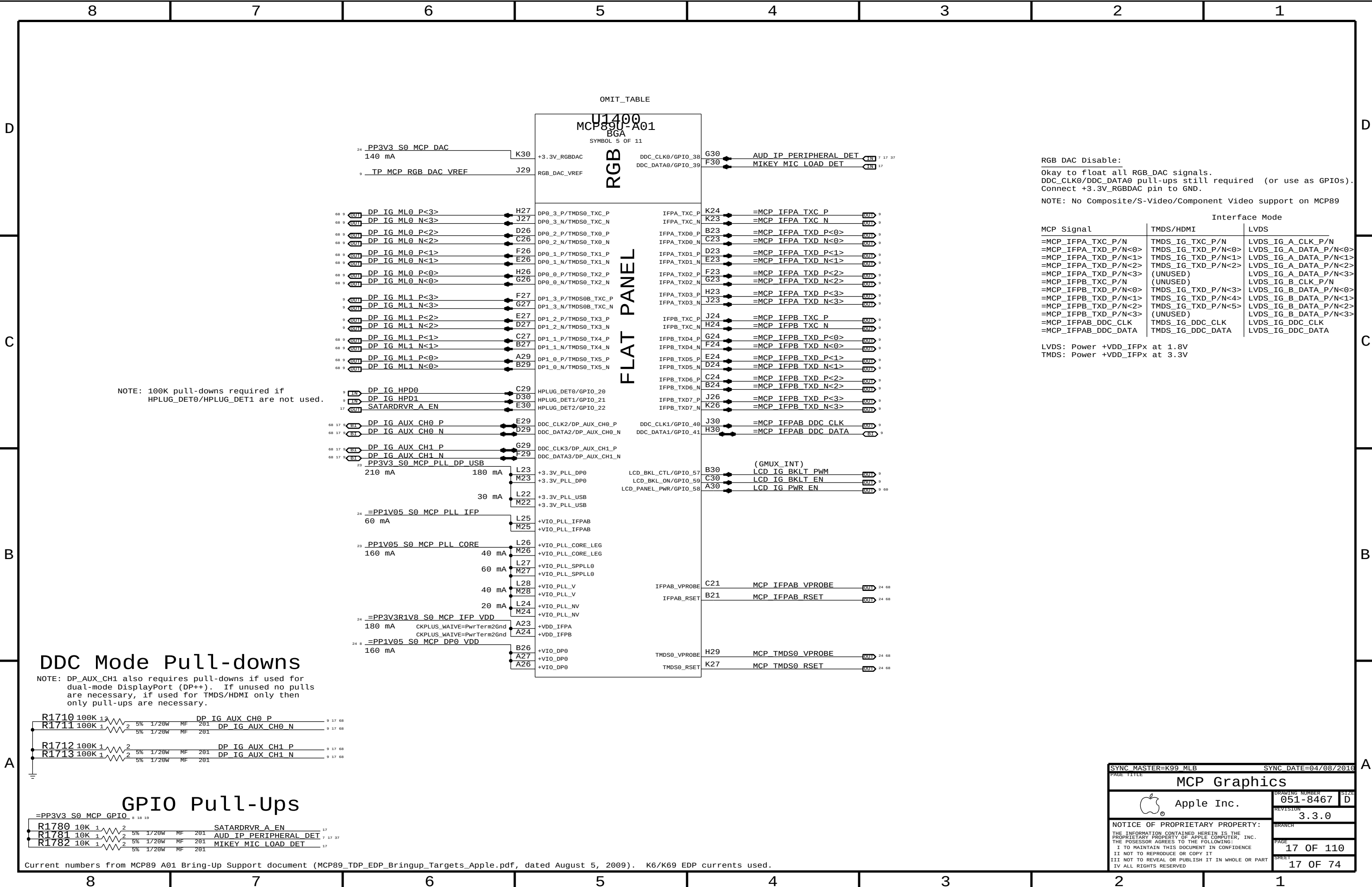
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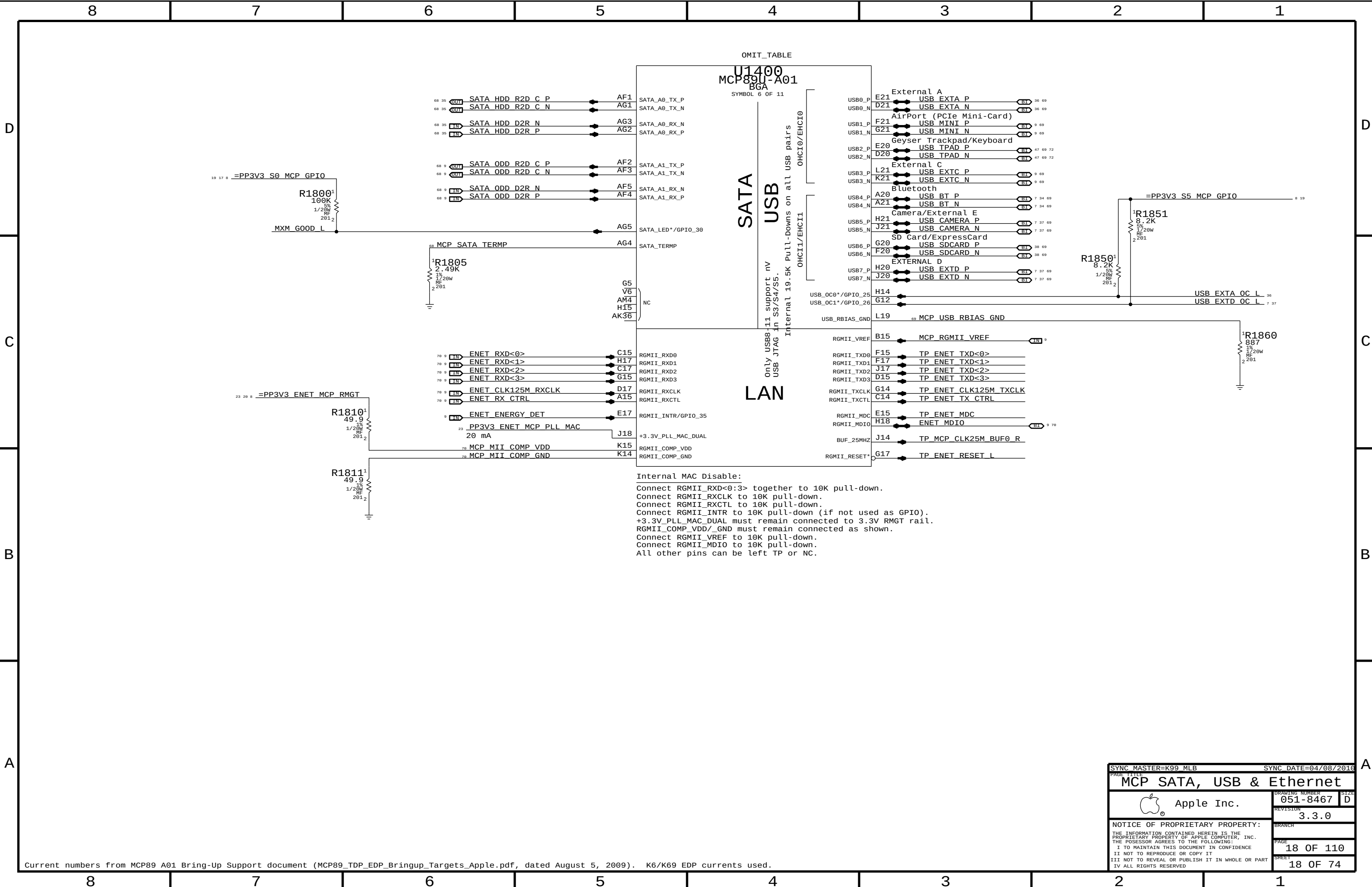
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RGB DAC Disable:  
Okay to float all RGB\_DAC signals.  
DDC\_CLK0/DDC\_DATA0 pull-ups still required (or use as GPIOs).  
Connect +3.3V\_RGBDAC pin to GND.  
NOTE: No Composite/S-Video/Component Video support on MCP89



Internal MAC Disable:  
Connect RGMII\_RXD<0:3> together to 10K pull-down.  
Connect RGMII\_RXCLK to 10K pull-down.  
Connect RGMII\_RXCTL to 10K pull-down.  
Connect RGMII\_INTR to 10K pull-down (if not used as GPIO).  
+3.3V\_PLL\_MAC\_DUAL must remain connected to 3.3V RMGT rail.  
RGMII\_COMP\_VDD/\_GND must remain connected as shown.  
Connect RGMII\_VREF to 10K pull-down.  
Connect RGMII\_MDIO to 10K pull-down.  
All other pins can be left TP or NC.

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SYNC DATE=04/08/2016

MCP SATA, USB & Ethernet

Apple Inc.

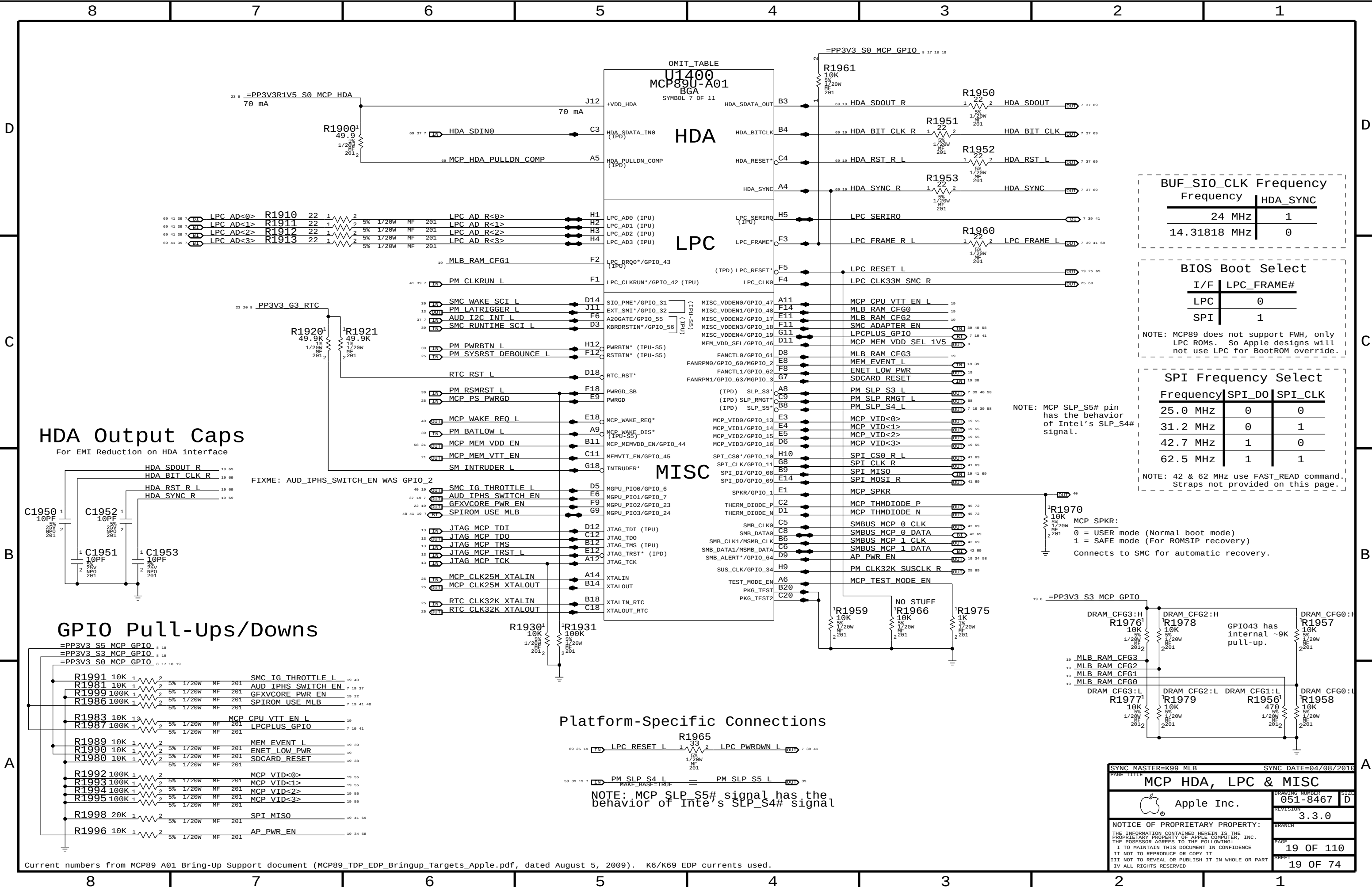
051-8467

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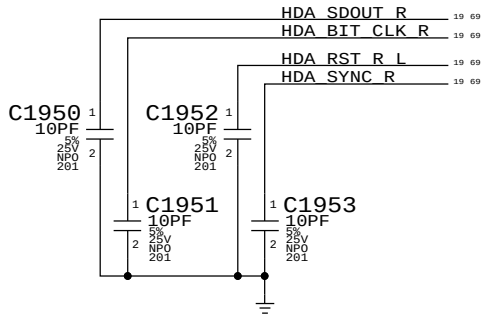
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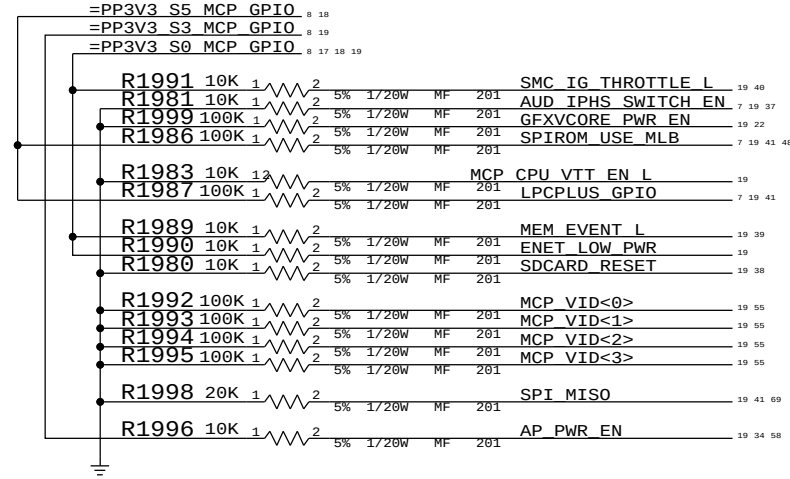


### HDA Output Caps

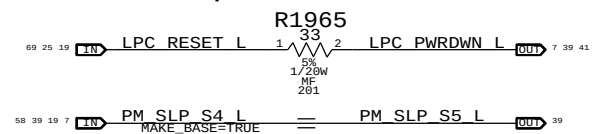
For EMI Reduction on HDA interface



### GPIO Pull-Ups/Downs



### Platform-Specific Connections



NOTE: MCP SLP\_S5# signal has the behavior of Intel's SLP\_S4# signal

| BUF_SIO_CLK Frequency |          |
|-----------------------|----------|
| Frequency             | HDA_SYNC |
| 24 MHz                | 1        |
| 14.31818 MHz          | 0        |

### BIOS Boot Select

| I/F | LPC_FRAME# |
|-----|------------|
| LPC | 0          |
| SPI | 1          |

NOTE: MCP89 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.

### SPI Frequency Select

| Frequency | SPI_DO | SPI_CLK |
|-----------|--------|---------|
| 25.0 MHz  | 0      | 0       |
| 31.2 MHz  | 0      | 1       |
| 42.7 MHz  | 1      | 0       |
| 62.5 MHz  | 1      | 1       |

NOTE: 42 & 62 MHz use FAST\_READ command. Straps not provided on this page.

NOTE: MCP SLP\_S5# pin has the behavior of Intel's SLP\_S4# signal.

MCP\_SPKR:  
0 = USER mode (Normal boot mode)  
1 = SAFE mode (For ROMSIP recovery)

Connects to SMC for automatic recovery.

SYNC MASTER=K99\_MLB

SYNC DATE=04/08/2016

MCP HDA, LPC & MISC

Apple Inc.

051-8467

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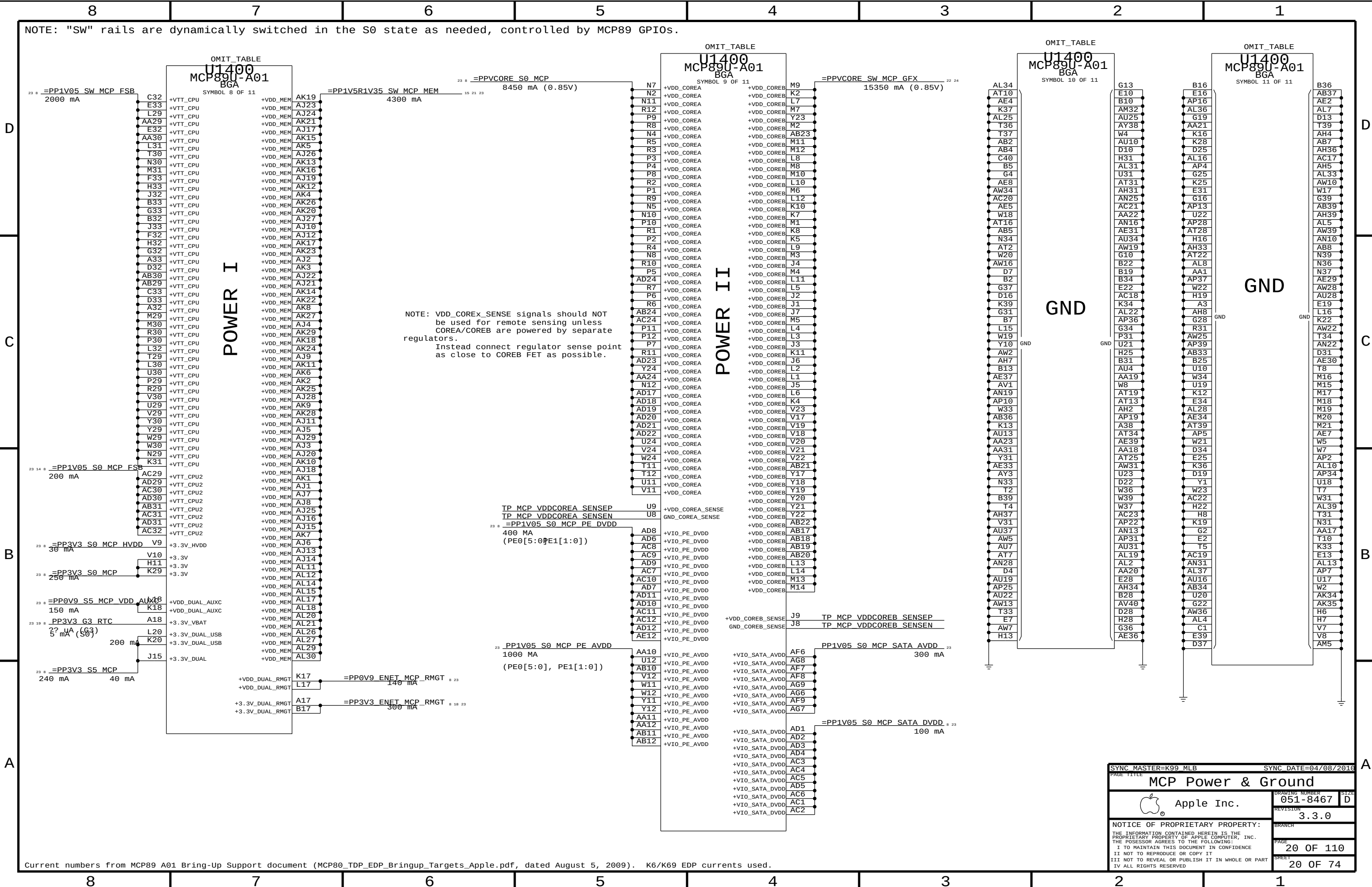
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MCP Power & Ground

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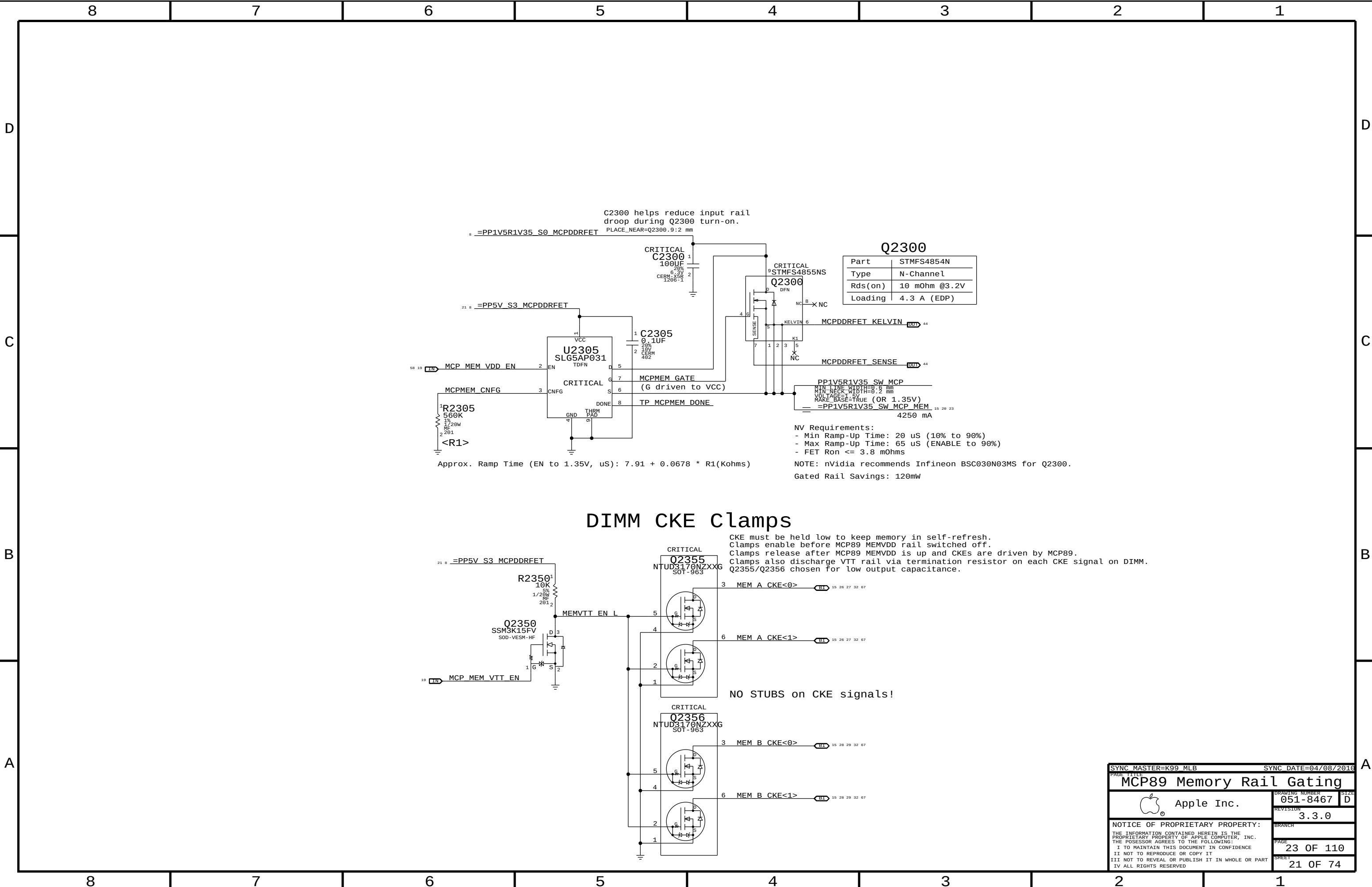
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DIMM CKE Clamps

CKE must be held low to keep memory in self-refresh.  
Clamps enable before MCP89 MEMVDD rail switched off.  
Clamps release after MCP89 MEMVDD is up and CKEs are driven by MCP89.  
Clamps also discharge VTT rail via termination resistor on each CKE signal on DIMM.  
Q2355/Q2356 chosen for low output capacitance.

NO STUBS on CKE signals!

SYNC MASTER=K99 MLB

SYNC DATE=04/08/2016

MCP89 Memory Rail Gating

Apple Inc.

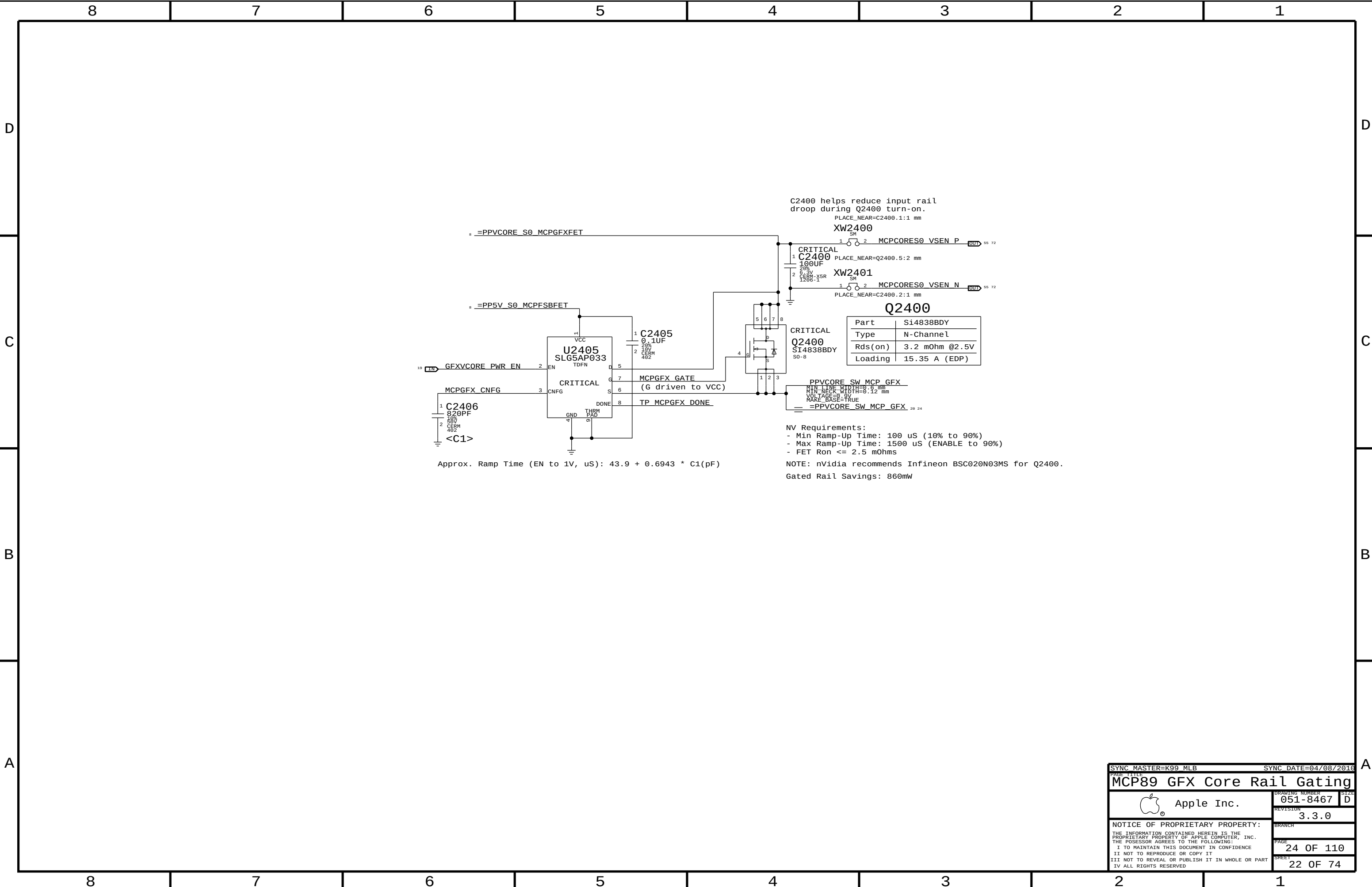
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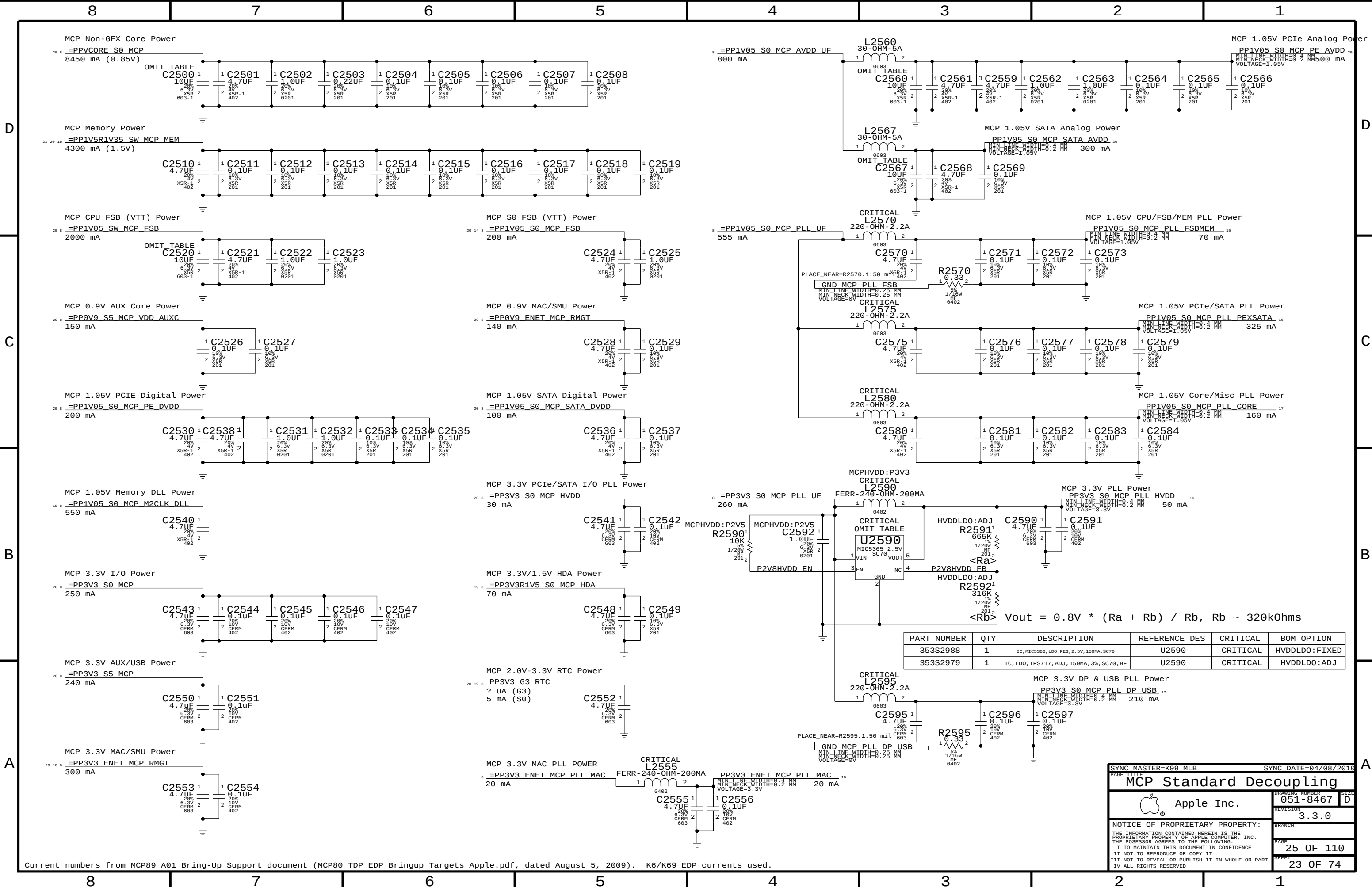
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3.3.0

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Current numbers from MCP89 A01 Bring-Up Support document (MCP80\_TDP\_EDP\_Bringup\_Targets\_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

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SYNC DATE=04/08/2016

MCP Standard Decoupling

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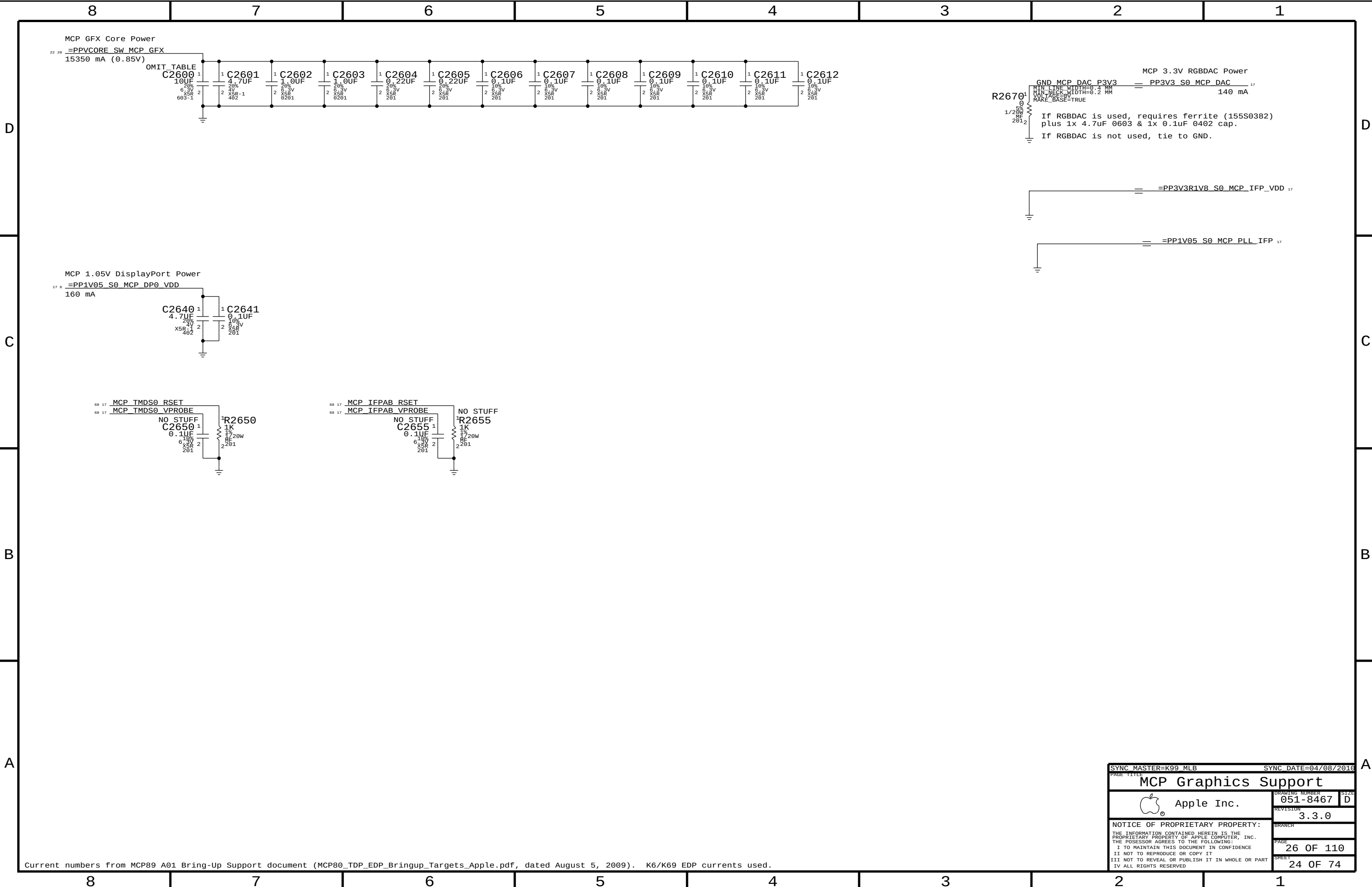
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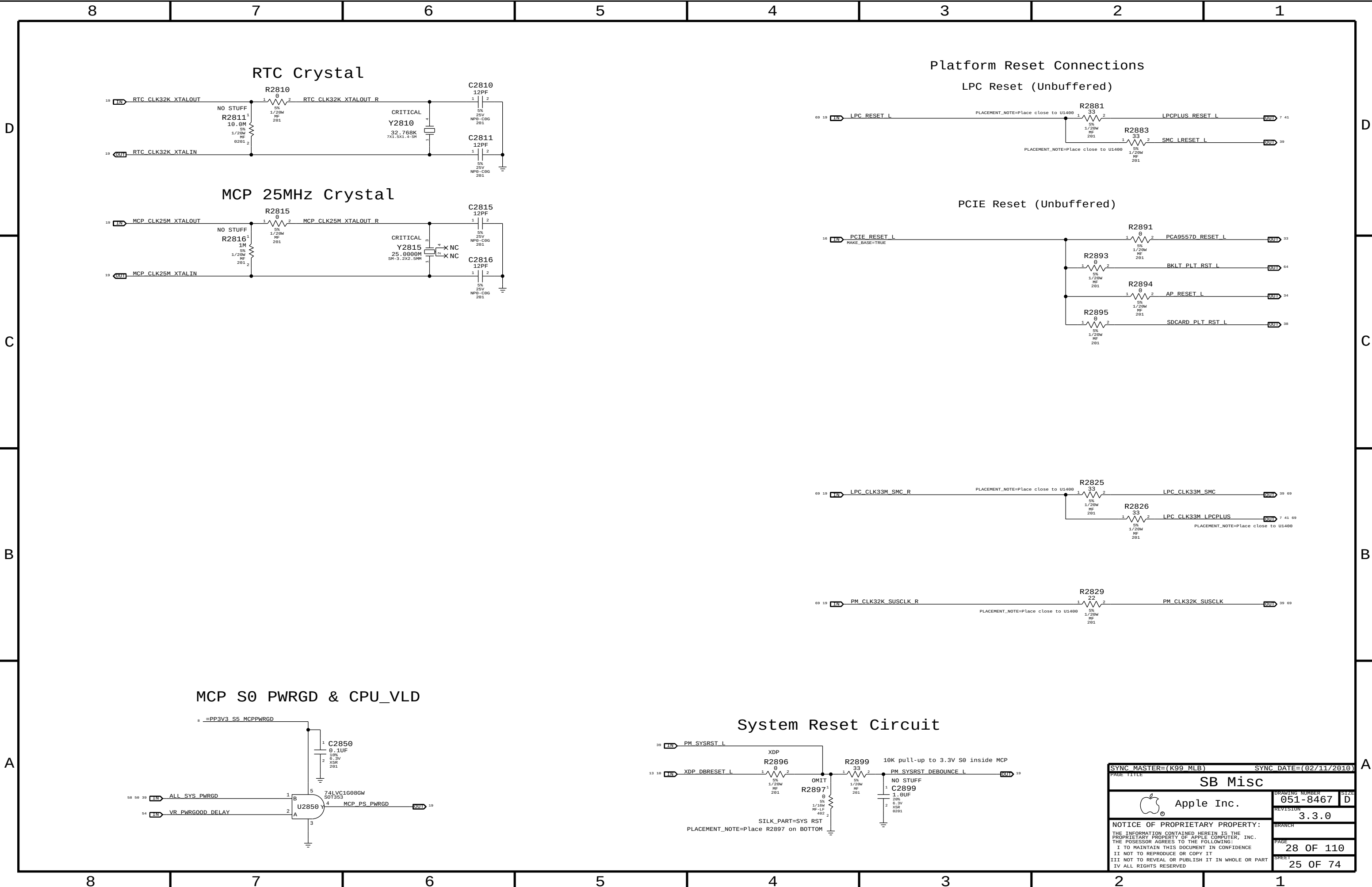
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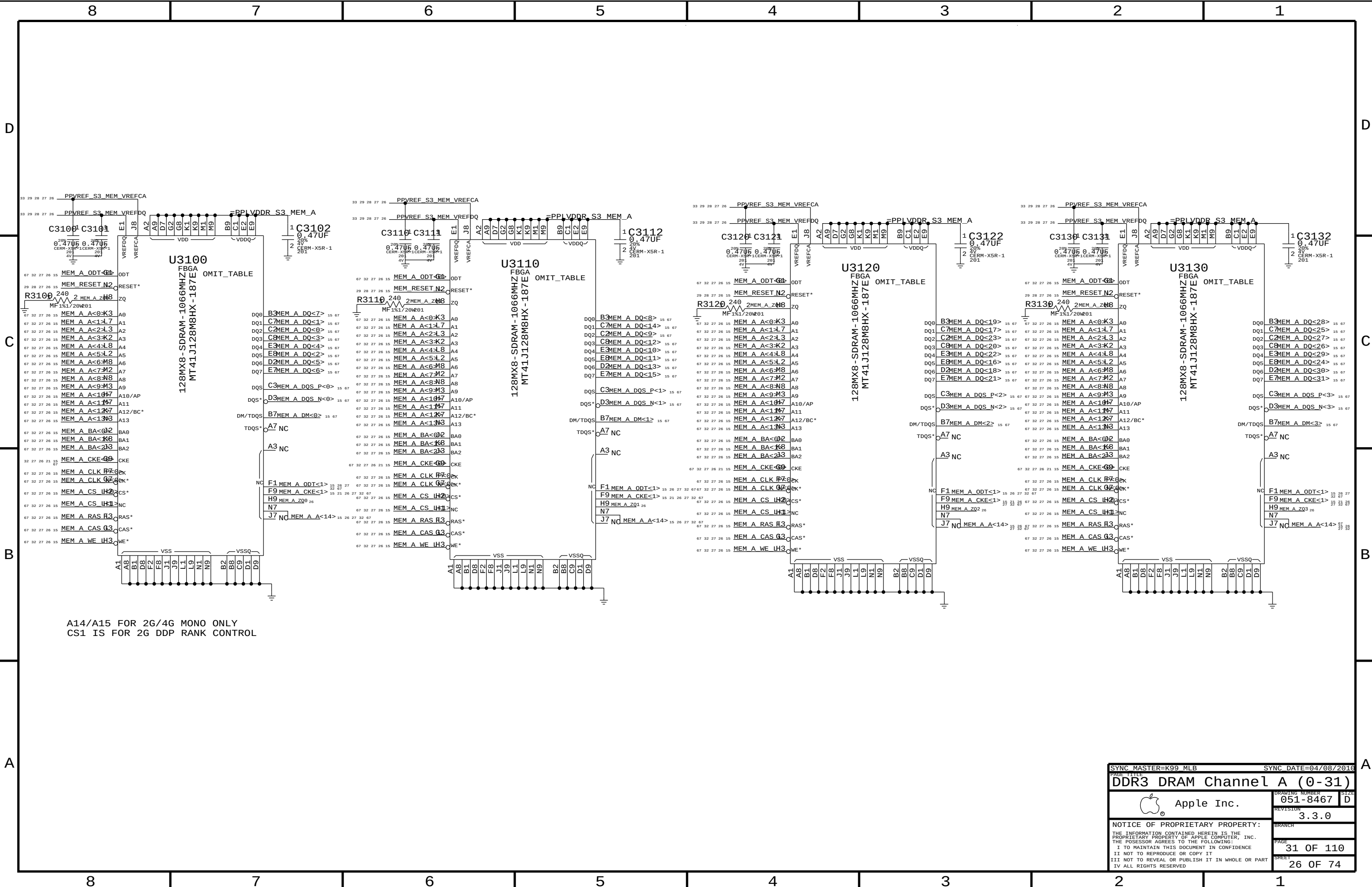


Current numbers from MCP89 A01 Bring-Up Support document (MCP80\_TDP\_EDP\_Bringup\_Targets\_Apple.pdf, dated August 5, 2009). K6/K69 EDP currents used.

|                                                                                                                                                                                                                                                                                                                                   |  |                      |           |
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| SYNC MASTER=K99 MLB                                                                                                                                                                                                                                                                                                               |  | SYNC DATE=04/08/2010 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                        |  |                      |           |
| MCP Graphics Support                                                                                                                                                                                                                                                                                                              |  |                      |           |
|  Apple Inc.                                                                                                                                                                                                                                  |  | DRAWING NUMBER       | 051-8467  |
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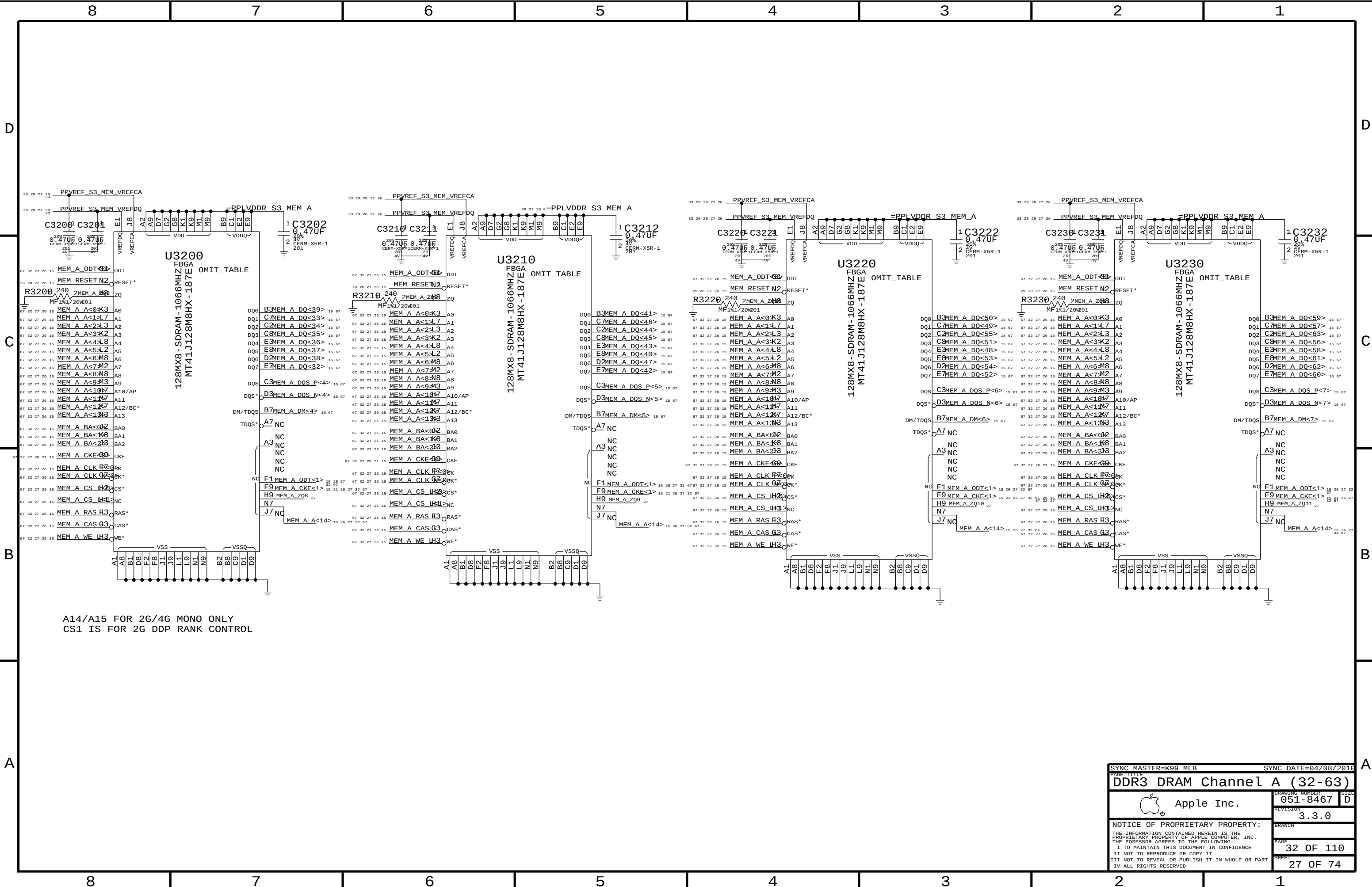






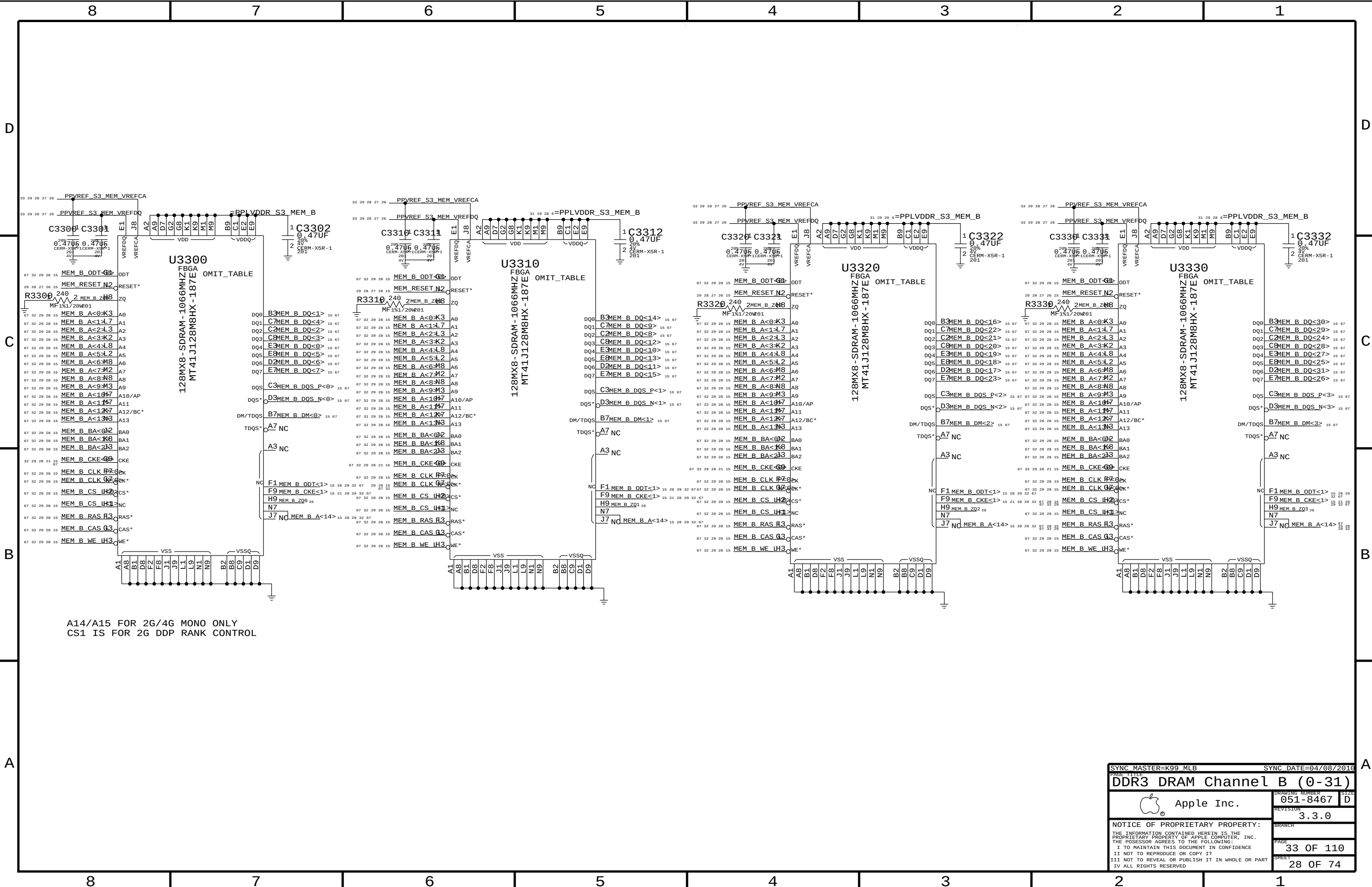
A14/A15 FOR 2G/4G MONO ONLY  
CS1 IS FOR 2G DDP RANK CONTROL

|                                                                                                                                                                                                                                                                                                                                   |  |                      |           |
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| PAGE TITLE                                                                                                                                                                                                                                                                                                                        |  |                      |           |
| DDR3 DRAM Channel A                                                                                                                                                                                                                                                                                                               |  | (0-31)               |           |
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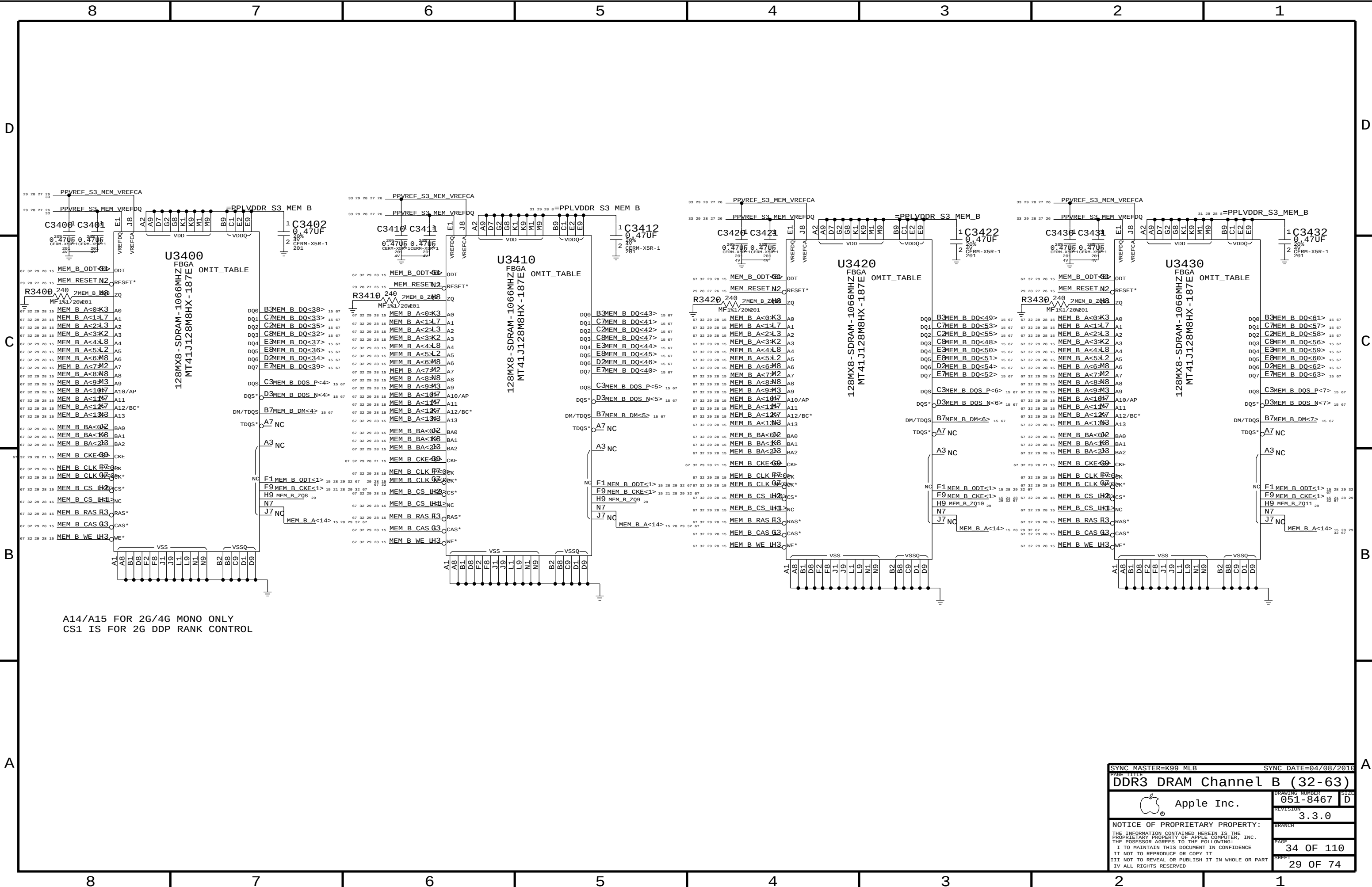
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CS1 IS FOR 2G DDP RANK CONTROL

|                                                                                                                                                                                                                                                                                                                                   |            |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------|-----------|
| SYNC MASTER=K99 MLB                                                                                                                                                                                                                                                                                                               |            | SYNC DATE=04/08/2016 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                        |            |                      |           |
| DDR3 DRAM Channel A                                                                                                                                                                                                                                                                                                               |            | A (32-63)            |           |
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|                                                                                                                                                                                                                                                                                                                                   |            | SHEET                | 27 OF 74  |
|                                                                                                                                                                                                                                                                                                                                   |            |                      |           |



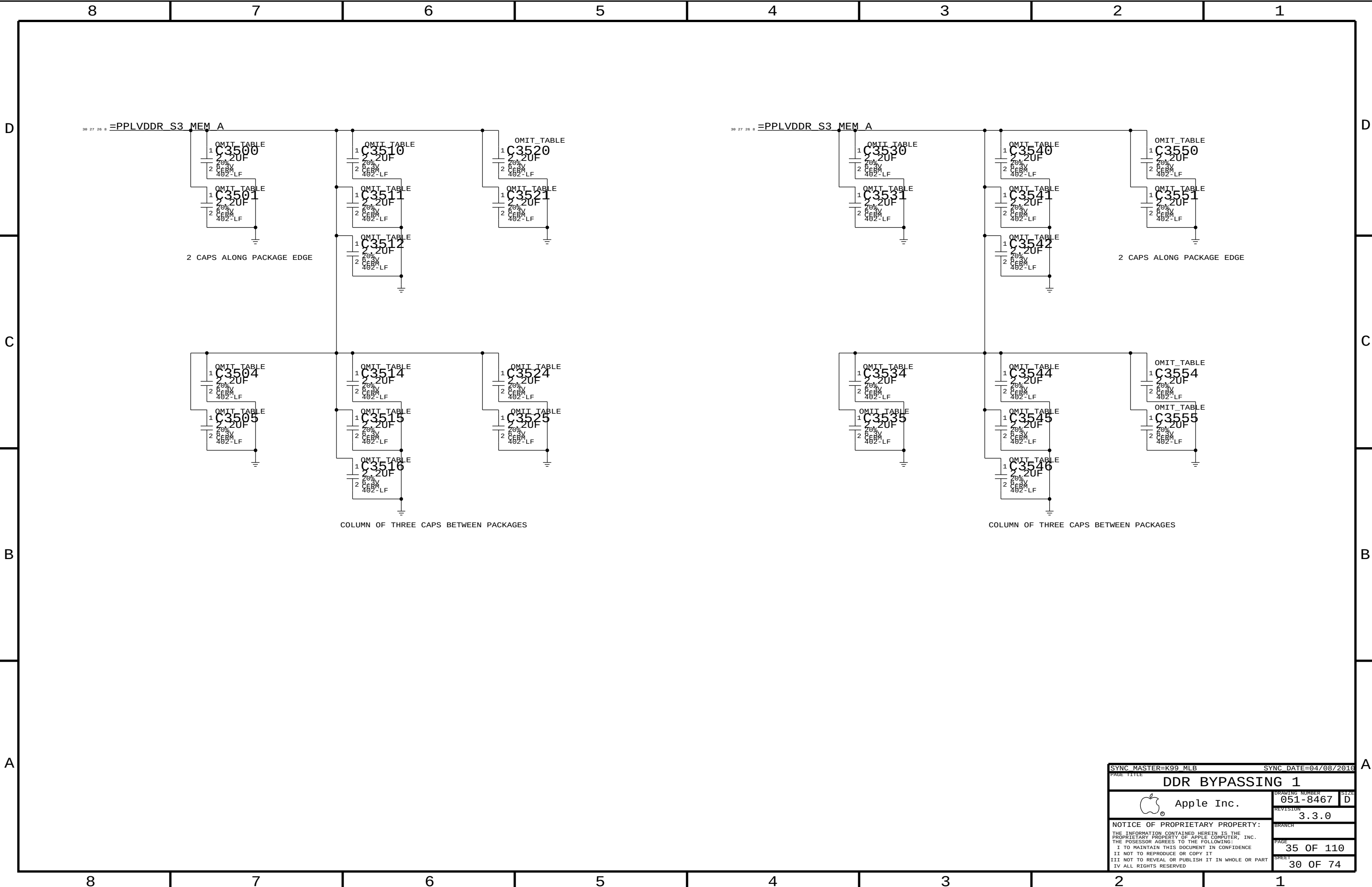
A14/A15 FOR 2G/4G MONO ONLY  
CS1 IS FOR 2G DDP RANK CONTROL

|                                                                                                                                                                                                                                                                                                                                          |            |                      |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------|-----------|
| SYNC MASTER=K99 MLB                                                                                                                                                                                                                                                                                                                      |            | SYNC DATE=04/08/2016 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                               |            |                      |           |
| DDR3 DRAM Channel B                                                                                                                                                                                                                                                                                                                      |            | B (0-31)             |           |
|                                                                                                                                                                                                                                                     | Apple Inc. | DRAWING NUMBER       | 051-8467  |
|                                                                                                                                                                                                                                                                                                                                          |            | SIZE                 | D         |
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|                                                                                                                                                                                                                                                                                                                                          |            | BRANCH               |           |
|                                                                                                                                                                                                                                                                                                                                          |            | PAGE                 | 33 OF 110 |
|                                                                                                                                                                                                                                                                                                                                          |            | SHEET                | 28 OF 74  |

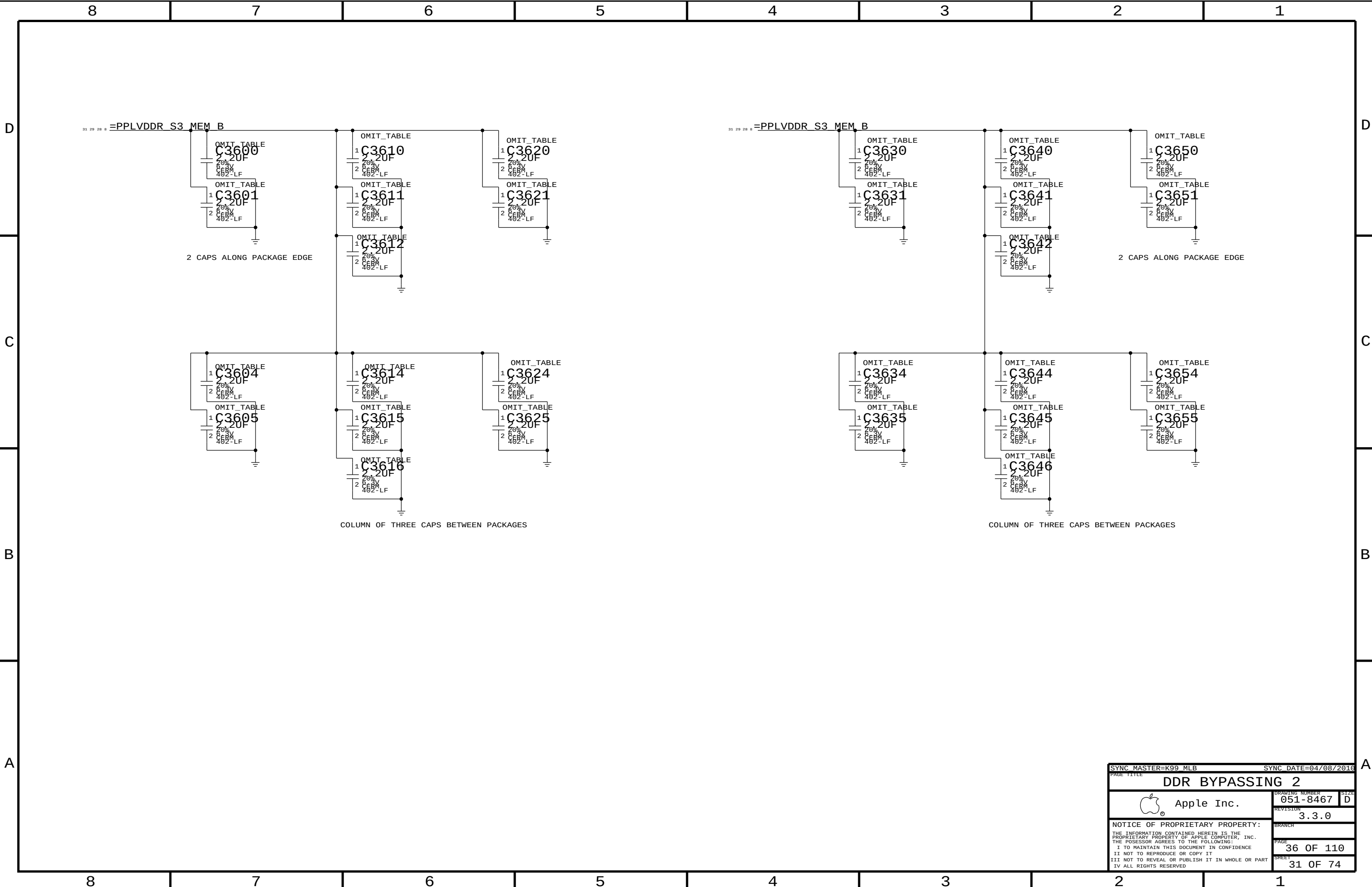


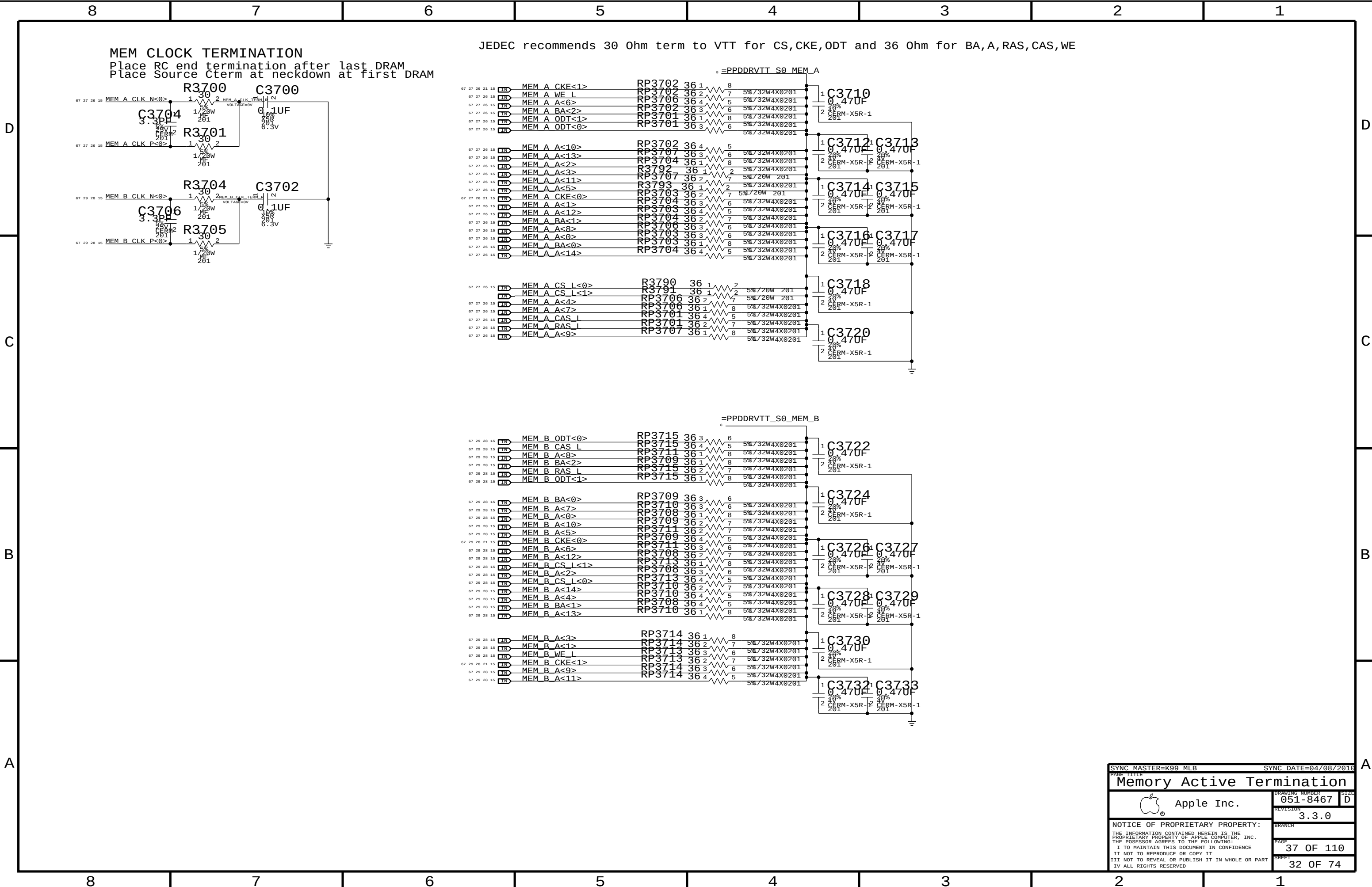
A14/A15 FOR 2G/4G MONO ONLY  
CS1 IS FOR 2G DDP RANK CONTROL

|                                                                                                                             |                |                      |          |
|-----------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|----------|
| SYNC MASTER=K99 MLB                                                                                                         |                | SYNC DATE=04/08/2016 |          |
| PAGE TITLE                                                                                                                  |                |                      |          |
| DDR3 DRAM Channel B                                                                                                         |                | B (32-63)            |          |
|  Apple Inc.                            | DRAWING NUMBER |                      | 051-8467 |
|                                                                                                                             | REVISION       |                      | 3.3.0    |
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| BRANCH                                                                                                                      |                |                      |          |
| PAGE                                                                                                                        |                | 34 OF 110            |          |
| SHEET                                                                                                                       |                | 29 OF 74             |          |

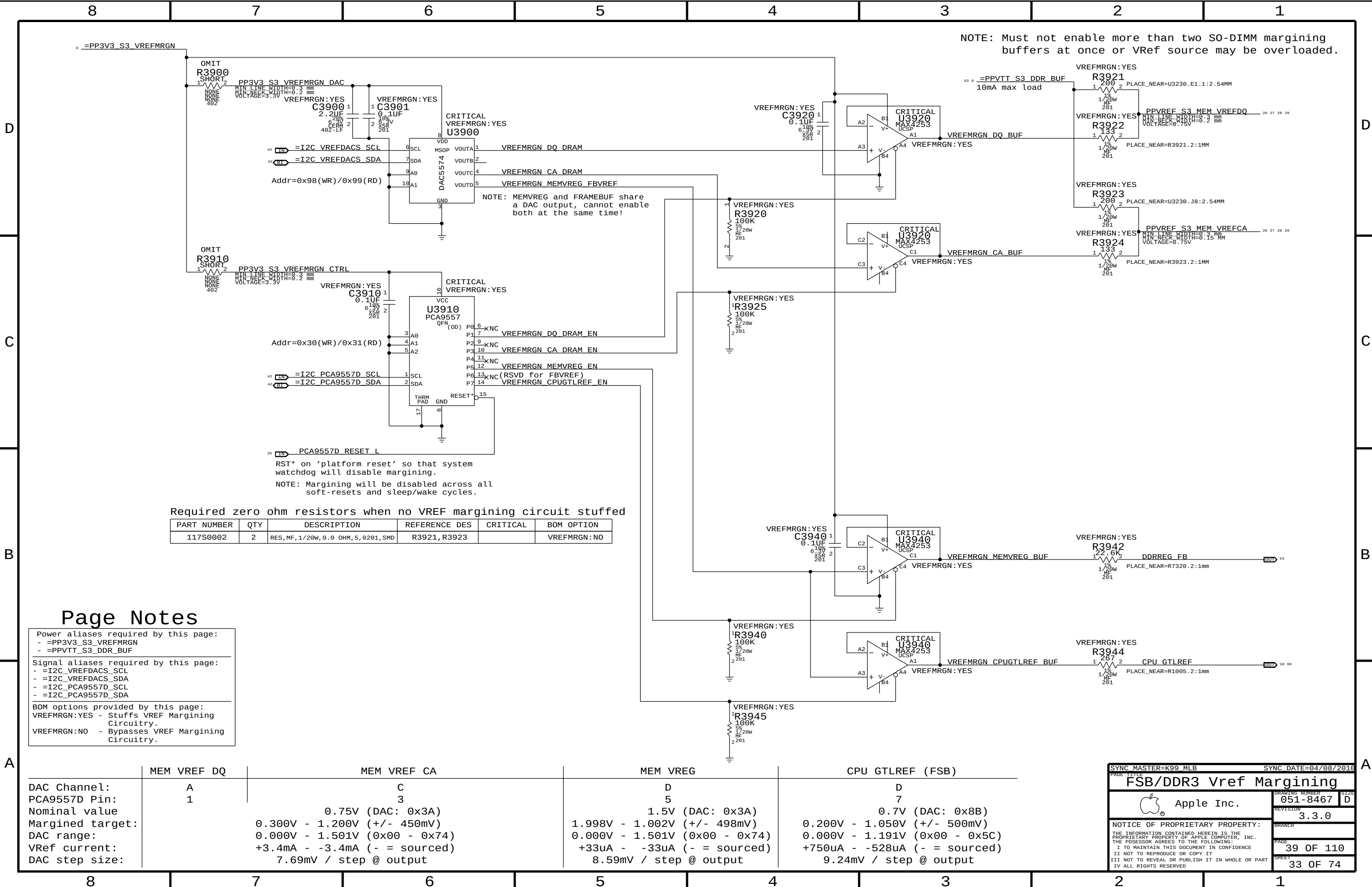












Page Notes

Power aliases required by this page:  
- =PP3V3\_S3\_VREFMRGN  
- =PPVTT\_S3\_DDR\_BUF

Signal aliases required by this page:  
- =I2C\_VREFDACS\_SCL  
- =I2C\_VREFDACS\_SDA  
- =I2C\_PCA9557D\_SCL  
- =I2C\_PCA9557D\_SDA

BOM options provided by this page:  
VREFMRGN:YES - Stuffs VREF Margining Circuitry.  
VREFMRGN:NO - Bypasses VREF Margining Circuitry.

|                  | MEM VREF DQ | MEM VREF CA                   | MEM VREG                      | CPU GTLREF (FSB)              |
|------------------|-------------|-------------------------------|-------------------------------|-------------------------------|
| DAC Channel:     | A           | C                             | D                             | D                             |
| PCA9557D Pin:    | 1           | 3                             | 5                             | 7                             |
| Nominal value    |             | 0.75V (DAC: 0x3A)             | 1.5V (DAC: 0x3A)              | 0.7V (DAC: 0x8B)              |
| Margined target: |             | 0.300V - 1.200V (+/- 450mV)   | 1.998V - 1.002V (+/- 498mV)   | 0.200V - 1.050V (+/- 500mV)   |
| DAC range:       |             | 0.000V - 1.501V (0x00 - 0x74) | 0.000V - 1.501V (0x00 - 0x74) | 0.000V - 1.191V (0x00 - 0x5C) |
| VRef current:    |             | +3.4mA - -3.4mA (- = sourced) | +33uA - -33uA (- = sourced)   | +750uA - -528uA (- = sourced) |
| DAC step size:   |             | 7.69mV / step @ output        | 8.59mV / step @ output        | 9.24mV / step @ output        |

SYNC MASTER=K99 MLB

SYNC DATE=04/08/2016

FSB/DDR3 Vref Margining

Apple Inc.

051-8467

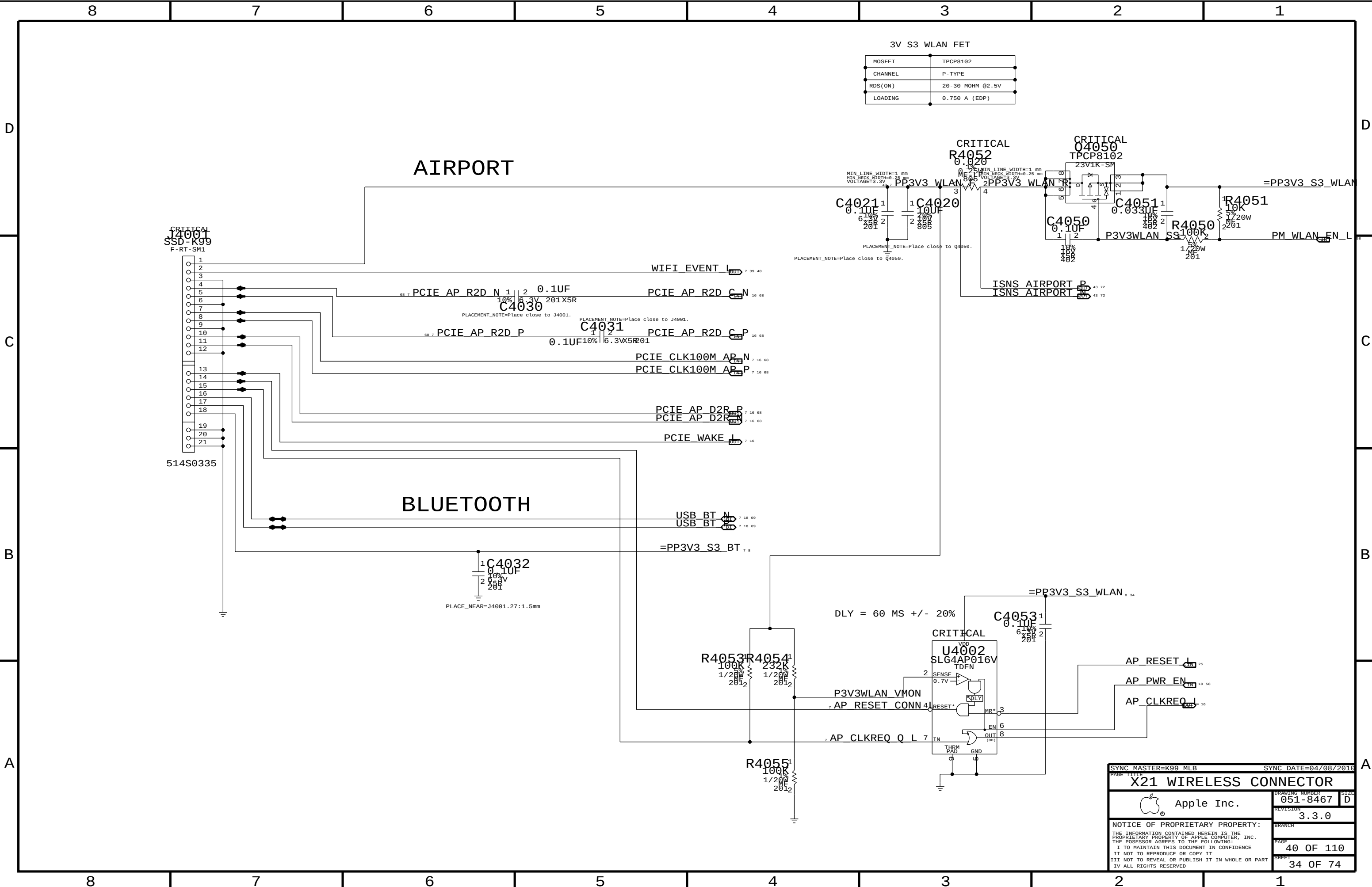
3.3.0

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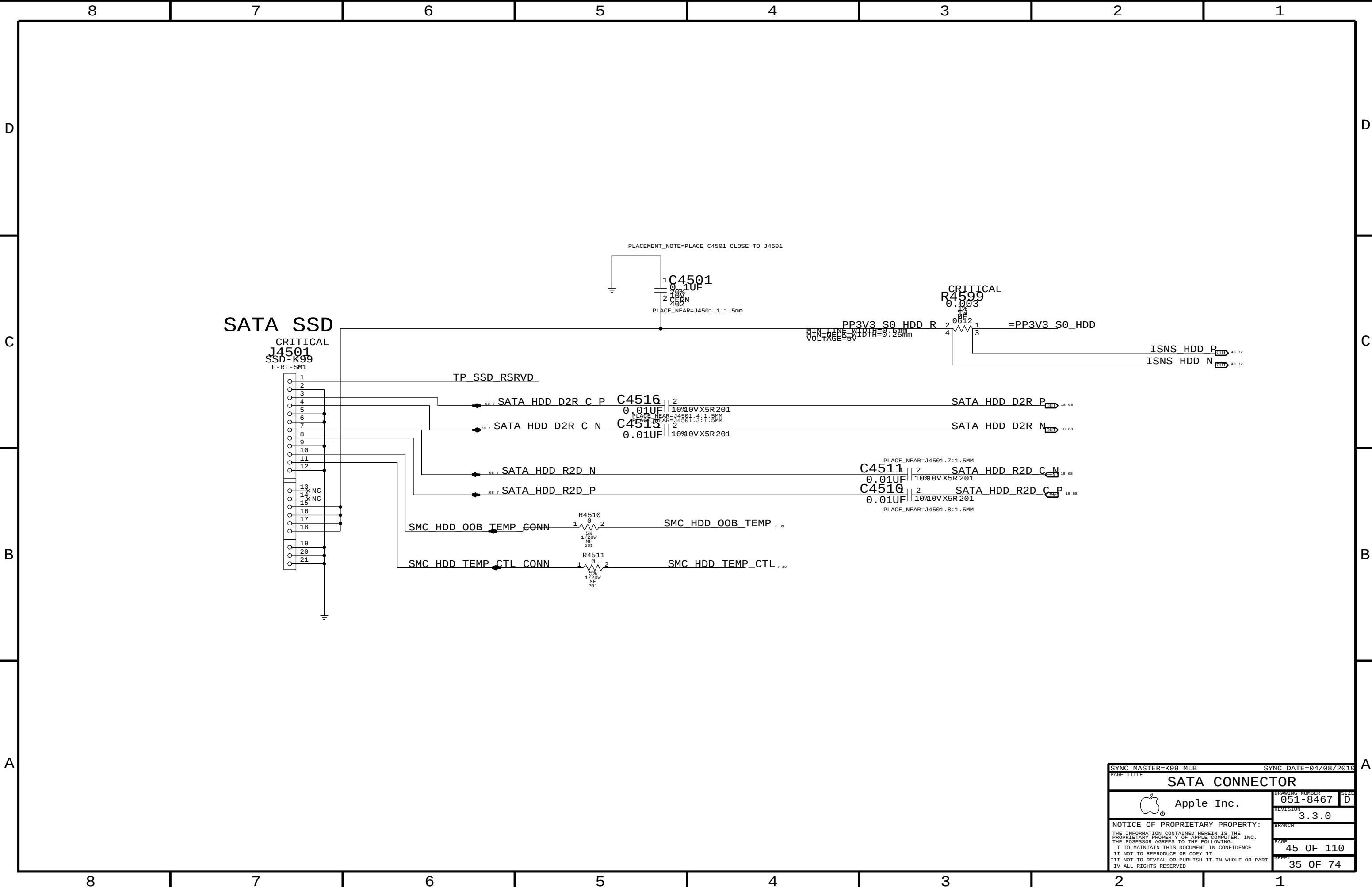
39 OF 110

33 OF 74



| 3V S3 WLAN FET |                  |
|----------------|------------------|
| MOSFET         | TPCP8102         |
| CHANNEL        | P-TYPE           |
| RDS(ON)        | 20-30 MOHM @2.5V |
| LOADING        | 0.750 A (EDP)    |

|                                                                                                                                                                                                                                                                                                                             |  |                        |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|-----------|
| SYNC MASTER=K99 MLB                                                                                                                                                                                                                                                                                                         |  | SYNC DATE=04/08/2016   |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  | X21 WIRELESS CONNECTOR |           |
| Apple Inc.                                                                                                                                                                                                                                                                                                                  |  | DRAWING NUMBER         | 051-8467  |
|                                                                                                                                                                                                                                                                                                                             |  | REVISION               | 3.3.0     |
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SYNC MASTER=K99 MLB

SYNC DATE=04/08/2016

|                                                                                                                                                                                                                                                                                                                             |  |                |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|-----------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  | SATA CONNECTOR |           |
|  Apple Inc.                                                                                                                                                                                                                            |  | DRAWING NUMBER | 051-8467  |
|                                                                                                                                                                                                                                                                                                                             |  | REVISION       | 3.3.0     |
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|                                                                                                                                                                                                                                                                                                                             |  | PAGE           | 45 OF 110 |
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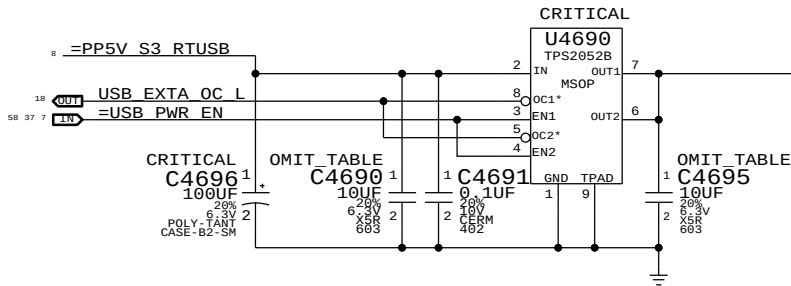
D

C

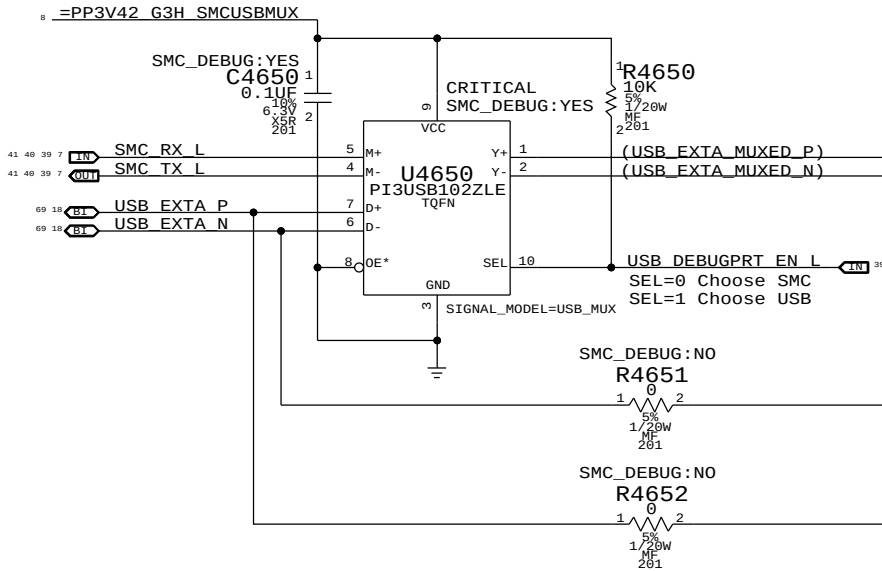
B

A

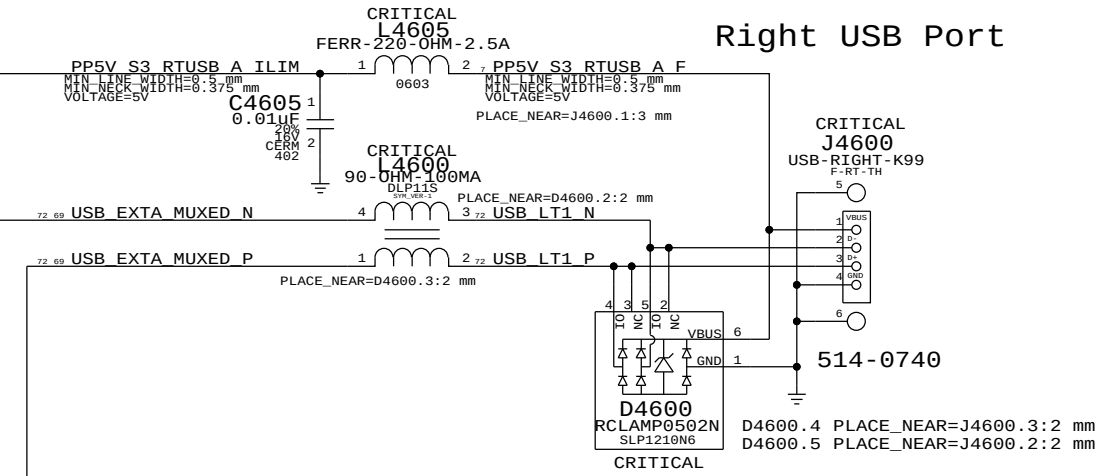
Port Power Switch



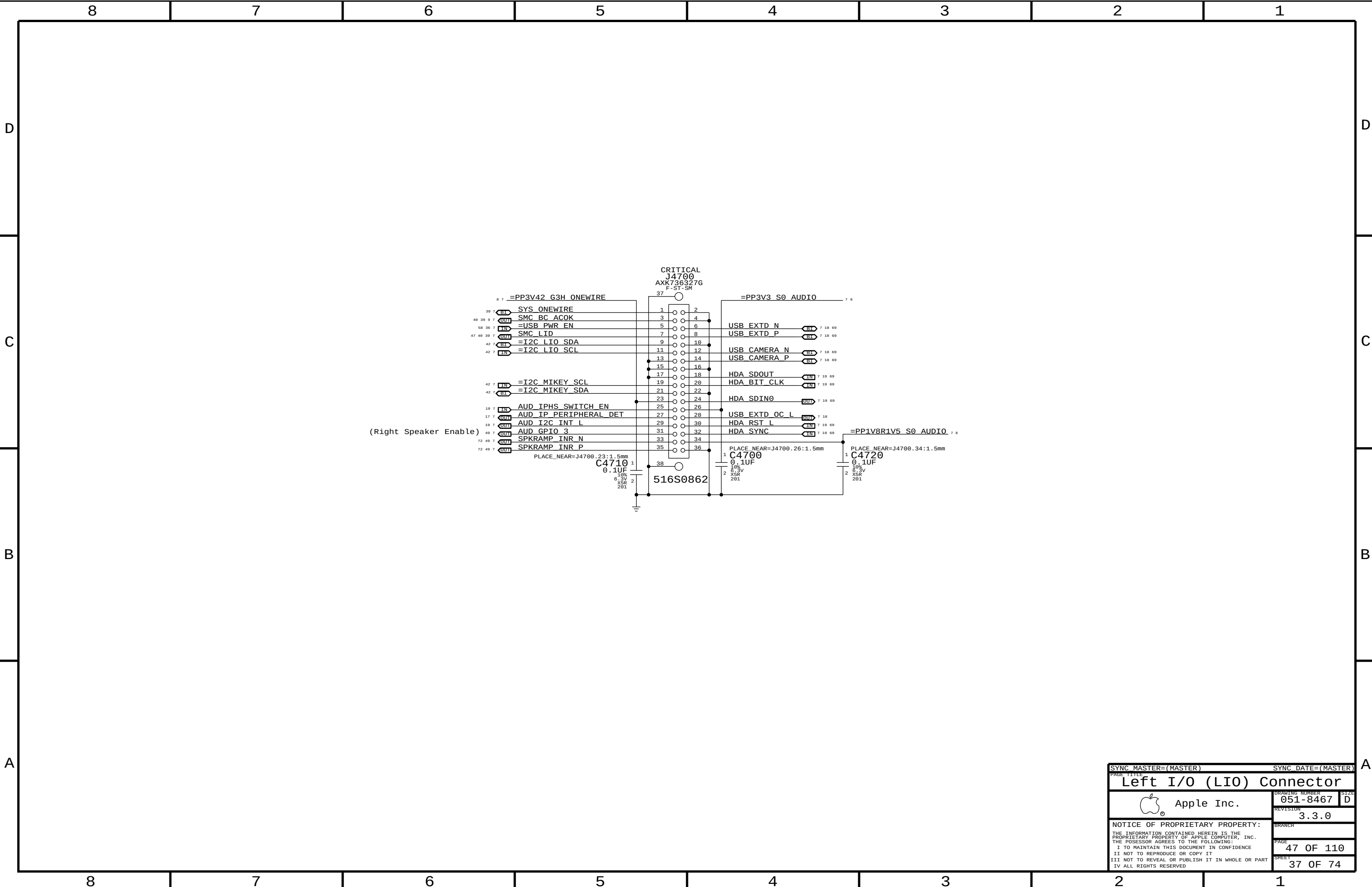
USB/SMC Debug Mux



Right USB Port

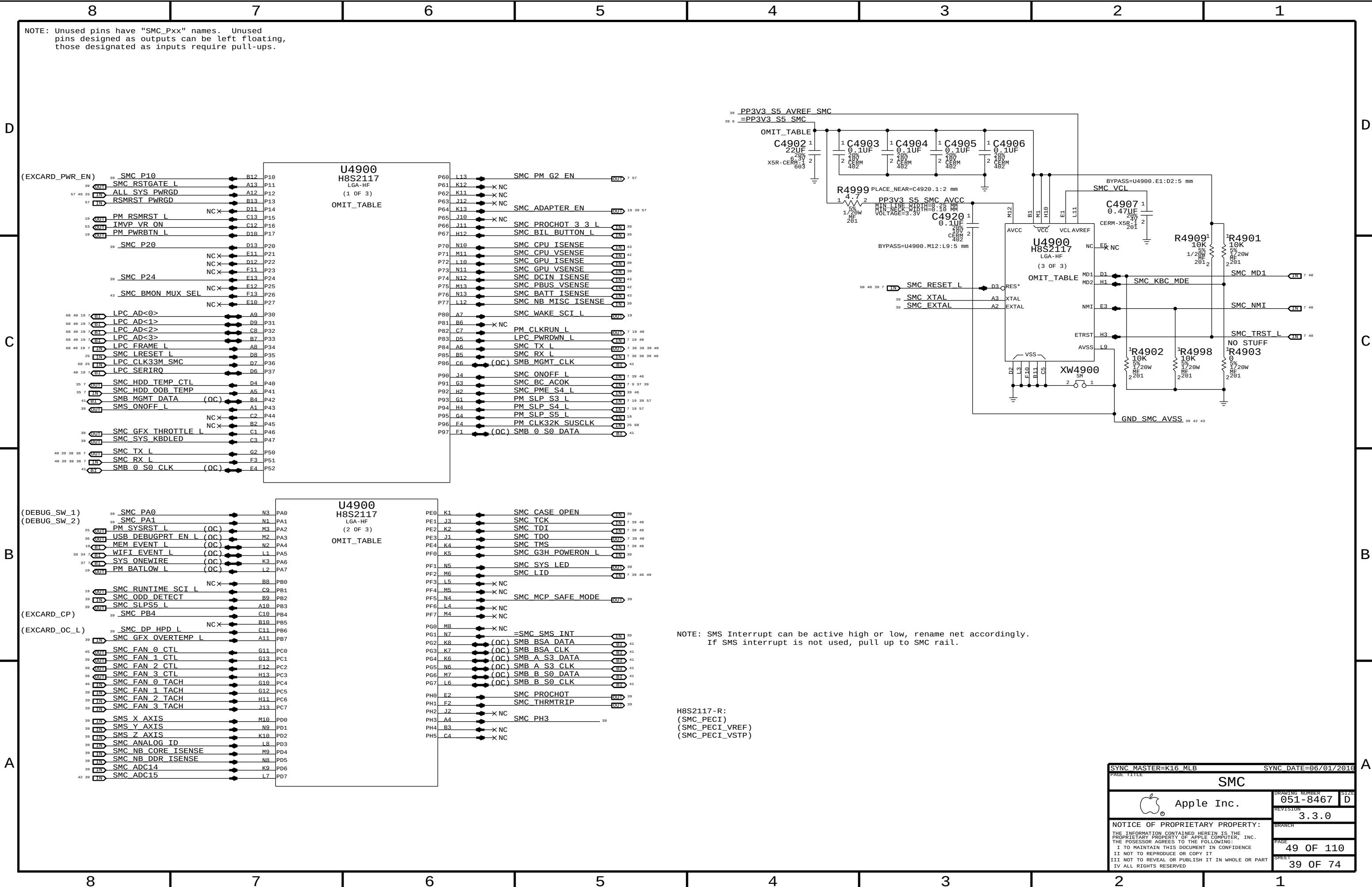


|                                                                                                                             |  |                         |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|-------------------------|--|
| SYNC MASTER=K99 MLB                                                                                                         |  | SYNC DATE=03/01/2016    |  |
| External USB Connectors                                                                                                     |  | DRAWING NUMBER 051-8467 |  |
| Apple Inc.                                                                                                                  |  | REVISION 3.3.0          |  |
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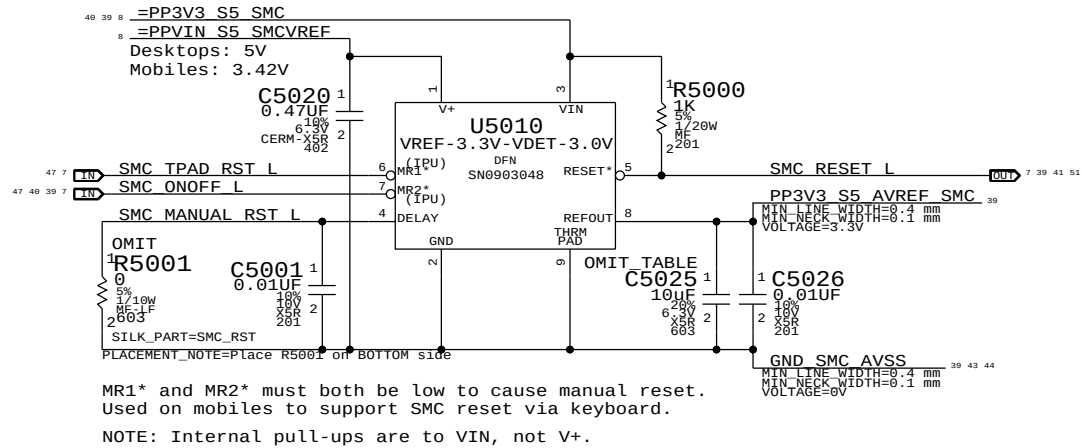




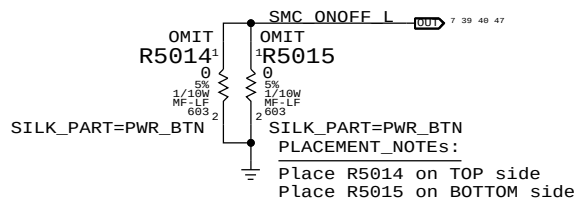




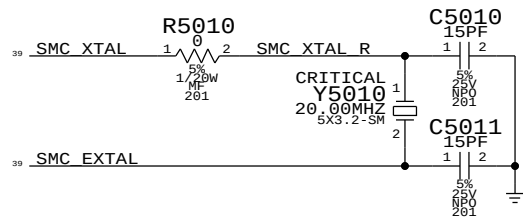
## SMC Reset "Button", Supervisor & AVREF Supply



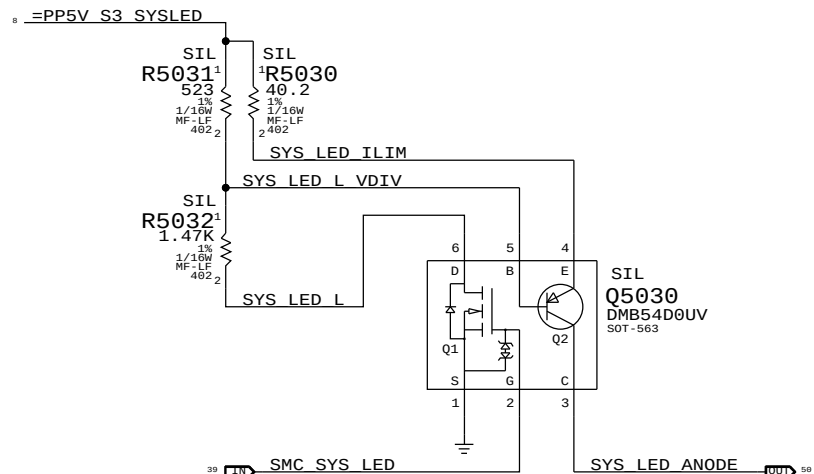
## Debug Power "Buttons"



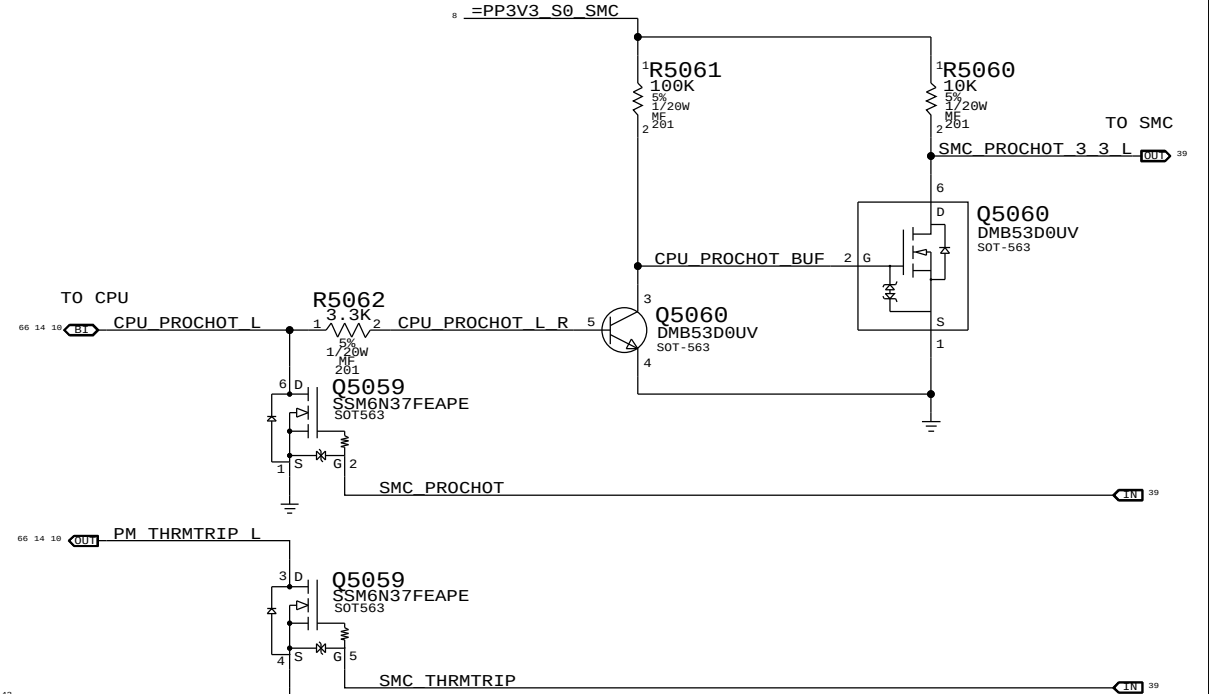
## SMC Crystal Circuit



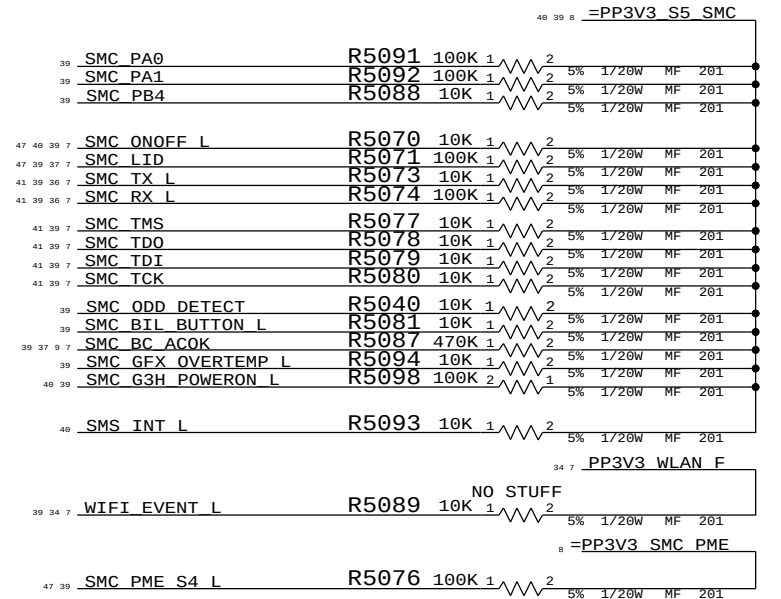
## System (Sleep) LED Circuit



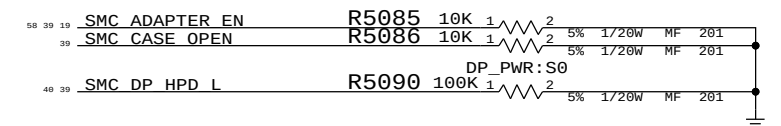
## SMC FSB to 3.3V Level Shifting



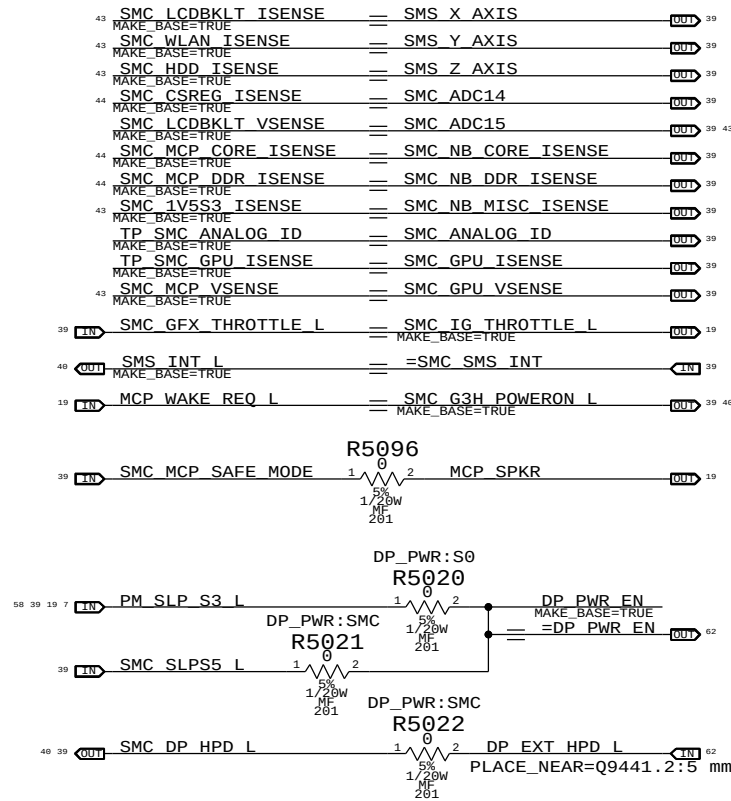
## SMC Pull-ups



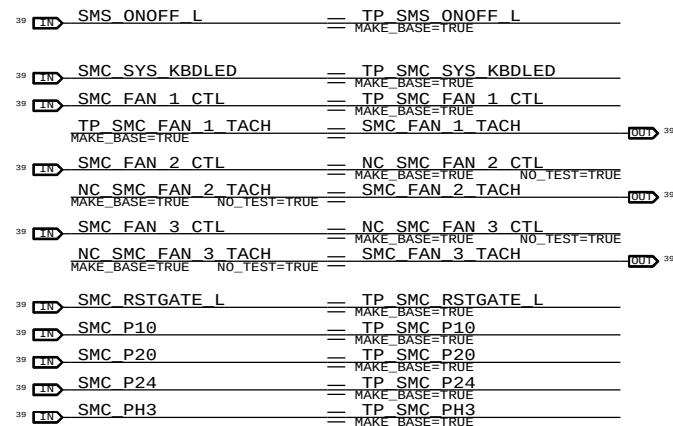
## SMC Pull-downs



## SMC Aliases

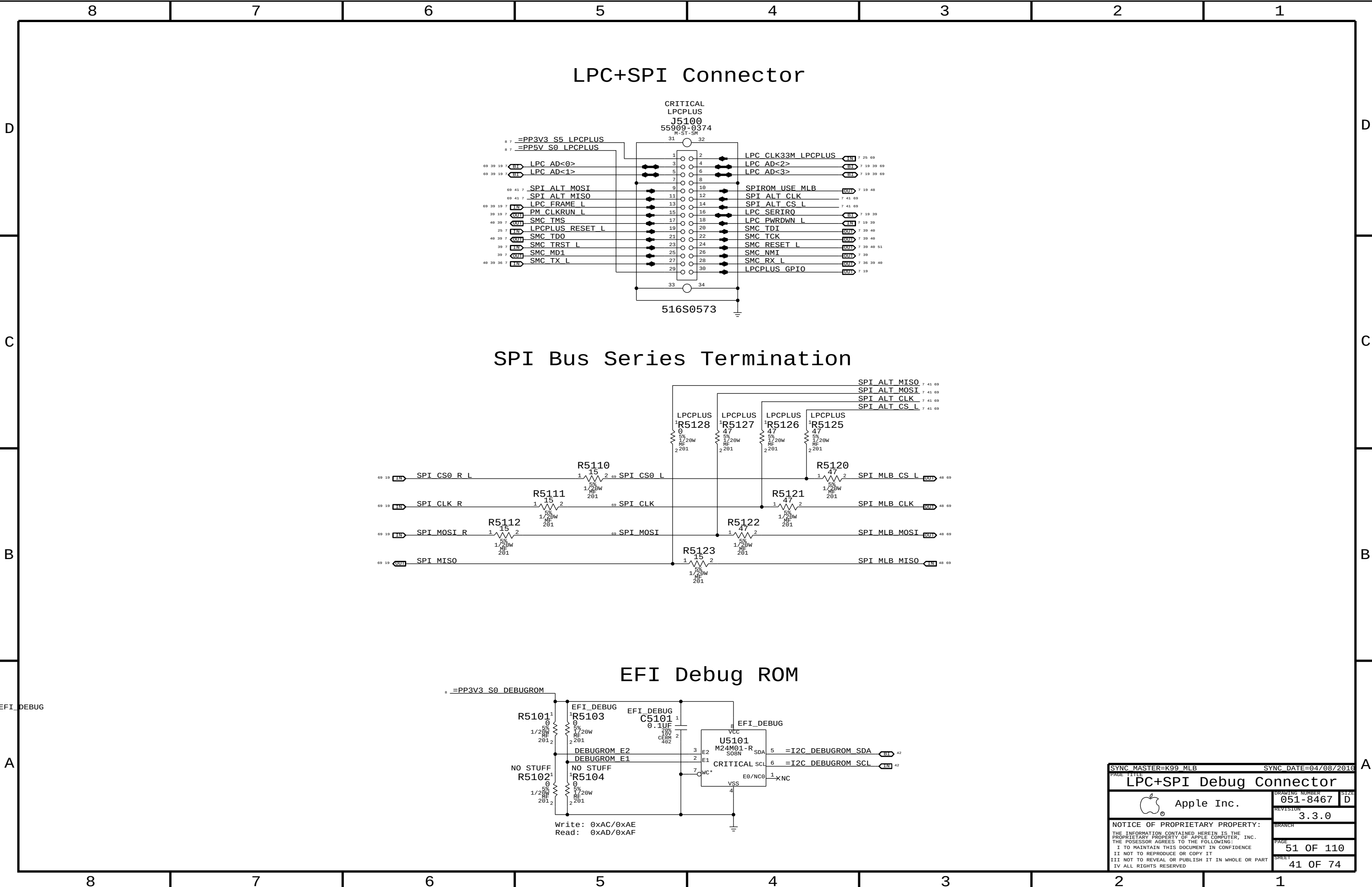


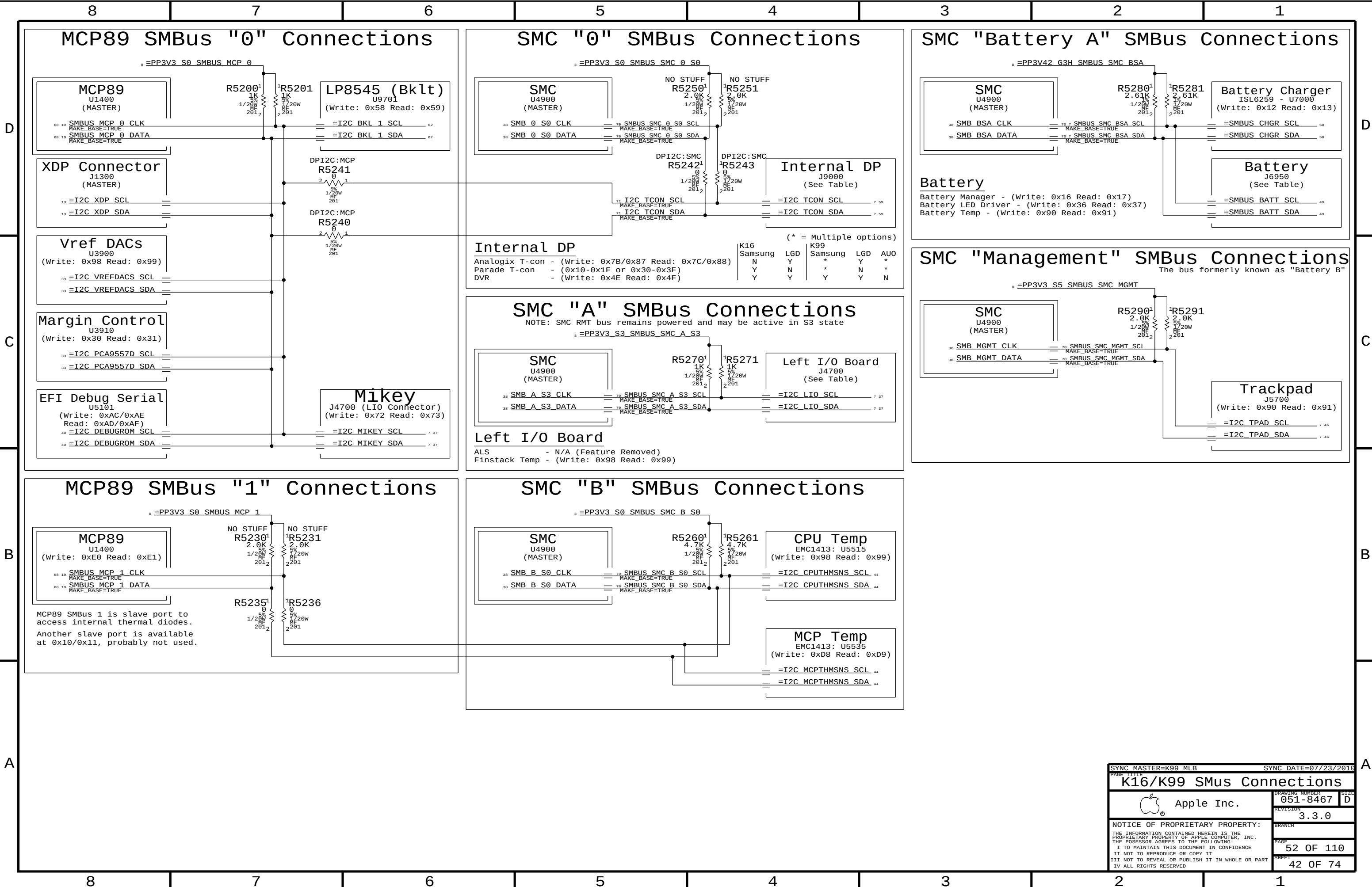
## Unused Pins



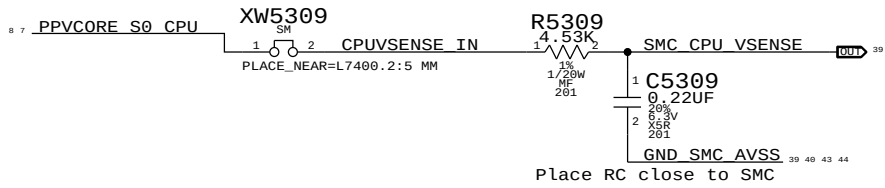
|                                                                                                                             |  |                        |      |
|-----------------------------------------------------------------------------------------------------------------------------|--|------------------------|------|
| SYNC MASTER=(K99 MLB)                                                                                                       |  | SYNC DATE=(03/01/2010) |      |
| PAGE TITLE                                                                                                                  |  |                        |      |
| SMC Support                                                                                                                 |  |                        |      |
| Apple Inc.                                                                                                                  |  | DRAWING NUMBER         | SIZE |
|                                                                                                                             |  | 051-8467               | D    |
|                                                                                                                             |  | REVISION               |      |
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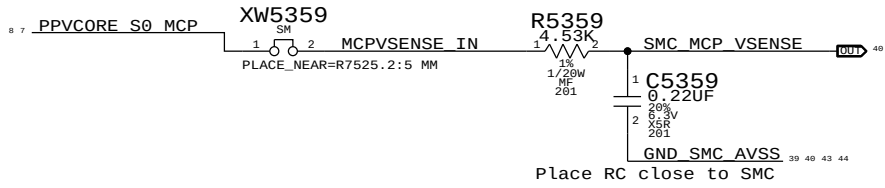




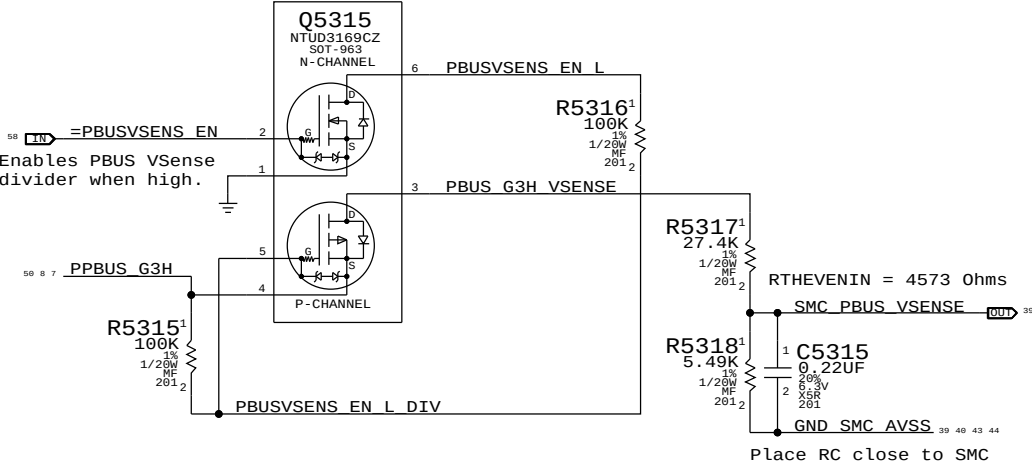
CPU Voltage Sense / Filter



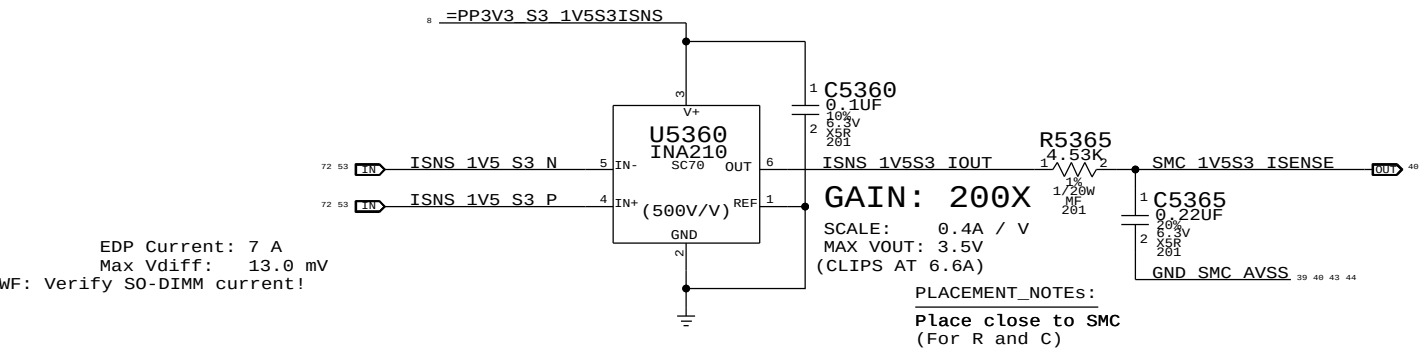
MCP Voltage Sense / Filter



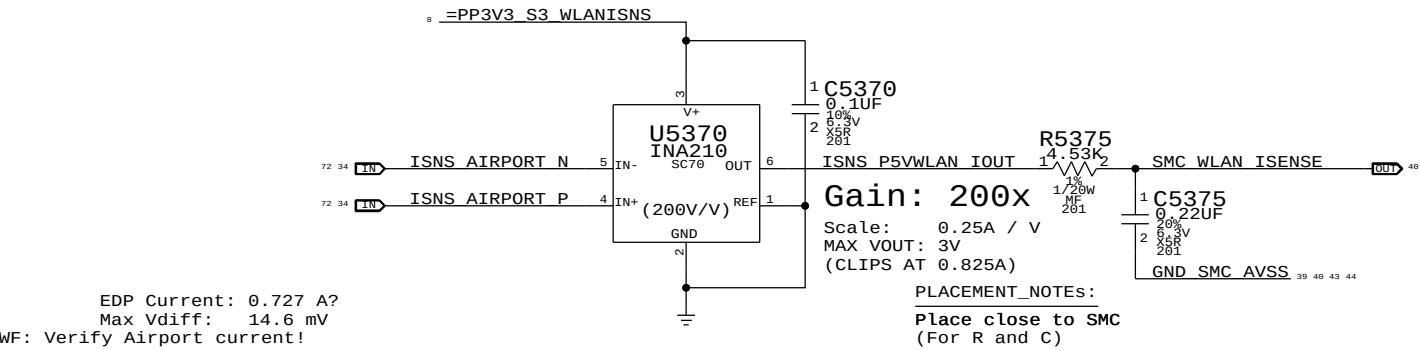
PBUS Voltage Sense Enable & Filter



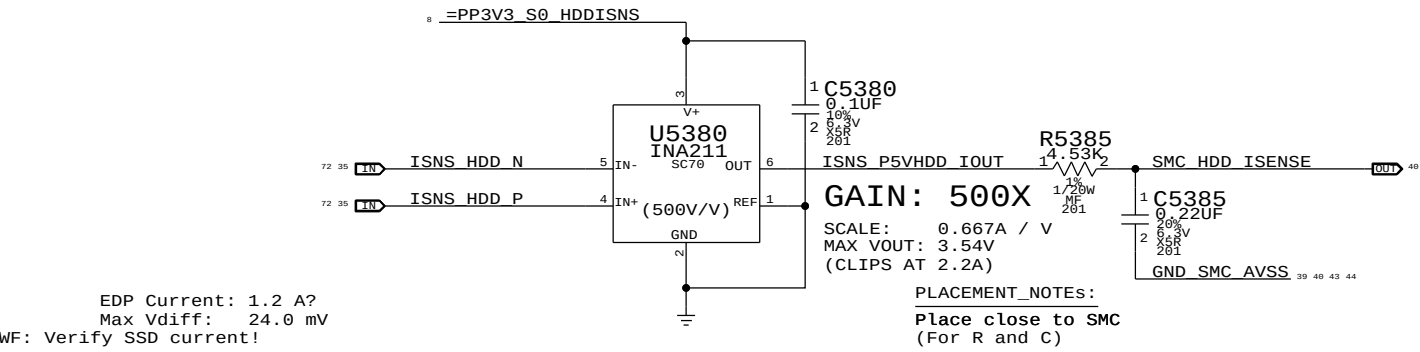
DDR3 1V5R1V35 Current Sense / Filter



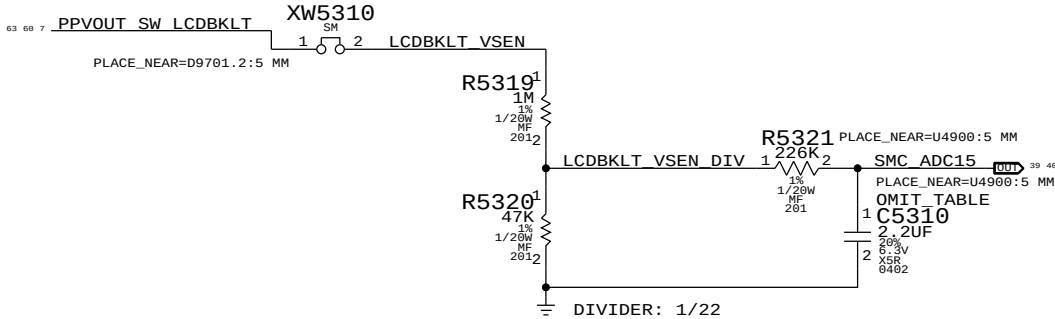
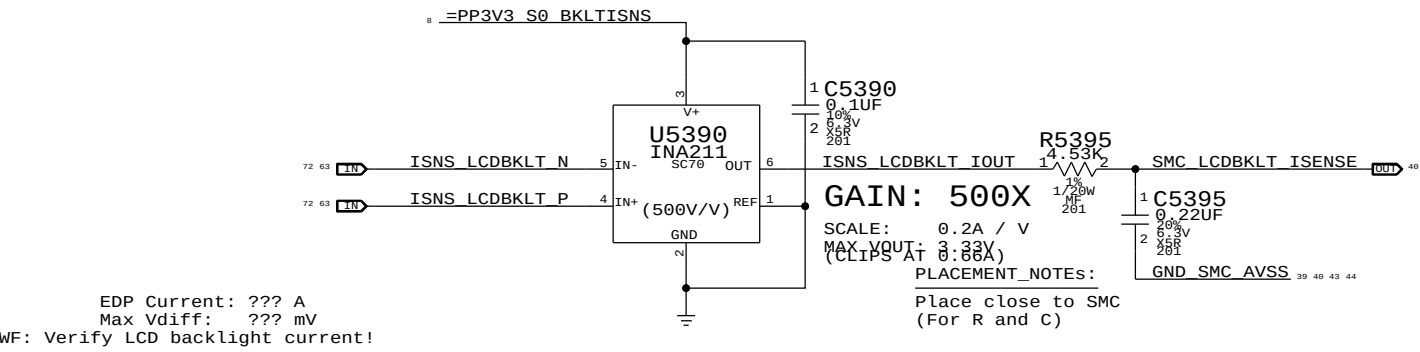
AirPort Current Sense / Filter



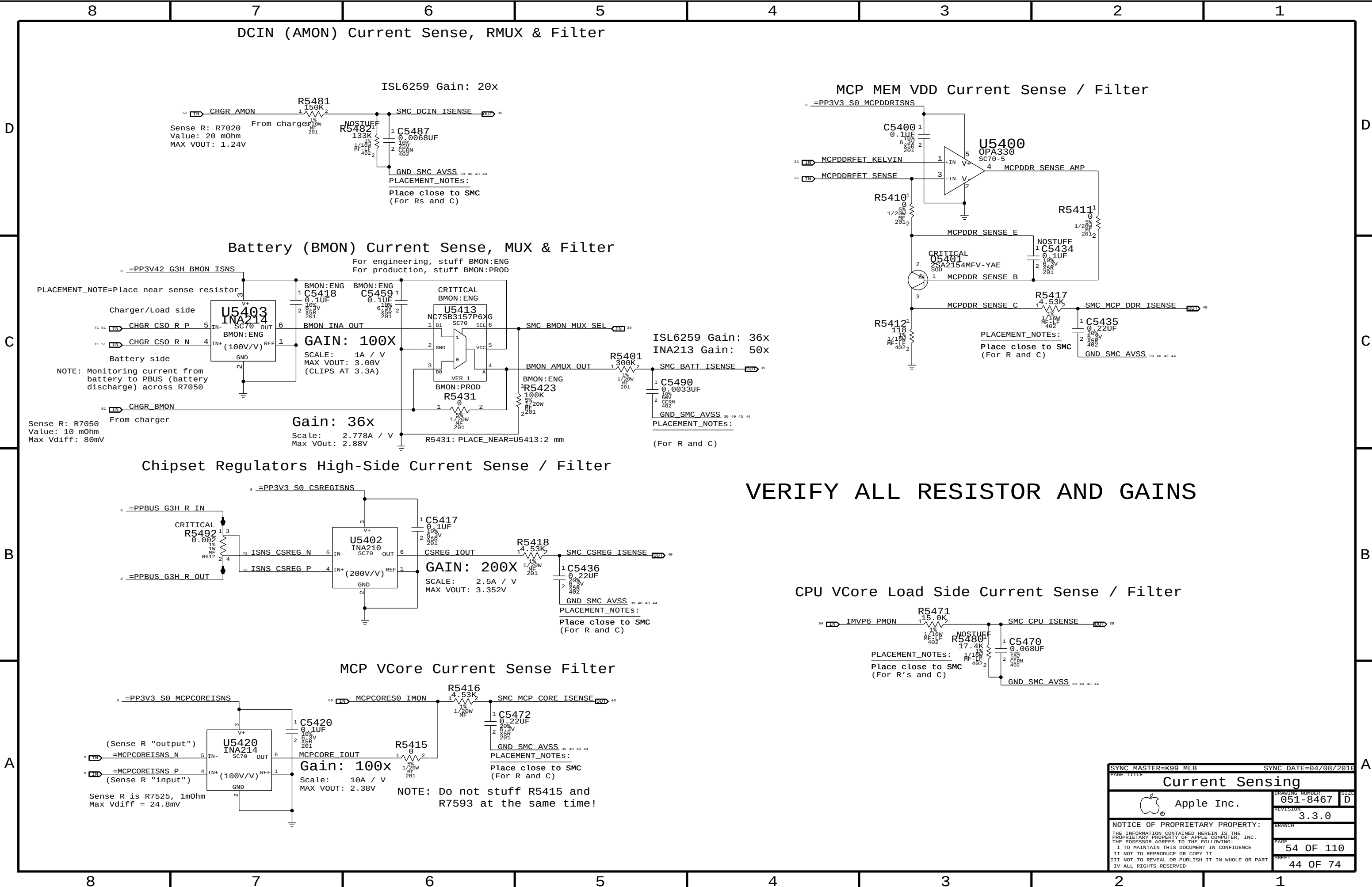
HDD Current Sense / Filter



LCD Backlight Driver Input Current Sense / Filter



|                                                                                                                             |  |                |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------|--|
| PAGE TITLE                                                                                                                  |  | DRAWING NUMBER |  |
| Voltage & Current Sensing                                                                                                   |  | 051-8467       |  |
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|                                                                                                                                                                                                                                                                                                                                   |  |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=K99 MLB                                                                                                                                                                                                                                                                                                               |  | SYNC DATE=04/08/2016 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                        |  |                      |           |
| Current Sensing                                                                                                                                                                                                                                                                                                                   |  |                      |           |
|  Apple Inc.                                                                                                                                                                                                                                  |  | DRAWING NUMBER       | 051-8467  |
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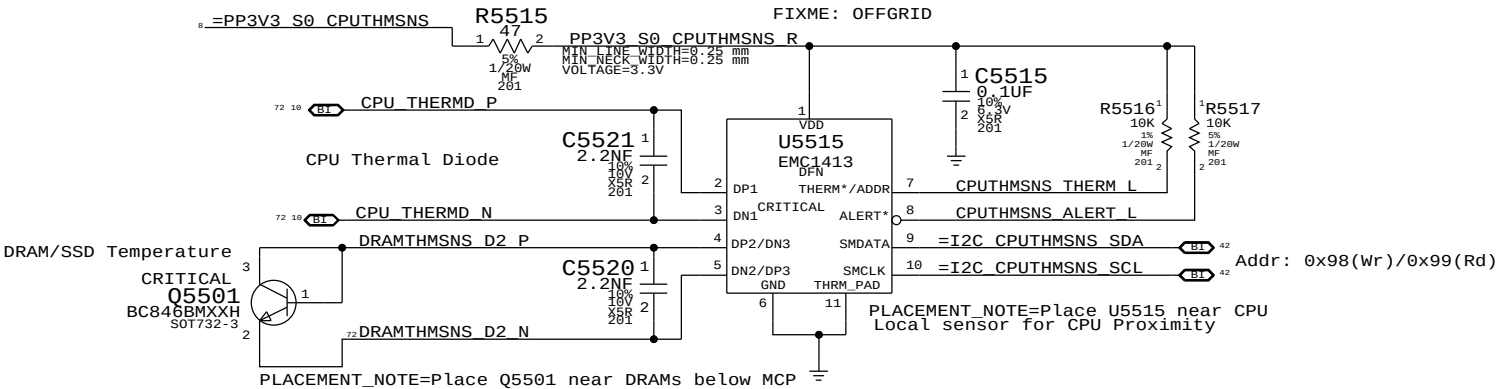
D

C

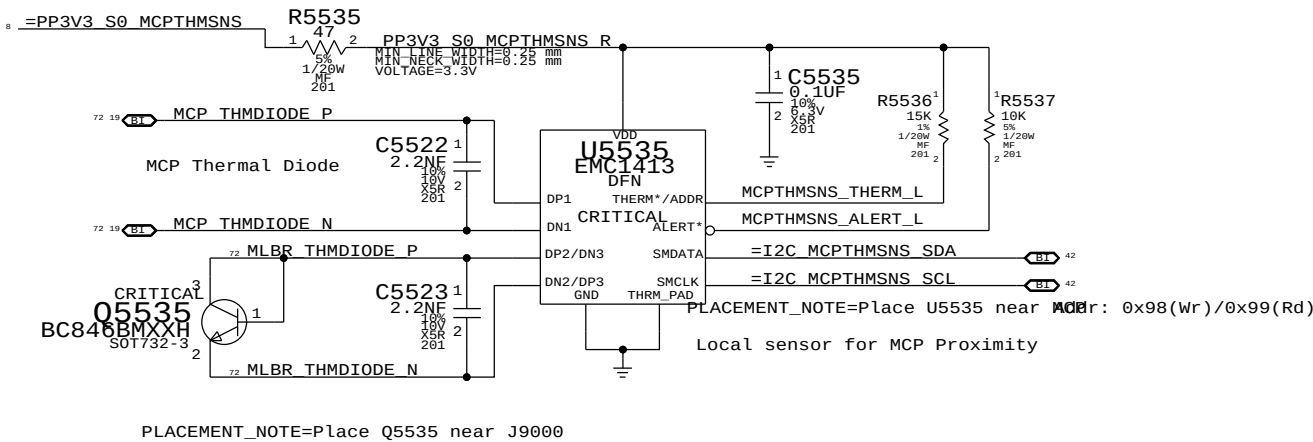
B

A

# CPU T-Diode Thermal Sensor

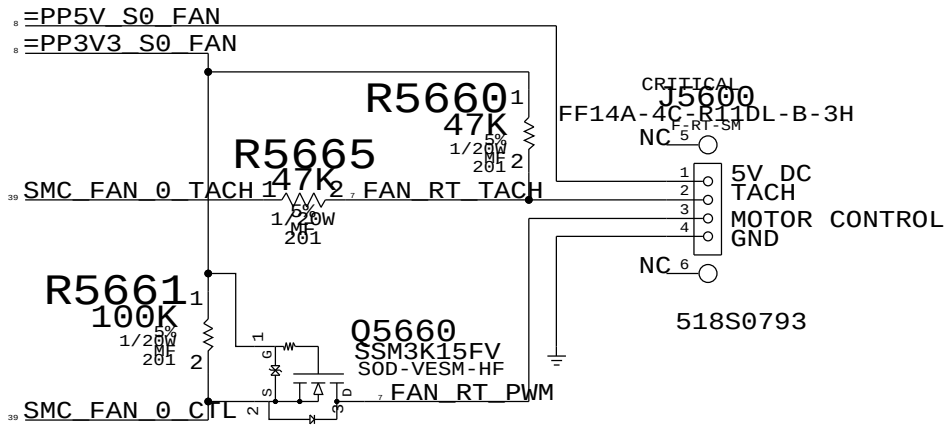


# MCP T-Diode Thermal Sensor



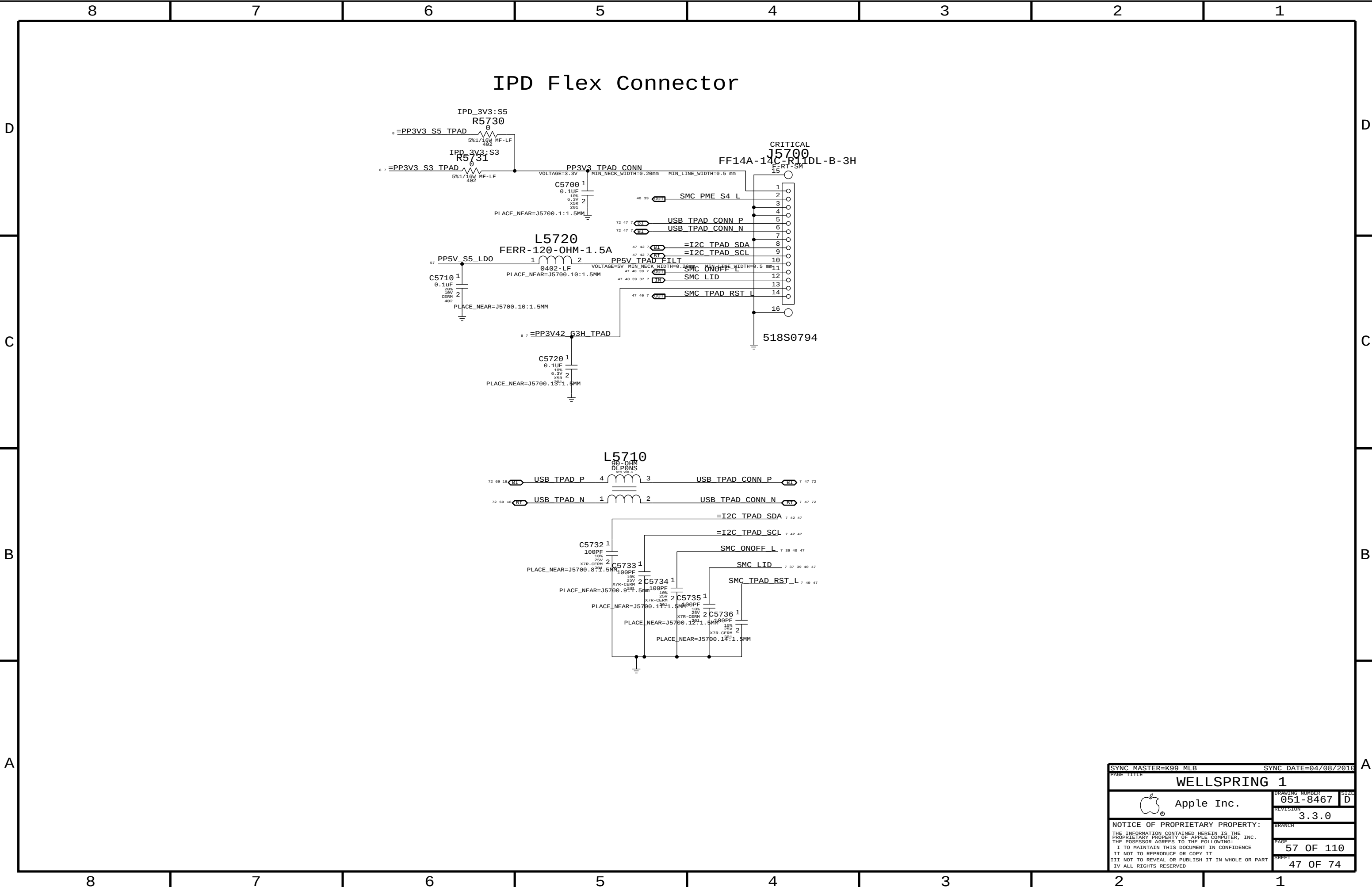
|                                                                                                                             |  |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=K99 MLB                                                                                                         |  | SYNC DATE=04/08/2016 |           |
| PAGE TITLE                                                                                                                  |  | Thermal Sensors      |           |
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# FAN CONNECTOR



|                                                                                                                                                                                                                                                                                                                             |  |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC_MASTER=K99_MLB                                                                                                                                                                                                                                                                                                         |  | SYNC_DATE=04/08/2016 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  | Fan                  |           |
|                                                                                                                                                                                                                                                                                                                             |  | DRAWING NUMBER       | 051-8467  |
|                                                                                                                                                                                                                                                                                                                             |  | REVISION             | 3.3.0     |
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|                                                                                                                                                                                                                                                                                                                             |  | PAGE                 | 56 OF 110 |
|                                                                                                                                                                                                                                                                                                                             |  | SHEET                | 46 OF 74  |









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## D



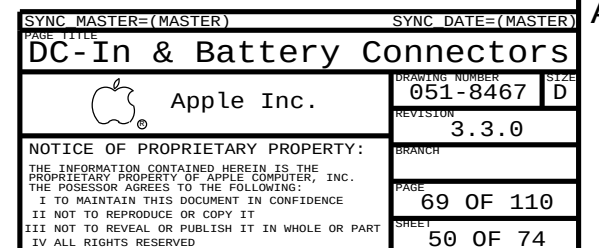
## B



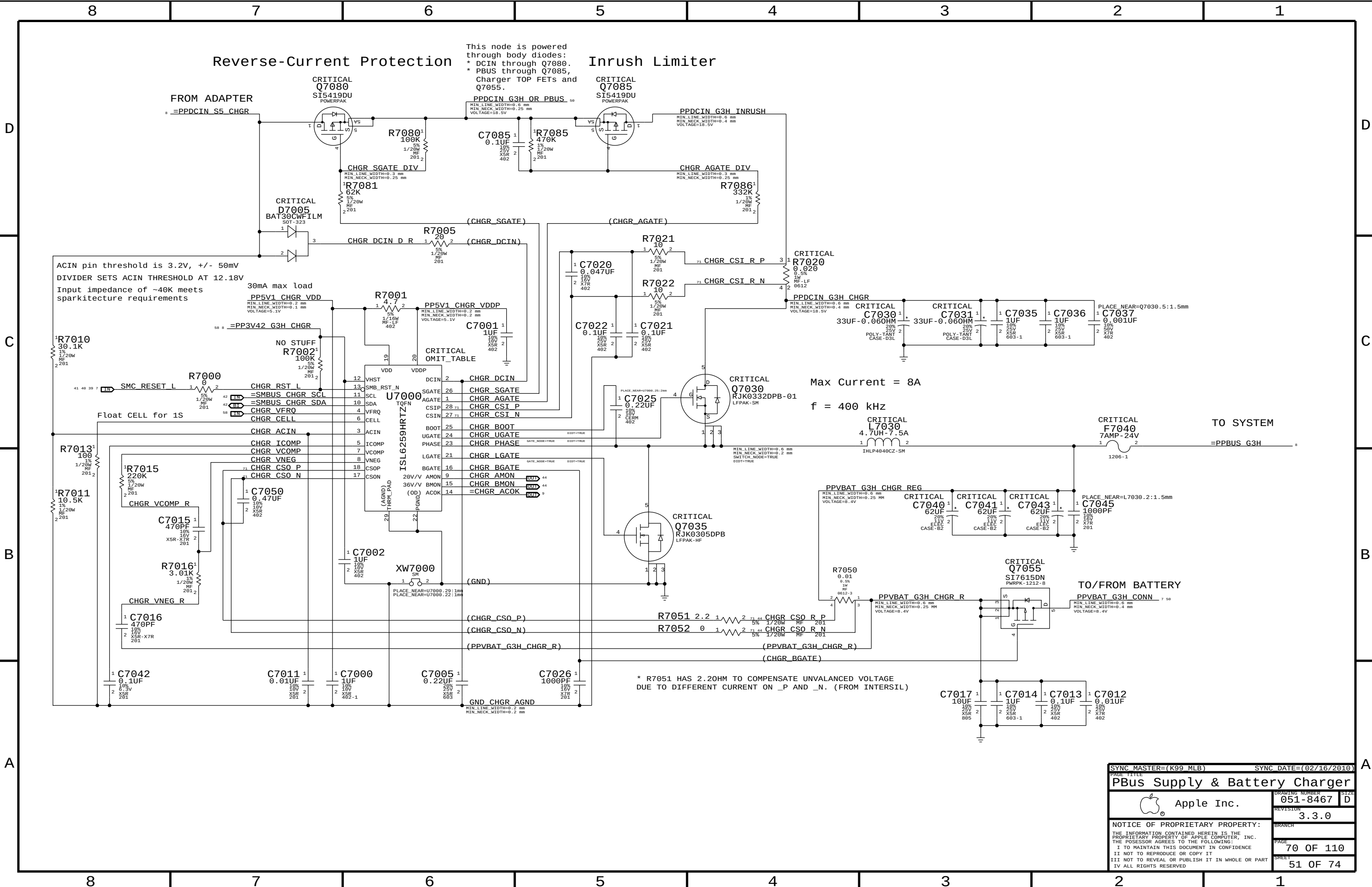
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This node is powered through body diodes:  
\* DCIN through Q7080.  
\* PBUS through Q7085, charger TOP FETs and Q7055.

**Inrush Limiter**

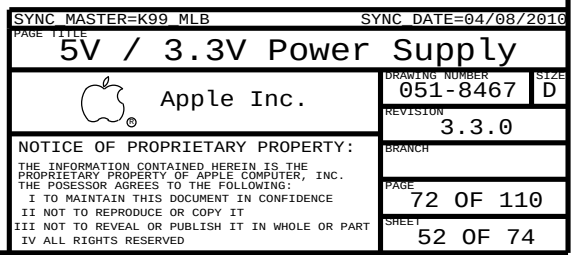
**Reverse-Current Protection**

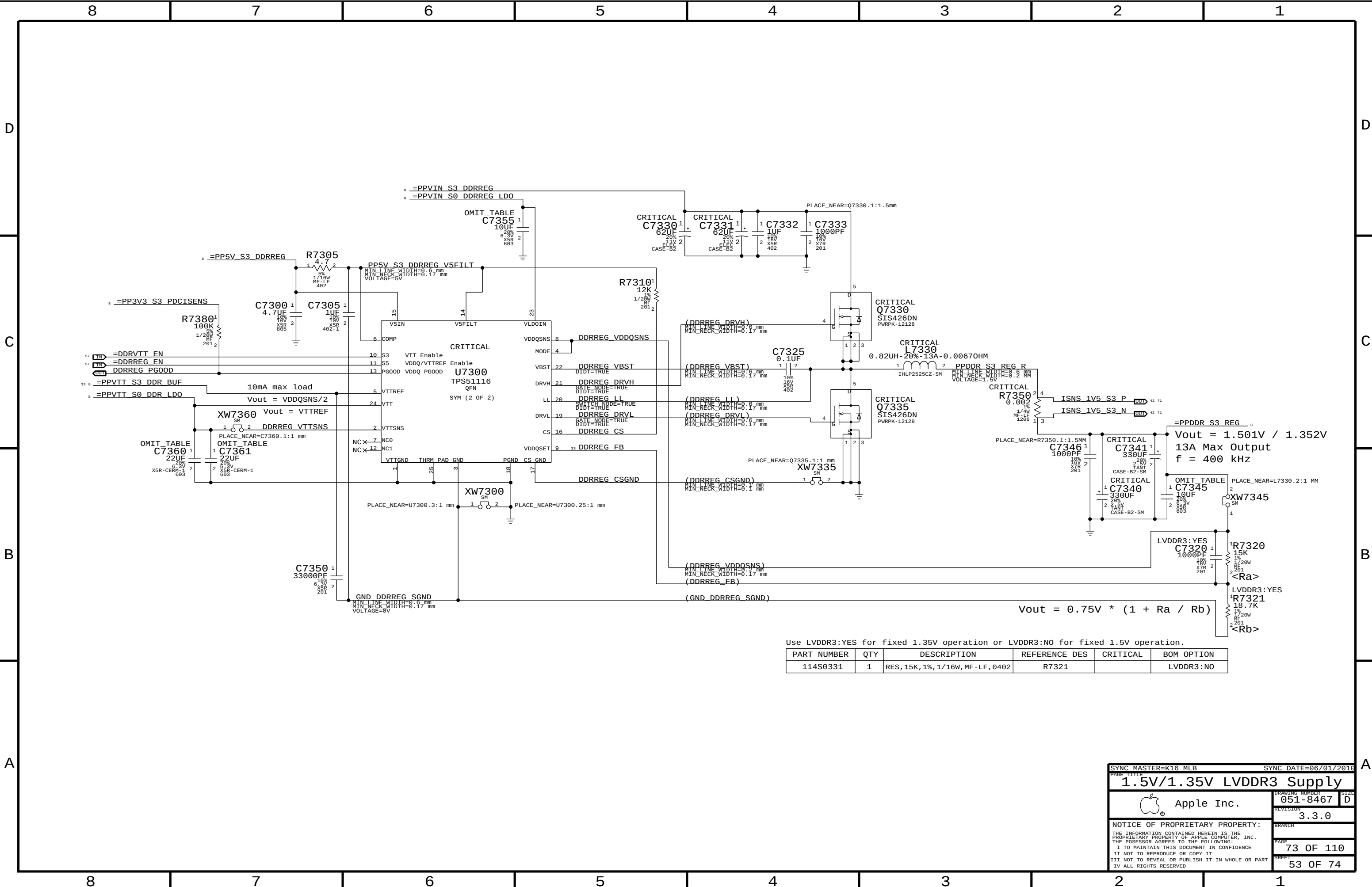
Max Current = 8A

f = 400 kHz

\* R7051 HAS 2.20HM TO COMPENSATE UNBALANCED VOLTAGE DUE TO DIFFERENT CURRENT ON \_P AND \_N. (FROM INTERSIL)

|                                                                                                                                                                                                                                                                                                                             |  |                               |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------|-----------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  | PAGE TITLE                    |           |
| PBus Supply & Battery Charger                                                                                                                                                                                                                                                                                               |  | PBus Supply & Battery Charger |           |
| Apple Inc.                                                                                                                                                                                                                                                                                                                  |  | DRAWING NUMBER                | 051-8467  |
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|                                                                                                                                                                                                                                                                                                                             |  | PAGE                          | 70 OF 110 |
|                                                                                                                                                                                                                                                                                                                             |  | SHEET                         | 51 OF 74  |





Use LVDDR3:YES for fixed 1.35V operation or LVDDR3:NO for fixed 1.5V operation.

| PART NUMBER | QTY | DESCRIPTION                      | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|----------------------------------|---------------|----------|------------|
| 114S0331    | 1   | RES, 15K, 1%, 1/16W, MF-LF, 0402 | R7321         |          | LVDDR3:NO  |

SYNC MASTER=K16 MLB

SYNC DATE=06/01/2016

1.5V/1.35V LVDDR3 Supply

Apple Inc.

051-8467

3.3.0

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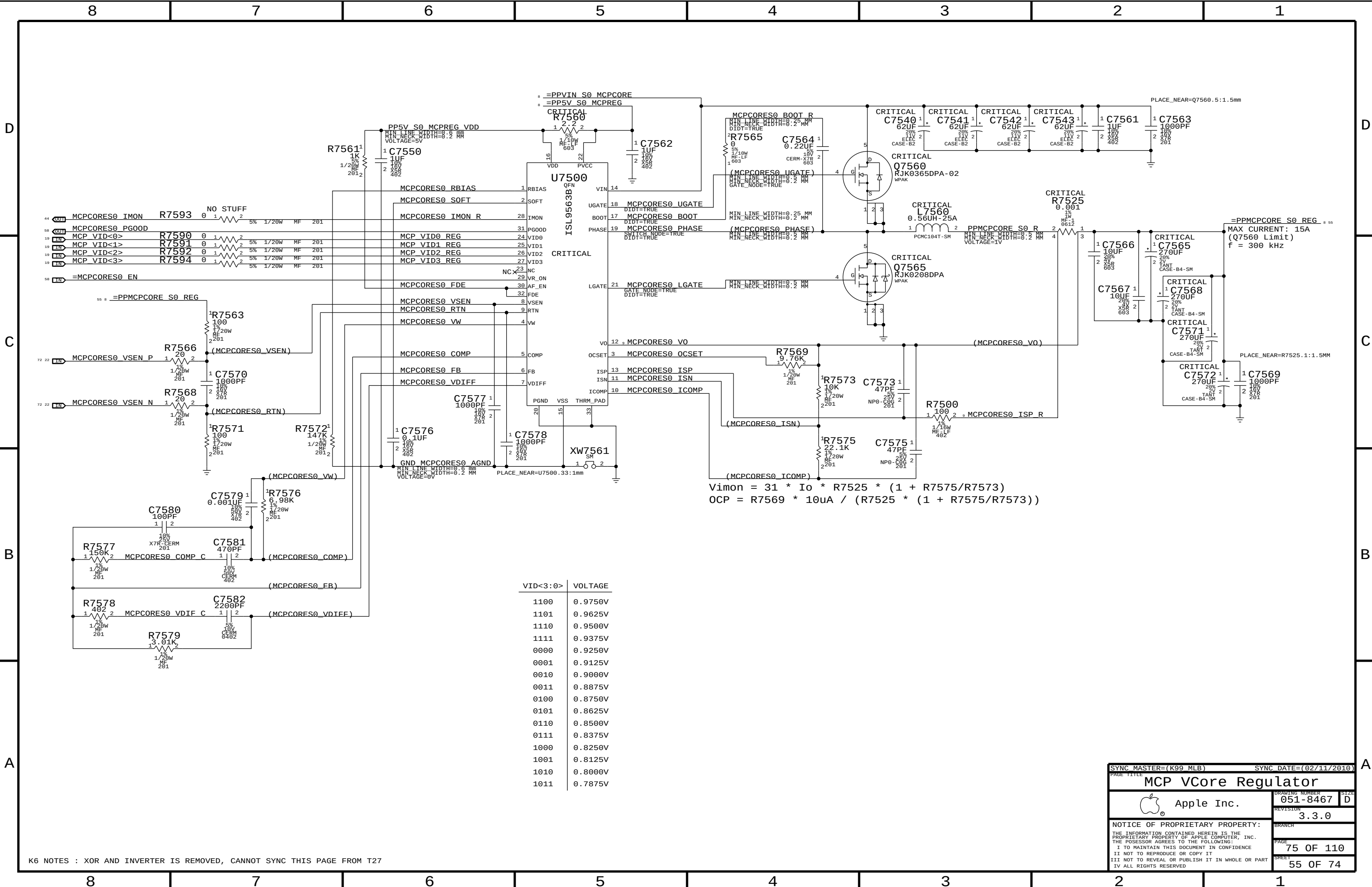
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K6 NOTES : XOR AND INVERTER IS REMOVED, CANNOT SYNC THIS PAGE FROM T27

SYNC\_MASTER=(K99\_MLB)

SYNC\_DATE=(02/11/2010)

MCP VCore Regulator

Apple Inc.

051-8467

3.3.0

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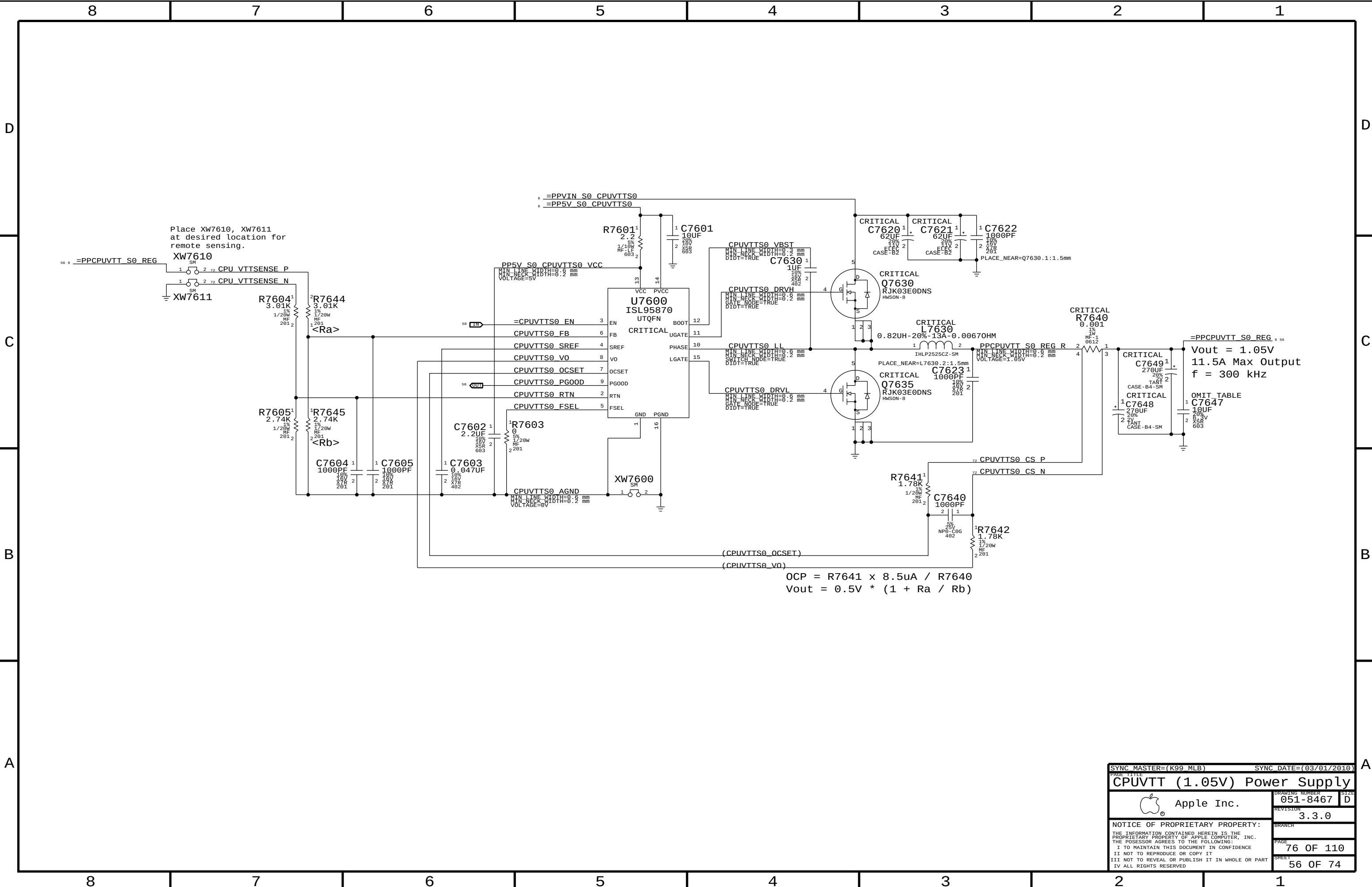
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| VID<3:0> | VOLTAGE |
|----------|---------|
| 1100     | 0.9750V |
| 1101     | 0.9625V |
| 1110     | 0.9500V |
| 1111     | 0.9375V |
| 0000     | 0.9250V |
| 0001     | 0.9125V |
| 0010     | 0.9000V |
| 0011     | 0.8875V |
| 0100     | 0.8750V |
| 0101     | 0.8625V |
| 0110     | 0.8500V |
| 0111     | 0.8375V |
| 1000     | 0.8250V |
| 1001     | 0.8125V |
| 1010     | 0.8000V |
| 1011     | 0.7875V |

$$V_{im} = 31 * I_o * R_{7525} * (1 + R_{7575}/R_{7573})$$
$$OCP = R_{7569} * 10\mu A / (R_{7525} * (1 + R_{7575}/R_{7573}))$$



|                             |  |                |  |
|-----------------------------|--|----------------|--|
| PAGE TITLE                  |  | DRAWING NUMBER |  |
| CPUVTT (1.05V) Power Supply |  | 051-8467       |  |
| REVISION                    |  | 3.3.0          |  |
| BRANCH                      |  | 76 OF 110      |  |
| SHEET                       |  | 56 OF 74       |  |

SYNC MASTER=(K99 MLB)

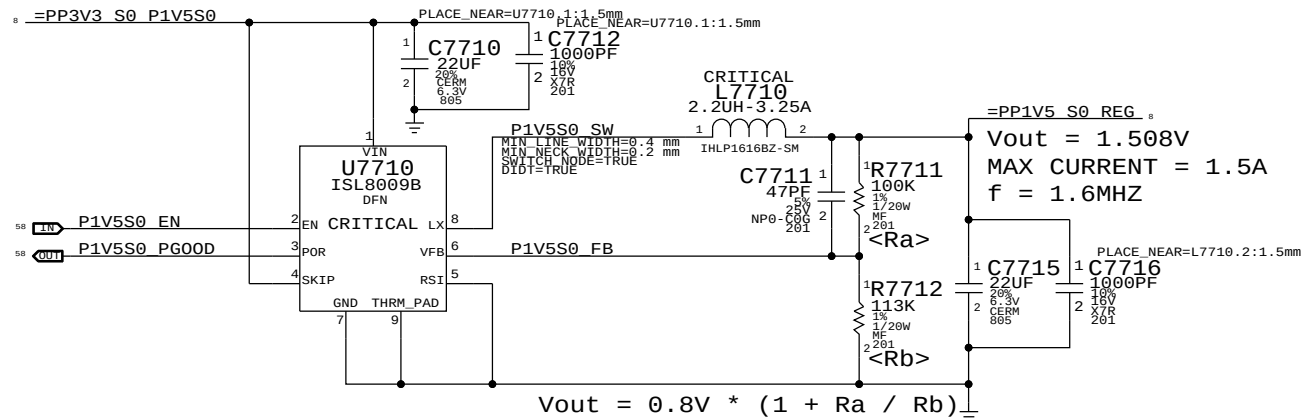
SYNC DATE=(03/01/2010)

Apple Inc.

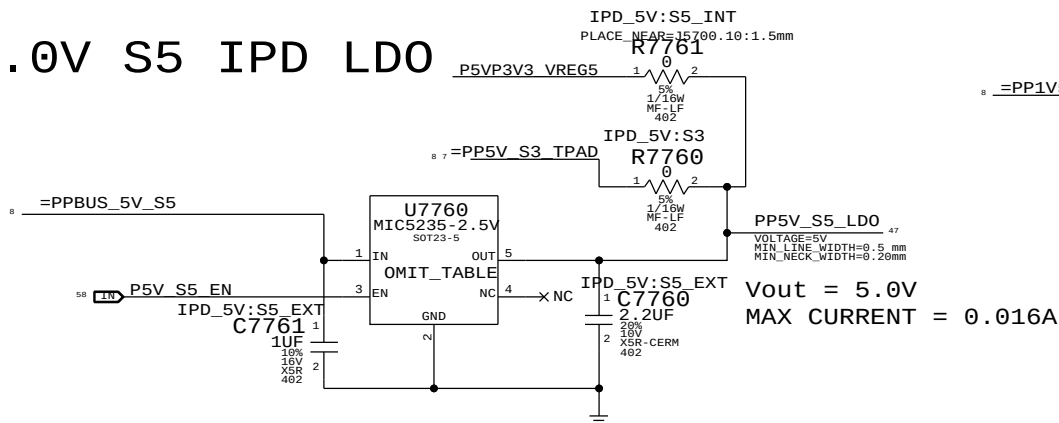
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## 1.5V S0 Regulator

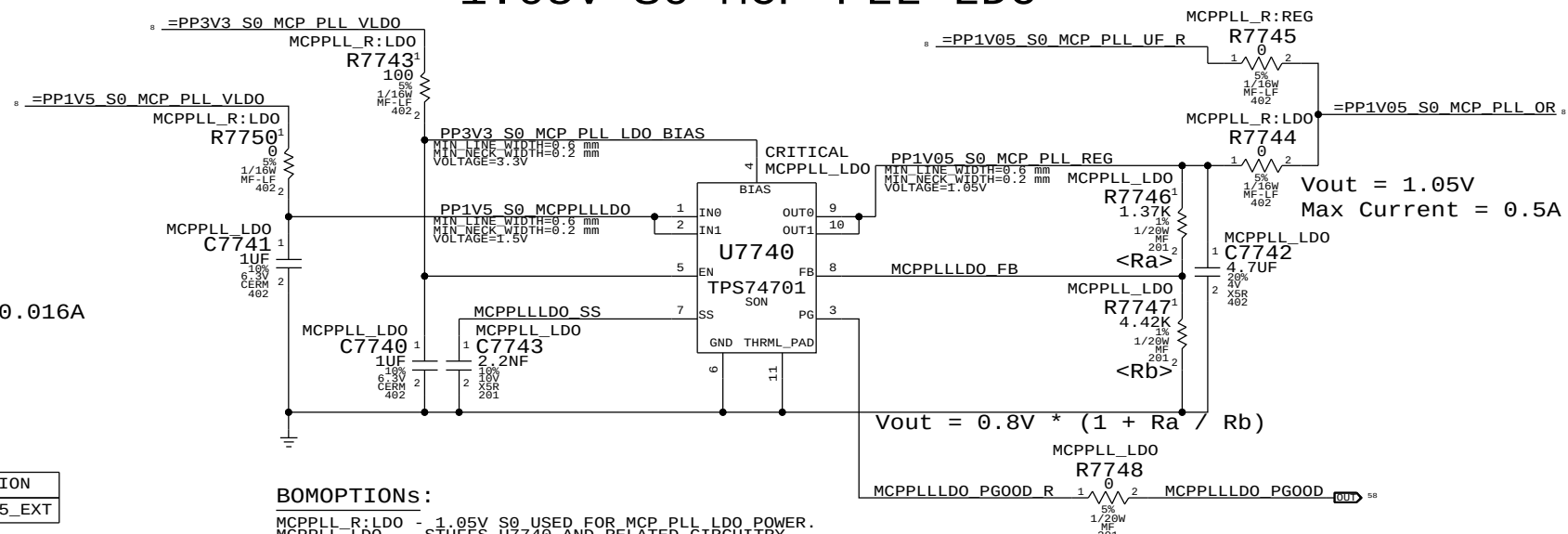


## 5.0V S5 IPD LDO



| PART NUMBER | QTY | DESCRIPTION                              | REFERENCE DES | CRITICAL | BOM OPTION    |
|-------------|-----|------------------------------------------|---------------|----------|---------------|
| 353S3034    | 1   | IC, LDO, MIC5235, 5V, 1%, 150mA, SOT23-5 | U7760         |          | IPD_5V:S5_EXT |

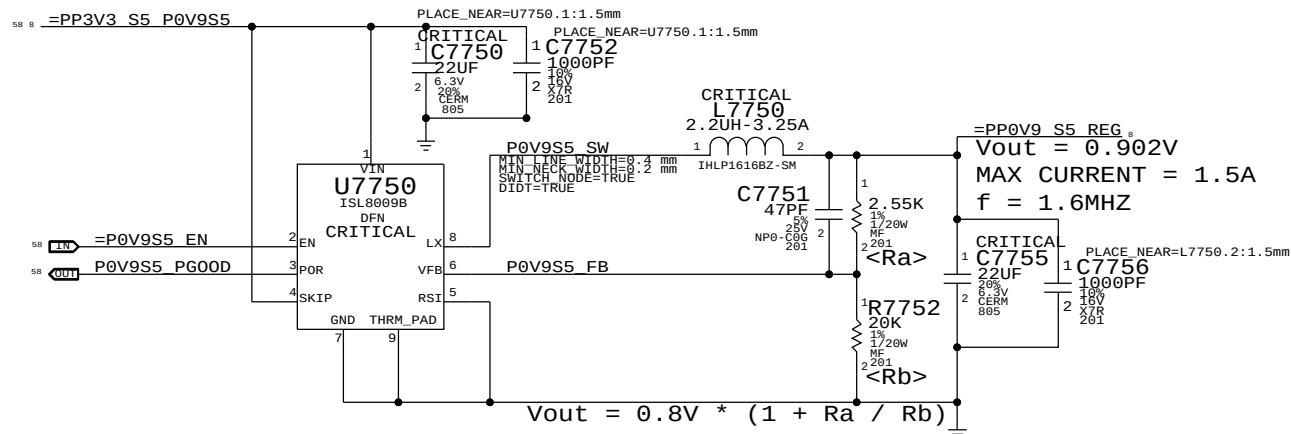
## 1.05V S0 MCP PLL LDO



### BOMOPTIONS:

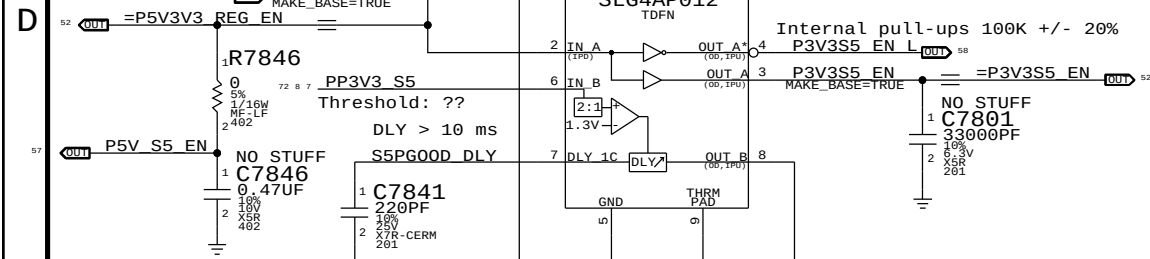
MCPPLL\_R:LD0 - 1.05V S0 USED FOR MCP PLL LDO POWER.  
MCPPLL\_LD0 - STUFFS U7740 AND RELATED CIRCUITRY.  
TO USE U7740, MCPPLL\_R:LD0 AND MCPPLL\_LD0 MUST BE ACTIVE.  
TO USE 1.05V S0, MCPPLL\_R:REG MUST BE ACTIVE, MCPPLL\_LD0 CAN BE ACTIVE, MCPPLL\_R:LD0 MUST BE INACTIVE.

## MCP 0.9V S5 (AUXC) Switcher

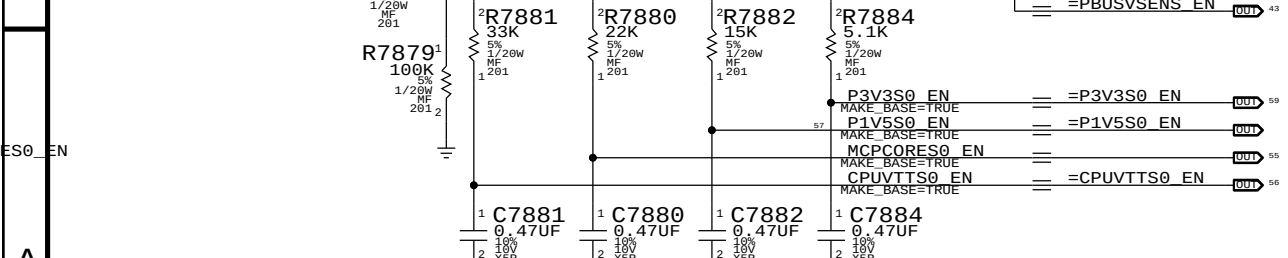
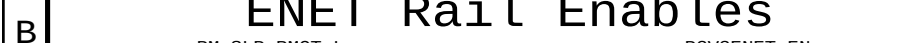
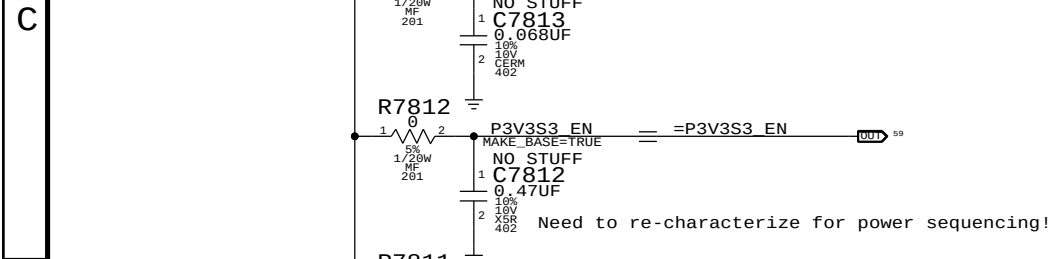


|                                                                                                                             |  |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=K99 MLB                                                                                                         |  | SYNC DATE=04/08/2016 |           |
| PAGE TITLE                                                                                                                  |  | Misc Power Supplies  |           |
| Apple Inc.                                                                                                                  |  | DRAWING NUMBER       | 051-8467  |
|                                                                                                                             |  | REVISION             | 3.3.0     |
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|   |   |  |
|---|---|--|
| 8 | 7 |  |
|---|---|--|

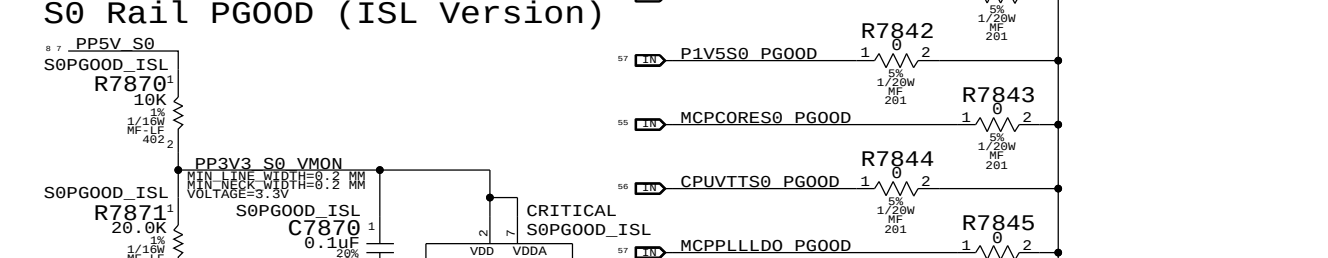


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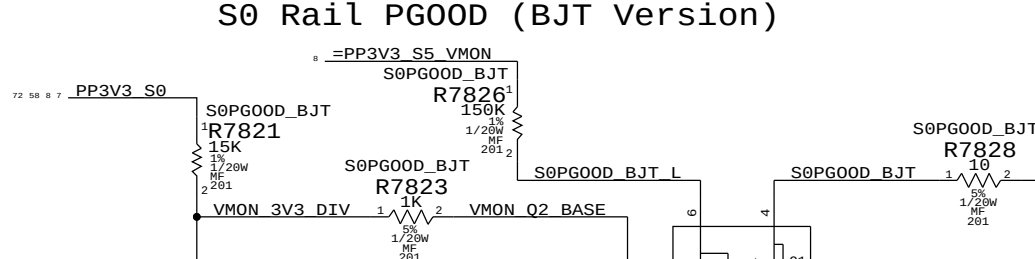


VTT rail must ramp up in about the same time as MEMVDD rail (Q2300).

|   |   |
|---|---|
| 5 | 4 |
|---|---|



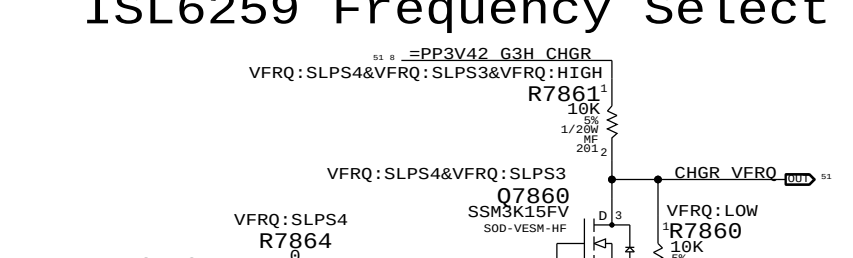
610V 



|   |   |
|---|---|
| 2 | 1 |
|---|---|

## Power Control Signals

| State               | SMC_PM_G2_ENABLE | PM_SLP_S4_L | PM_SLP_S3_L |
|---------------------|------------------|-------------|-------------|
| Run (S0)            | 1                | 1           | 1           |
| Sleep (S3)          | 1                | 1           | 0           |
| Soft-Off (S5)       | 1                | 0           | 0           |
| Battery Off (G3Hot) | 0                | 0           | 0           |



1000

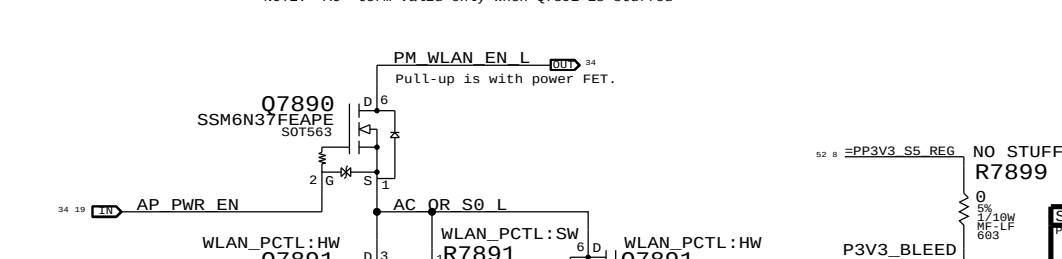
## Enable Genera

## WLAN Enable Generation

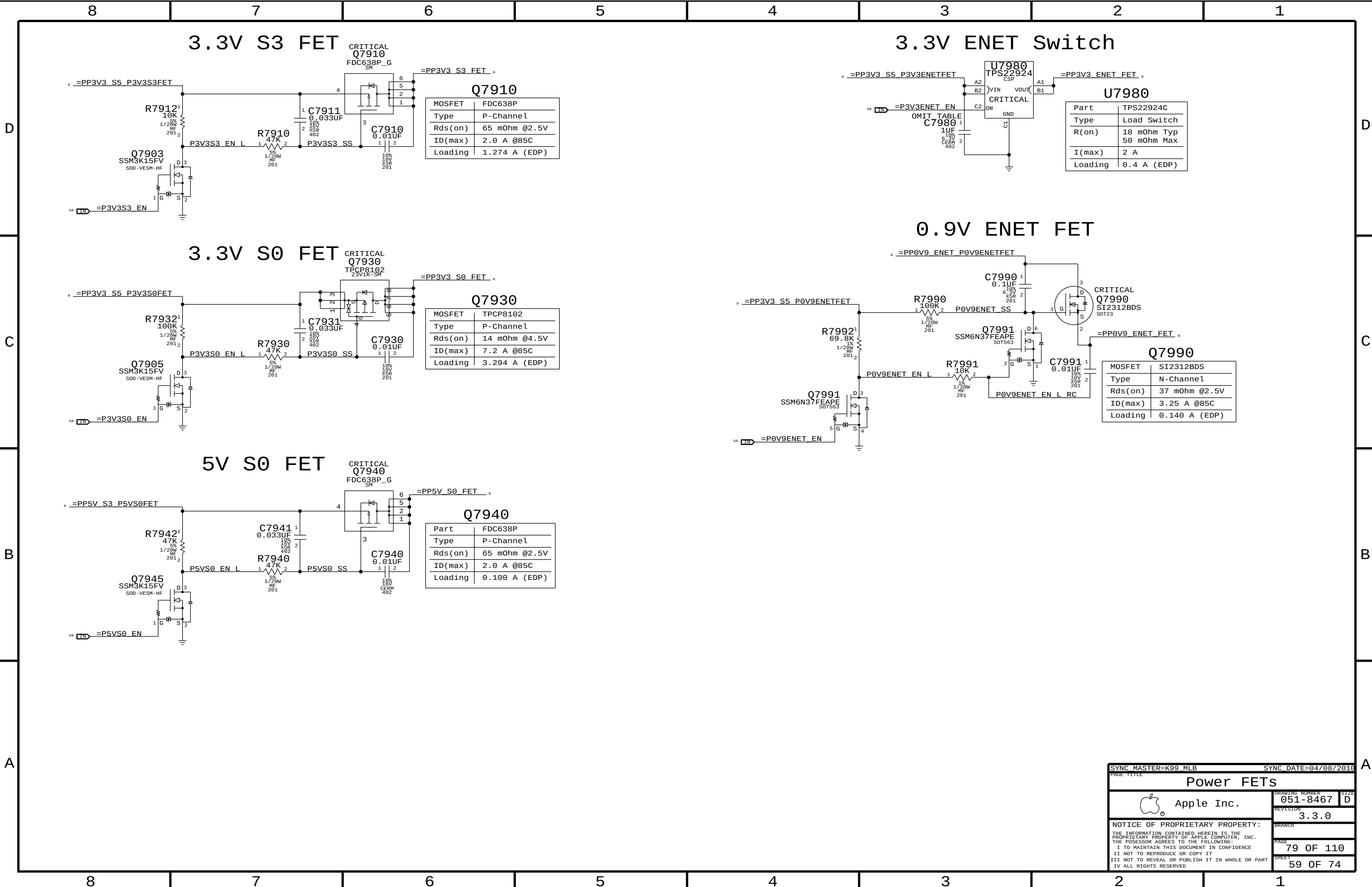
```
"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))
```

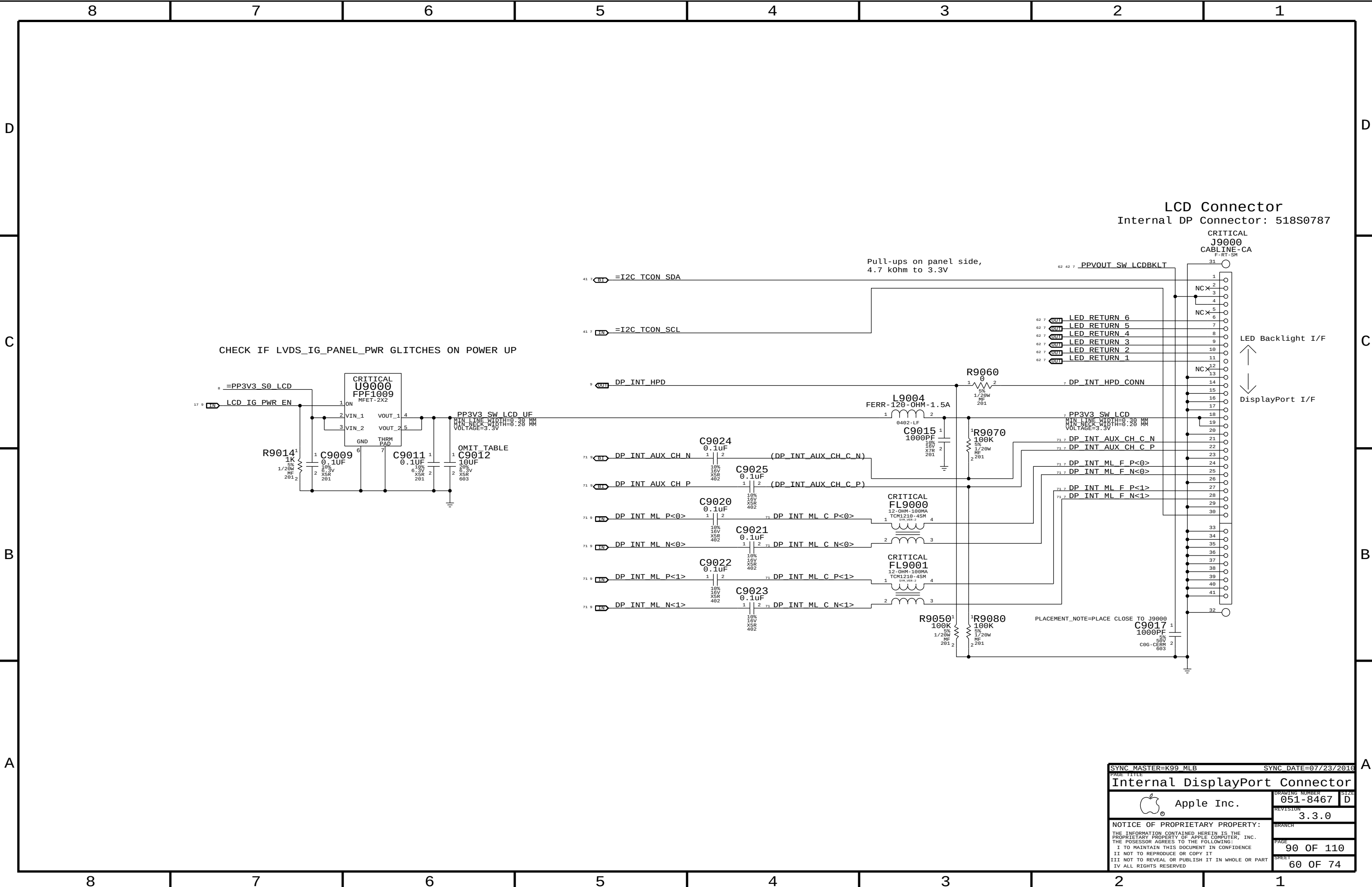
NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP\_PWR\_EN signal.

NOTE: "AC" term valid only when Q7891 is stuffed



|                                                                                                                                                                                                                                                                                                                                   |  |                     |  |                      |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------|--|----------------------|--|
| OFF<br>9                                                                                                                                                                                                                                                                                                                          |  | SYNC MASTER=K99 MLB |  | SYNC DATE=04/08/2010 |  |
| PAGE TITLE<br><b>Power Sequencing</b>                                                                                                                                                                                                                                                                                             |  |                     |  |                      |  |
|  Apple Inc.                                                                                                                                                                                                                                  |  | DRAWING NUMBER      |  | SIZE                 |  |
|                                                                                                                                                                                                                                                                                                                                   |  | 051-8467            |  | D                    |  |
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|                                                                                                                                                                                                                                                                                                                                   |  | PAGE                |  | 78 OF 110            |  |
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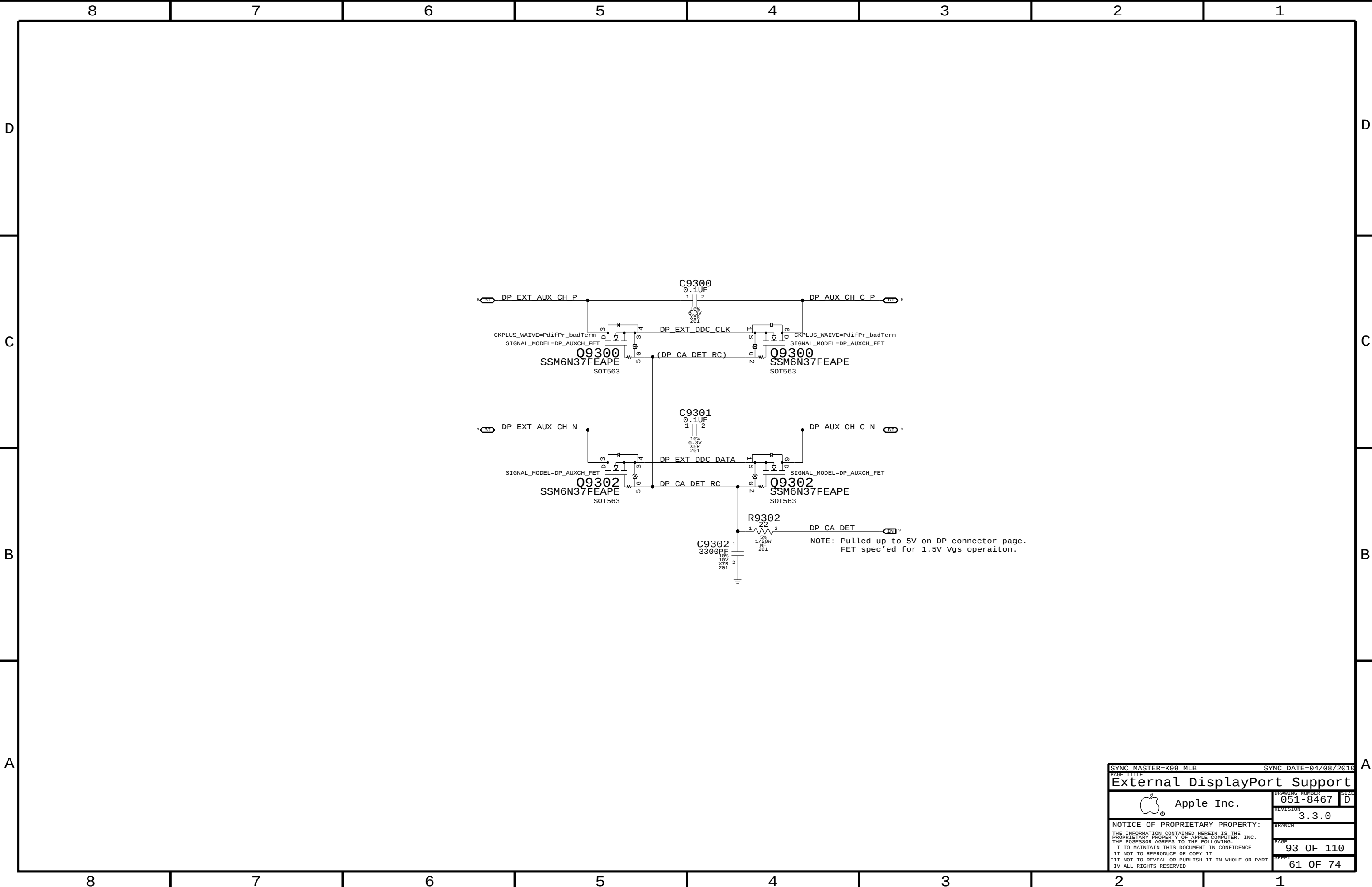


LCD Connector  
Internal DP Connector: 518S0787

CRITICAL  
J9000  
CABLINE-CA  
F-RT-SM

LED Backlight I/F  
↑  
↓  
DisplayPort I/F

|                                                                                                                                                                                                                                                                                                                                       |  |                      |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=K99 MLB                                                                                                                                                                                                                                                                                                                   |  | SYNC DATE=07/23/2016 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                            |  |                      |           |
| Internal DisplayPort Connector                                                                                                                                                                                                                                                                                                        |  |                      |           |
|  Apple Inc.                                                                                                                                                                                                                                      |  | DRAWING NUMBER       | 051-8467  |
|                                                                                                                                                                                                                                                                                                                                       |  | SIZE                 | D         |
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|                                                                                                                                                                                                                                                                                                                                       |  | PAGE                 | 90 OF 110 |
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SYNC MASTER=K99 MLB

SYNC DATE=04/08/2016

PAGE TITLE

External DisplayPort Support

 Apple Inc.

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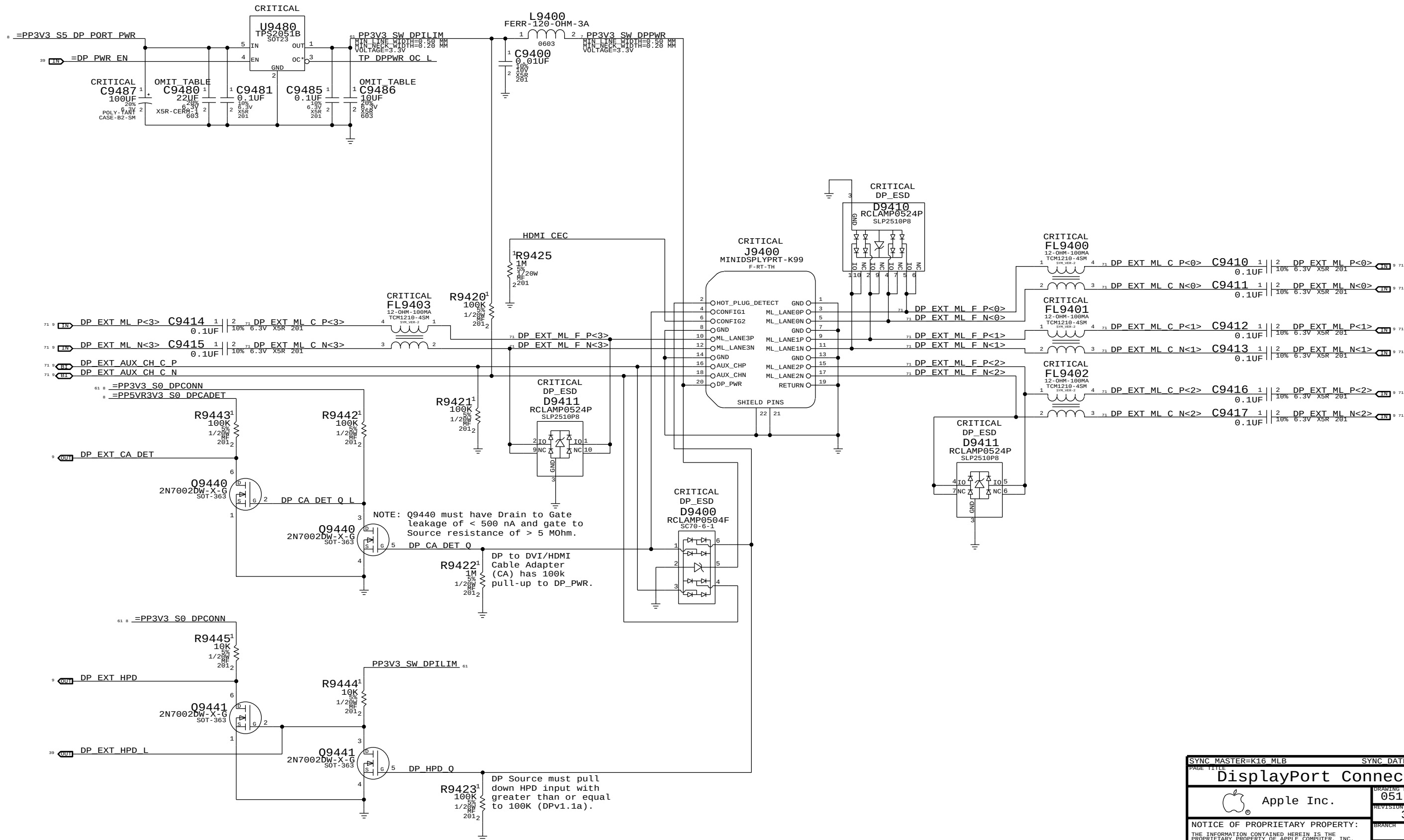
PAGE

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SHEET

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# Port Power Switch



|                                                                                                                             |  |                      |          |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|----------|
| SYNC MASTER=K16 MLB                                                                                                         |  | SYNC DATE=06/01/2016 |          |
| PAGE TITLE                                                                                                                  |  |                      |          |
| DisplayPort Connector                                                                                                       |  |                      |          |
|  Apple Inc.                            |  | DRAWING NUMBER       | 051-8467 |
|                                                                                                                             |  | SIZE                 | D        |
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D

C

B

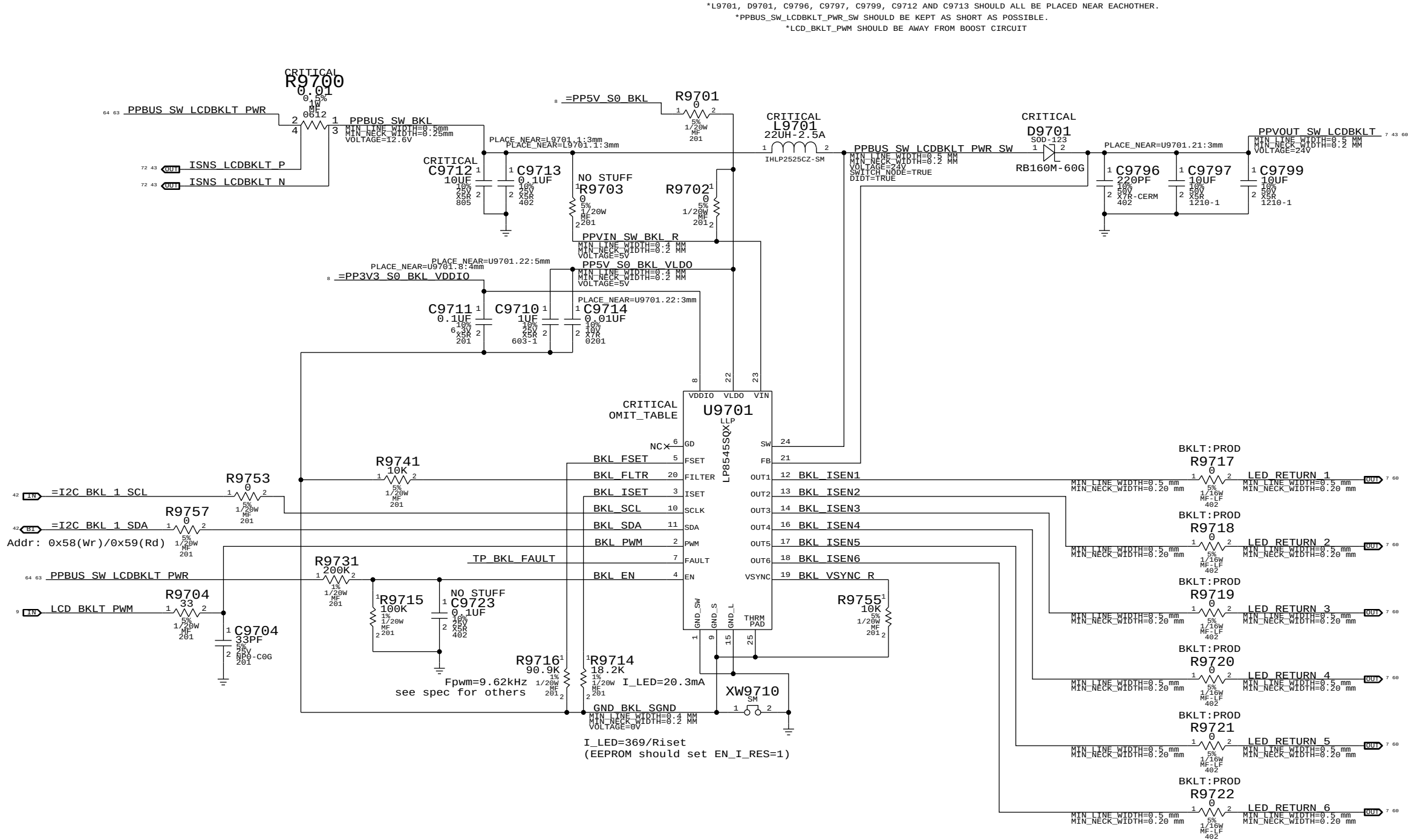
A

D

C

B

A

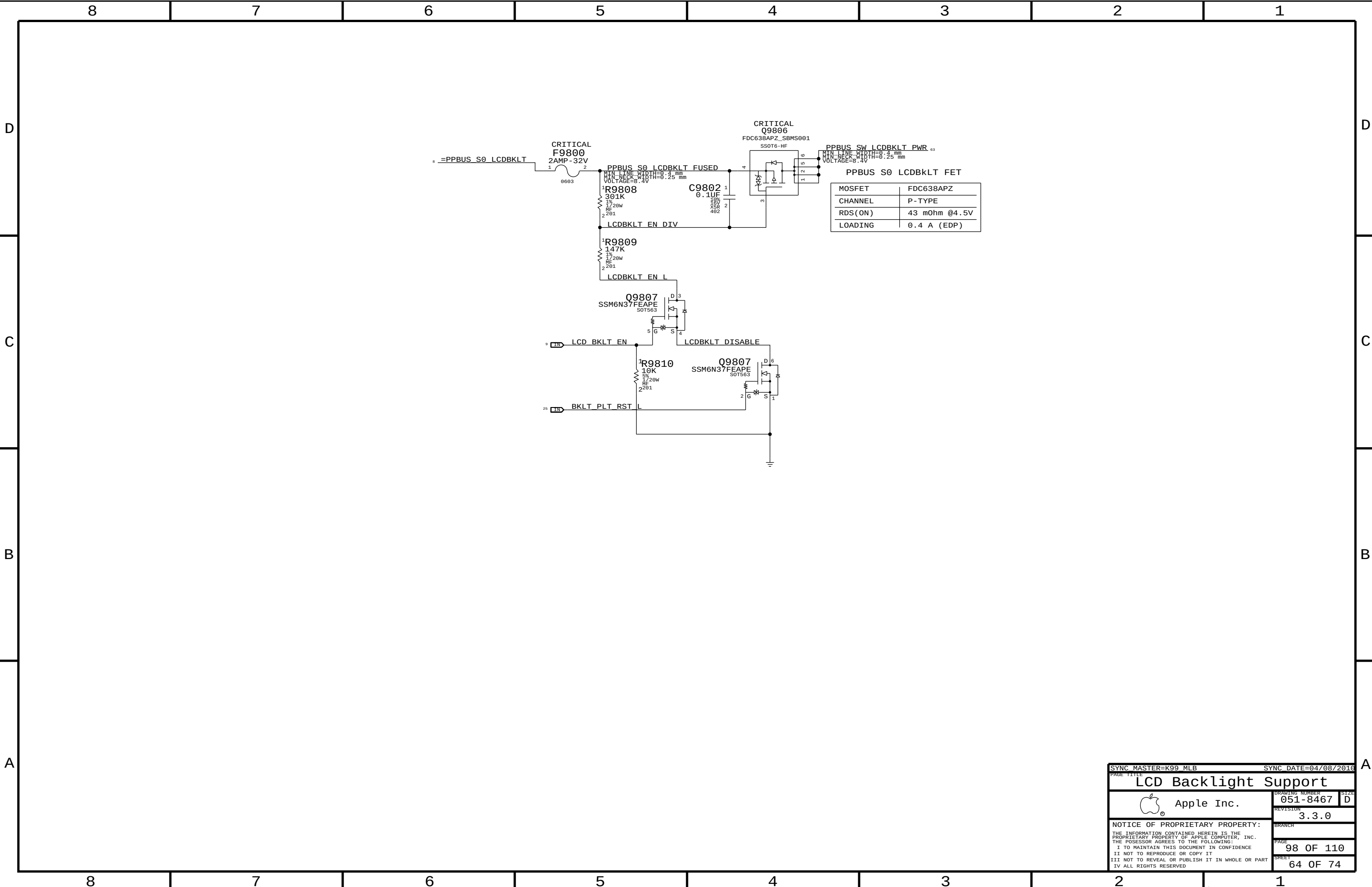


FOR LP8543:  
STUFF R9741  
NO STUFF R9740, C9740, C9741, R9754

| PART NUMBER | QTY |                                                | REFERENCE DES       | CRITICAL | BOM OPTION |
|-------------|-----|------------------------------------------------|---------------------|----------|------------|
| 103S0198    | 3   | RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM | R9717, R9718, R9719 |          | BKLT:ENG   |
| 103S0198    | 3   | RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM | R9720, R9721, R9722 |          | BKLT:ENG   |
| 353S2896    | 1   | IC, LP8545, LED BKLT CTRLR, PRODUCTIO, LLP24   | U9701               | CRITICAL | PROJ:K16   |
| 353S2967    | 1   | IC, LP8545, LED BKLT CTRLR, LLP24, K99 VER     | U9701               | CRITICAL | PROJ:K99   |

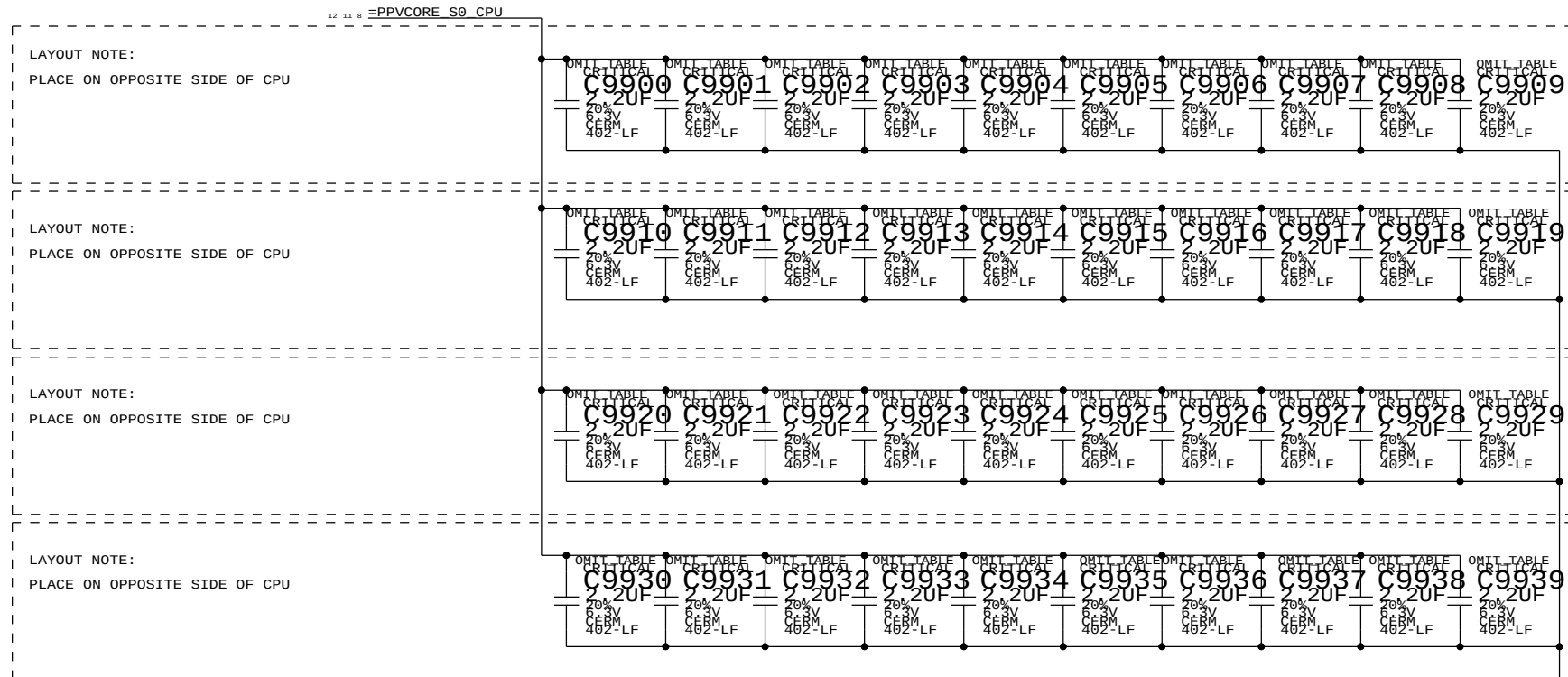
10.2 ohm resistors for current measurement on LED strings.

|                                                                                                                                                                                                                                                                                                                                   |  |                        |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------|-----------|
| SYNC MASTER=(K99 MLB)                                                                                                                                                                                                                                                                                                             |  | SYNC DATE=(03/01/2010) |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                        |  |                        |           |
| LCD Backlight Driver                                                                                                                                                                                                                                                                                                              |  |                        |           |
|  Apple Inc.                                                                                                                                                                                                                                  |  | DRAWING NUMBER         | 051-8467  |
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|                                                                                                                                                                                                                                                                                                                                   |  | PAGE                   | 97 OF 110 |
|                                                                                                                                                                                                                                                                                                                                   |  | SHEET                  | 63 OF 74  |





ADDITIONAL CPU VCORE HF DECOUPLING  
40x 1uF 0402





## Memory Bus Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| MEM_50S           | *     | =50_OHM_SE            | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          | =STANDARD            | =STANDARD         |
| MEM_55S           | *     | =55_OHM_SE            | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =STANDARD            | =STANDARD         |
| MEM_70D           | *     | =70_OHM_DIFF          | =70_OHM_DIFF       | =70_OHM_DIFF       | =70_OHM_DIFF        | =70_OHM_DIFF         | =70_OHM_DIFF      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| MEM_CLK2MEM      | *     | =4:1_SPACING         | ?      |
| MEM_CTRL2CTRL    | *     | =2:1_SPACING         | ?      |
| MEM_CTRL2MEM     | *     | =2.5:1_SPACING       | ?      |
| MEM_CMD2CMD      | *     | =1.5:1_SPACING       | ?      |
| MEM_CMD2MEM      | *     | =3:1_SPACING         | ?      |
| MEM_DATA2DATA    | *     | =1.5:1_SPACING       | ?      |
| MEM_DATA2MEM     | *     | =3:1_SPACING         | ?      |
| MEM_DQS2MEM      | *     | =3:1_SPACING         | ?      |
| MEM_20THER       | *     | 25 MIL               | ?      |

NV DG says 3x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 2x inner, 4x outer

NV DG says 4x inner, 5x outer

## Memory Bus Spacing Group Assignments

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CLK           | MEM_CLK           | *         | MEM_CLK2MEM      |
| MEM_CLK           | MEM_CTRL          | *         | MEM_CLK2MEM      |
| MEM_CLK           | MEM_CMD           | *         | MEM_CLK2MEM      |
| MEM_CLK           | MEM_DATA          | *         | MEM_CLK2MEM      |
| MEM_CLK           | MEM_DQS           | *         | MEM_CLK2MEM      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CMD           | MEM_CLK           | *         | MEM_CMD2MEM      |
| MEM_CMD           | MEM_CTRL          | *         | MEM_CMD2MEM      |
| MEM_CMD           | MEM_CMD           | *         | MEM_CMD2CMD      |
| MEM_CMD           | MEM_DATA          | *         | MEM_CMD2MEM      |
| MEM_CMD           | MEM_DQS           | *         | MEM_CMD2MEM      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CTRL          | MEM_CLK           | *         | MEM_CTRL2MEM     |
| MEM_CTRL          | MEM_CTRL          | *         | MEM_CTRL2CTRL    |
| MEM_CTRL          | MEM_CMD           | *         | MEM_CTRL2MEM     |
| MEM_CTRL          | MEM_DATA          | *         | MEM_CTRL2MEM     |
| MEM_CTRL          | MEM_DQS           | *         | MEM_CTRL2MEM     |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_DATA          | MEM_CLK           | *         | MEM_DATA2MEM     |
| MEM_DATA          | MEM_CTRL          | *         | MEM_DATA2MEM     |
| MEM_DATA          | MEM_CMD           | *         | MEM_DATA2MEM     |
| MEM_DATA          | MEM_DATA          | *         | MEM_DATA2DATA    |
| MEM_DATA          | MEM_DQS           | *         | MEM_DATA2MEM     |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_DQS           | MEM_CLK           | *         | MEM_DQS2MEM      |
| MEM_DQS           | MEM_CTRL          | *         | MEM_DQS2MEM      |
| MEM_DQS           | MEM_CMD           | *         | MEM_DQS2MEM      |
| MEM_DQS           | MEM_DATA          | *         | MEM_DQS2MEM      |
| MEM_DQS           | MEM_DQS           | *         | MEM_DQS2MEM      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CLK           | *                 | *         | MEM_20THER       |
| MEM_CTRL          | *                 | *         | MEM_20THER       |
| MEM_CMD           | *                 | *         | MEM_20THER       |
| MEM_DATA          | *                 | *         | MEM_20THER       |
| MEM_DQS           | *                 | *         | MEM_20THER       |

DDR3:

DQ signals should be matched within 5 ps of associated DQS pair.

DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 360 ps

No DQS to clock matching requirement.

CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.

CMD/CTRL signals should be matched within 150 ps.

All memory signals maximum length is 1.030 ps.

SOURCE: MCP89 Interface DG (DG-04625-001\_v0.9), Section 2.2.3

SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

## MCP MEM COMP Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| MCP_MEM_COMP      | *     | =40_OHM_SE            | =40_OHM_SE         | =40_OHM_SE         | =40_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| MCP_MEM_COMP     | *     | =2x_DIELECTRIC       | ?      |

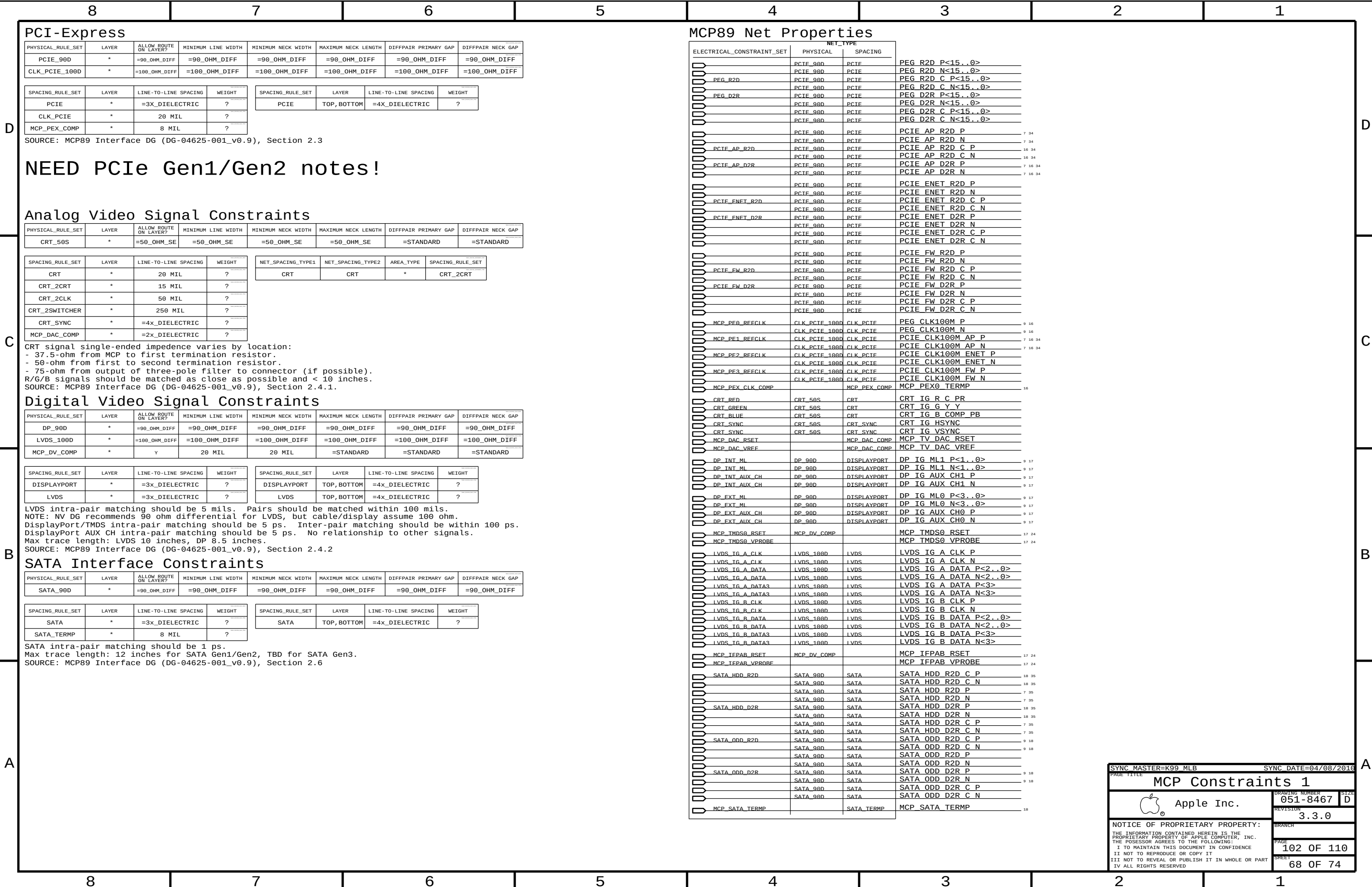
SOURCE: MCP89 Interface DG (DG-04625-001\_v0.9), Section 2.2.2

## Memory Net Properties

| ELECTRICAL_CONSTRAINT_SET | NET_TYPE     |              |                   |                |
|---------------------------|--------------|--------------|-------------------|----------------|
|                           | PHYSICAL     | SPACING      |                   |                |
| MEM_A_CLK                 | MEM_70D      | MEM_CLK      | MEM A CLK P<5..0> | 9 15 26 27 32  |
| MEM_A_CLK                 | MEM_70D      | MEM_CLK      | MEM A CLK N<5..0> | 9 15 26 27 32  |
| MEM_A_CKE                 | MEM_50S      | MEM_CTRL     | MEM A CKE<3..0>   | 15 21 26 27 32 |
| MEM_A_CNTL                | MEM_50S      | MEM_CTRL     | MEM A CS L<3..0>  | 15 26 27 32    |
| MEM_A_CNTL                | MEM_50S      | MEM_CTRL     | MEM A ODT<3..0>   | 15 26 27 32    |
| MEM_A_CMD                 | MEM_50S      | MEM_CMD      | MEM A A<15..0>    | 9 15 26 27 32  |
| MEM_A_CMD                 | MEM_50S      | MEM_CMD      | MEM A BA<2..0>    | 15 26 27 32    |
| MEM_A_CMD                 | MEM_50S      | MEM_CMD      | MEM A RAS L       | 15 26 27 32    |
| MEM_A_CMD                 | MEM_50S      | MEM_CMD      | MEM A CAS L       | 15 26 27 32    |
| MEM_A_CMD                 | MEM_50S      | MEM_CMD      | MEM A WE L        | 15 26 27 32    |
| MEM_A_DQ_BYTE0            | MEM_55S      | MEM_DATA     | MEM A DQ<7..0>    | 15 26          |
| MEM_A_DQ_BYTE1            | MEM_55S      | MEM_DATA     | MEM A DQ<15..8>   | 15 26          |
| MEM_A_DQ_BYTE2            | MEM_55S      | MEM_DATA     | MEM A DQ<23..16>  | 15 26          |
| MEM_A_DQ_BYTE3            | MEM_55S      | MEM_DATA     | MEM A DQ<31..24>  | 15 26          |
| MEM_A_DQ_BYTE4            | MEM_55S      | MEM_DATA     | MEM A DQ<39..32>  | 15 27          |
| MEM_A_DQ_BYTE5            | MEM_55S      | MEM_DATA     | MEM A DQ<47..40>  | 15 27          |
| MEM_A_DQ_BYTE6            | MEM_55S      | MEM_DATA     | MEM A DQ<55..48>  | 15 27          |
| MEM_A_DQ_BYTE7            | MEM_55S      | MEM_DATA     | MEM A DQ<63..56>  | 15 27          |
| MEM_A_DQ_BYTE0            | MEM_55S      | MEM_DATA     | MEM A DM<0>       | 15 26          |
| MEM_A_DQ_BYTE1            | MEM_55S      | MEM_DATA     | MEM A DM<1>       | 15 26          |
| MEM_A_DQ_BYTE2            | MEM_55S      | MEM_DATA     | MEM A DM<2>       | 15 26          |
| MEM_A_DQ_BYTE3            | MEM_55S      | MEM_DATA     | MEM A DM<3>       | 15 26          |
| MEM_A_DQ_BYTE4            | MEM_55S      | MEM_DATA     | MEM A DM<4>       | 15 27          |
| MEM_A_DQ_BYTE5            | MEM_55S      | MEM_DATA     | MEM A DM<5>       | 15 27          |
| MEM_A_DQ_BYTE6            | MEM_55S      | MEM_DATA     | MEM A DM<6>       | 15 27          |
| MEM_A_DQ_BYTE7            | MEM_55S      | MEM_DATA     | MEM A DM<7>       | 15 27          |
| MEM_A_DQS0                | MEM_70D      | MEM_DQS      | MEM A DQS P<0>    | 15 26          |
| MEM_A_DQS0                | MEM_70D      | MEM_DQS      | MEM A DQS N<0>    | 15 26          |
| MEM_A_DQS1                | MEM_70D      | MEM_DQS      | MEM A DQS P<1>    | 15 26          |
| MEM_A_DQS1                | MEM_70D      | MEM_DQS      | MEM A DQS N<1>    | 15 26          |
| MEM_A_DQS2                | MEM_70D      | MEM_DQS      | MEM A DQS P<2>    | 15 26          |
| MEM_A_DQS2                | MEM_70D      | MEM_DQS      | MEM A DQS N<2>    | 15 26          |
| MEM_A_DQS3                | MEM_70D      | MEM_DQS      | MEM A DQS P<3>    | 15 26          |
| MEM_A_DQS3                | MEM_70D      | MEM_DQS      | MEM A DQS N<3>    | 15 26          |
| MEM_A_DQS4                | MEM_70D      | MEM_DQS      | MEM A DQS P<4>    | 15 27          |
| MEM_A_DQS4                | MEM_70D      | MEM_DQS      | MEM A DQS N<4>    | 15 27          |
| MEM_A_DQS5                | MEM_70D      | MEM_DQS      | MEM A DQS P<5>    | 15 27          |
| MEM_A_DQS5                | MEM_70D      | MEM_DQS      | MEM A DQS N<5>    | 15 27          |
| MEM_A_DQS6                | MEM_70D      | MEM_DQS      | MEM A DQS P<6>    | 15 27          |
| MEM_A_DQS6                | MEM_70D      | MEM_DQS      | MEM A DQS N<6>    | 15 27          |
| MEM_A_DQS7                | MEM_70D      | MEM_DQS      | MEM A DQS P<7>    | 15 27          |
| MEM_A_DQS7                | MEM_70D      | MEM_DQS      | MEM A DQS N<7>    | 15 27          |
| MEM_B_CLK                 | MEM_70D      | MEM_CLK      | MEM B CLK P<5..0> | 9 15 28 29 32  |
| MEM_B_CLK                 | MEM_70D      | MEM_CLK      | MEM B CLK N<5..0> | 9 15 28 29 32  |
| MEM_B_CKE                 | MEM_50S      | MEM_CTRL     | MEM B CKE<3..0>   | 15 21 28 29 32 |
| MEM_B_CNTL                | MEM_50S      | MEM_CTRL     | MEM B CS L<3..0>  | 15 28 29 32    |
| MEM_B_CNTL                | MEM_50S      | MEM_CTRL     | MEM B ODT<3..0>   | 15 28 29 32    |
| MEM_B_CMD                 | MEM_50S      | MEM_CMD      | MEM B A<15..0>    | 9 15 28 29 32  |
| MEM_B_CMD                 | MEM_50S      | MEM_CMD      | MEM B BA<2..0>    | 15 28 29 32    |
| MEM_B_CMD                 | MEM_50S      | MEM_CMD      | MEM B RAS L       | 15 28 29 32    |
| MEM_B_CMD                 | MEM_50S      | MEM_CMD      | MEM B CAS L       | 15 28 29 32    |
| MEM_B_CMD                 | MEM_50S      | MEM_CMD      | MEM B WE L        | 15 28 29 32    |
| MEM_B_DQ_BYTE0            | MEM_55S      | MEM_DATA     | MEM B DQ<7..0>    | 15 28          |
| MEM_B_DQ_BYTE1            | MEM_55S      | MEM_DATA     | MEM B DQ<15..8>   | 15 28          |
| MEM_B_DQ_BYTE2            | MEM_55S      | MEM_DATA     | MEM B DQ<23..16>  | 15 28          |
| MEM_B_DQ_BYTE3            | MEM_55S      | MEM_DATA     | MEM B DQ<31..24>  | 15 28          |
| MEM_B_DQ_BYTE4            | MEM_55S      | MEM_DATA     | MEM B DQ<39..32>  | 15 29          |
| MEM_B_DQ_BYTE5            | MEM_55S      | MEM_DATA     | MEM B DQ<47..40>  | 15 29          |
| MEM_B_DQ_BYTE6            | MEM_55S      | MEM_DATA     | MEM B DQ<55..48>  | 15 29          |
| MEM_B_DQ_BYTE7            | MEM_55S      | MEM_DATA     | MEM B DQ<63..56>  | 15 29          |
| MEM_B_DQ_BYTE0            | MEM_55S      | MEM_DATA     | MEM B DM<0>       | 15 28          |
| MEM_B_DQ_BYTE1            | MEM_55S      | MEM_DATA     | MEM B DM<1>       | 15 28          |
| MEM_B_DQ_BYTE2            | MEM_55S      | MEM_DATA     | MEM B DM<2>       | 15 28          |
| MEM_B_DQ_BYTE3            | MEM_55S      | MEM_DATA     | MEM B DM<3>       | 15 28          |
| MEM_B_DQ_BYTE4            | MEM_55S      | MEM_DATA     | MEM B DM<4>       | 15 29          |
| MEM_B_DQ_BYTE5            | MEM_55S      | MEM_DATA     | MEM B DM<5>       | 15 29          |
| MEM_B_DQ_BYTE6            | MEM_55S      | MEM_DATA     | MEM B DM<6>       | 15 29          |
| MEM_B_DQ_BYTE7            | MEM_55S      | MEM_DATA     | MEM B DM<7>       | 15 29          |
| MEM_B_DQS0                | MEM_70D      | MEM_DQS      | MEM B DQS P<0>    | 15 28          |
| MEM_B_DQS0                | MEM_70D      | MEM_DQS      | MEM B DQS N<0>    | 15 28          |
| MEM_B_DQS1                | MEM_70D      | MEM_DQS      | MEM B DQS P<1>    | 15 28          |
| MEM_B_DQS1                | MEM_70D      | MEM_DQS      | MEM B DQS N<1>    | 15 28          |
| MEM_B_DQS2                | MEM_70D      | MEM_DQS      | MEM B DQS P<2>    | 15 28          |
| MEM_B_DQS2                | MEM_70D      | MEM_DQS      | MEM B DQS N<2>    | 15 28          |
| MEM_B_DQS3                | MEM_70D      | MEM_DQS      | MEM B DQS P<3>    | 15 28          |
| MEM_B_DQS3                | MEM_70D      | MEM_DQS      | MEM B DQS N<3>    | 15 28          |
| MEM_B_DQS4                | MEM_70D      | MEM_DQS      | MEM B DQS P<4>    | 15 29          |
| MEM_B_DQS4                | MEM_70D      | MEM_DQS      | MEM B DQS N<4>    | 15 29          |
| MEM_B_DQS5                | MEM_70D      | MEM_DQS      | MEM B DQS P<5>    | 15 29          |
| MEM_B_DQS5                | MEM_70D      | MEM_DQS      | MEM B DQS N<5>    | 15 29          |
| MEM_B_DQS6                | MEM_70D      | MEM_DQS      | MEM B DQS P<6>    | 15 29          |
| MEM_B_DQS6                | MEM_70D      | MEM_DQS      | MEM B DQS N<6>    | 15 29          |
| MEM_B_DQS7                | MEM_70D      | MEM_DQS      | MEM B DQS P<7>    | 15 29          |
| MEM_B_DQS7                | MEM_70D      | MEM_DQS      | MEM B DQS N<7>    | 15 29          |
| MCP_MEM_COMP              | MCP_MEM_COMP | MCP_MEM_COMP | MEM B DQS P<0>    | 15             |
| MCP_MEM_COMP              | MCP_MEM_COMP | MCP_MEM_COMP | MEM B DQS N<0>    | 15             |

MEM\_A/B\_CKE EC SET NAME IS CHANGED ON K6, CANNOT SYNC THIS PAGE FROM T27

|                                                                                                                             |  |                      |            |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|------------|
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## D

C

B

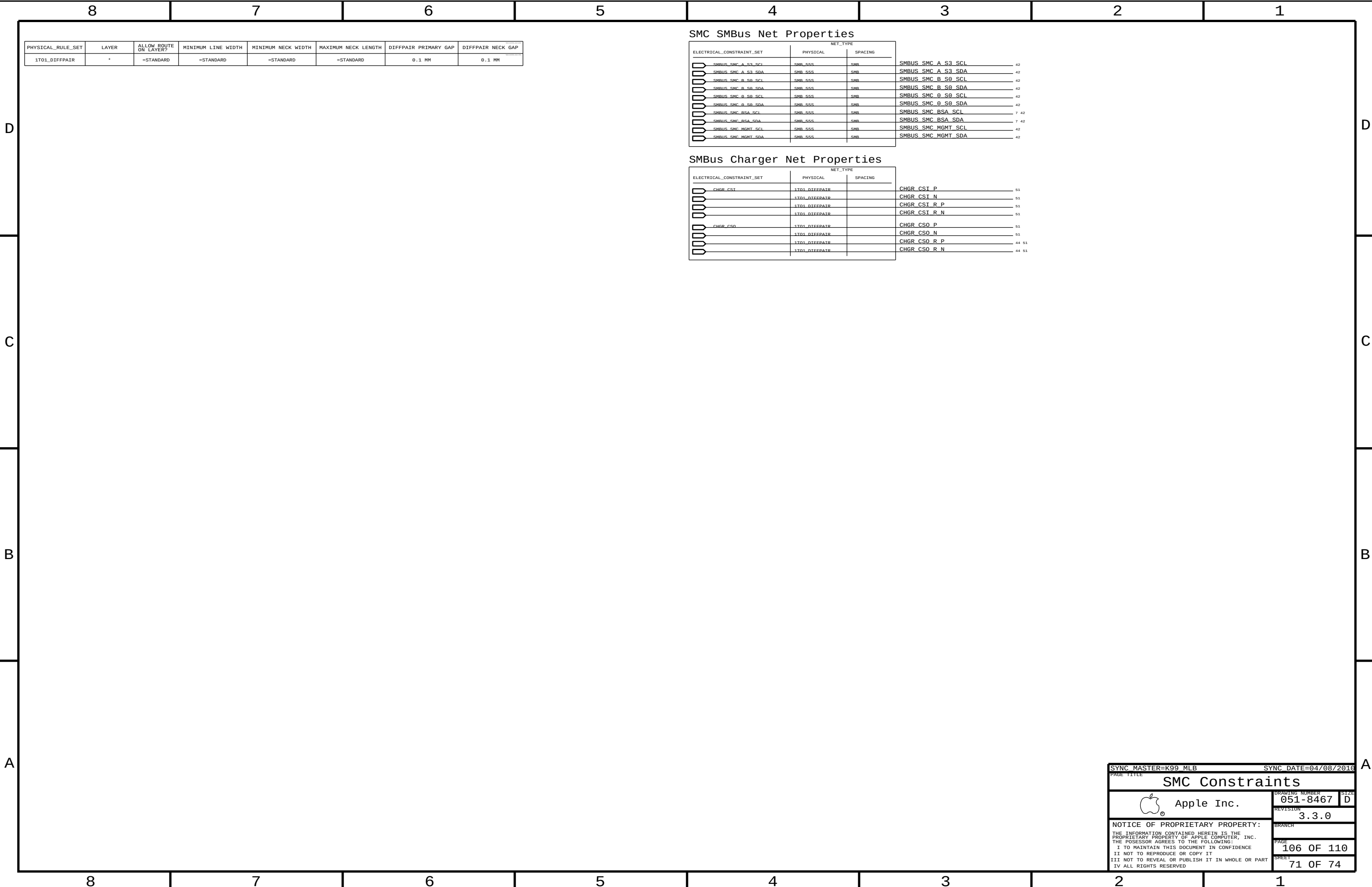
A

## D

CB







D

C

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SMC Constraints



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SHEET

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| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SENSE_1T01_55S    | *     | =1:1_DIFFPAIR         | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =1:1_DIFFPAIR        | =1:1_DIFFPAIR     |
| THERM_1T01_55S    | *     | =1:1_DIFFPAIR         | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =1:1_DIFFPAIR        | =1:1_DIFFPAIR     |
| DIFFPAIR          | *     | =1:1_DIFFPAIR         |                    |                    | =1:1_DIFFPAIR       | =1:1_DIFFPAIR        | =1:1_DIFFPAIR     |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SENSE            | *     | =1:1_SPACING         | ?      |
| THERM            | *     | =1:1_SPACING         | ?      |
| AUDIO            | *     | =1:1_SPACING         | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| ENETCONN         | *     | 25 MILS              | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| GND              | *     | =STANDARD            | ?      |
| MEM_POWER        | *     | =STANDARD            | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| GND_P2MM         | *     | 0.20 MM              | 1000   |
| PWR_P2MM         | *     | 0.20 MM              | 1000   |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CLK           | GND               | *         | GND_P2MM         |
| MEM_CMD           | GND               | *         | GND_P2MM         |
| MEM_CTRL          | GND               | *         | GND_P2MM         |
| MEM_DATA          | GND               | *         | GND_P2MM         |
| MEM_DQS           | GND               | *         | GND_P2MM         |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CLK           | MEM_POWER         | *         | PWR_P2MM         |
| MEM_CMD           | MEM_POWER         | *         | PWR_P2MM         |
| MEM_CTRL          | MEM_POWER         | *         | PWR_P2MM         |
| MEM_DATA          | MEM_POWER         | *         | PWR_P2MM         |
| MEM_DQS           | MEM_POWER         | *         | PWR_P2MM         |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CLK_PCIE          | GND               | *         | GND_P2MM         |
| PCIE              | GND               | *         | GND_P2MM         |
| SATA              | GND               | *         | GND_P2MM         |
| USB               | GND               | *         | GND_P2MM         |
| CLK_PCIE          | SB_POWER          | *         | PWR_P2MM         |
| SATA              | SB_POWER          | *         | PWR_P2MM         |
| USB               | SB_POWER          | *         | PWR_P2MM         |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CLK_FSB           | GND               | *         | GND_P2MM         |
| CPU_COMP          | GND               | *         | GND_P2MM         |
| CPU_GTLREF        | GND               | *         | GND_P2MM         |
| CPU_VCCSENSE      | GND               | *         | GND_P2MM         |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| LVDS              | GND               | *         | GND_P2MM         |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| ENET_MDI          | GND               | *         | GND_P2MM         |

## SD CARD READER LAYOUT RELAXATIONS

| PHYSICAL_RULE_SET  | LAYER         | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH    | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|--------------------|---------------|-----------------------|-----------------------|--------------------|---------------------|----------------------|-------------------|
| SD_55S<br>OVERRIDE | *<br>OVERRIDE | OVERRIDE              | =STANDARD<br>OVERRIDE | OVERRIDE           | OVERRIDE            | OVERRIDE             | OVERRIDE          |

## MCP Fanout Constraint Relaxations

| PHYSICAL_RULE_SET         | LAYER           | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH  | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|---------------------------|-----------------|-----------------------|--------------------|---------------------|---------------------|----------------------|-------------------|
| MEM_40S<br>OVERRIDE       | *               | OVERRIDE              | OVERRIDE           | 0.09 MM<br>OVERRIDE | 5.8 MM<br>OVERRIDE  | OVERRIDE             | OVERRIDE          |
| MCP_DV_COMP<br>OVERRIDE   | TOP<br>OVERRIDE | OVERRIDE              | OVERRIDE           | 0.1 MM<br>OVERRIDE  | 500 MIL<br>OVERRIDE | OVERRIDE             | OVERRIDE          |
| MCP_MEM_COMP<br>OVERRIDE  | TOP<br>OVERRIDE | OVERRIDE              | OVERRIDE           | 0.1 MM<br>OVERRIDE  | 500 MIL<br>OVERRIDE | OVERRIDE             | OVERRIDE          |
| MCP_MII_COMP<br>OVERRIDE  | TOP<br>OVERRIDE | OVERRIDE              | OVERRIDE           | 0.1 MM<br>OVERRIDE  | 500 MIL<br>OVERRIDE | OVERRIDE             | OVERRIDE          |
| MCP_USB_RBIAS<br>OVERRIDE | TOP<br>OVERRIDE | OVERRIDE              | OVERRIDE           | 0.1 MM<br>OVERRIDE  | 500 MIL<br>OVERRIDE | OVERRIDE             | OVERRIDE          |
| MCP_DV_COMP<br>OVERRIDE   | *<br>OVERRIDE   | OVERRIDE              | OVERRIDE           | 0.25 MM<br>OVERRIDE | 250 MIL<br>OVERRIDE | OVERRIDE             | OVERRIDE          |

## Misc Net Properties

| ELECTRICAL_CONSTRAINT_SET                                                           |                    | NET_TYPE |         |                   |          |
|-------------------------------------------------------------------------------------|--------------------|----------|---------|-------------------|----------|
|                                                                                     |                    | PHYSICAL | SPACING |                   |          |
|  | (USB_EXT_A)        | USB_90D  | USB     | USB_EXT_A_MUXED_P | 36 69    |
|  | (USB_EXT_A)        | USB_90D  | USB     | USB_EXT_A_MUXED_N | 36 69    |
|  | (USB_EXT_A)        | USB_90D  | USB     | USB_LT1_P         | 36       |
|  | (USB_EXT_A)        | USB_90D  | USB     | USB_LT1_N         | 36       |
|  | (USB_TPAD)         | USB_90D  | USB     | USB_TPAD_P        | 18 47 69 |
|  | (USB_TPAD)         | USB_90D  | USB     | USB_TPAD_N        | 18 47 69 |
|  | (USB_TPAD)         | USB_90D  | USB     | USB_TPAD_CONN_P   | 7 47     |
|  | (USB_TPAD)         | USB_90D  | USB     | USB_TPAD_CONN_N   | 7 47     |
|  | SMBUS_SMC_MGMT_SDA | SMR_55S  | SMR     | I2C_SMC_SMS_SDA_R |          |
|  | SMBUS_SMC_MGMT_SCL | SMR_55S  | SMR     | I2C_SMC_SMS_SCL_R |          |
|  |                    | SMR_55S  | SMR     | I2C_TCON_SCL      | 42       |
|  |                    | SMR_55S  | SMR     | I2C_TCON_SDA      | 42       |
|  |                    | SMR_55S  | SMR     | I2C_TCON_SCL_CONN | 68       |
|  |                    | SMR_55S  | SMR     | I2C_TCON_SDA_CONN | 68       |

## Graphics Net Properties

| ELECTRICAL_CONSTRAINT_SET |                 | NET_TYPE |                |                     |      |
|---------------------------|-----------------|----------|----------------|---------------------|------|
|                           |                 | PHYSICAL | SPACING        |                     |      |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT ML P<1..0>   | 9 60 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT ML N<1..0>   | 9 60 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT ML C P<1..0> | 60   |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT ML C N<1..0> | 60   |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT ML F P<1..0> | 7 60 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT ML F N<1..0> | 7 60 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT AUX CH C P   | 7 60 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT AUX CH C N   | 7 60 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT AUX CH P     | 9 60 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP INT AUX CH N     | 9 60 |
|                           | (DP_EXT_ML)     | DP 99D   | DTSDISPLAYPORT | DP EXT ML P<3..0>   | 9 62 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP EXT ML N<3..0>   | 9 62 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP EXT ML C P<3..0> | 62   |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP EXT ML C N<3..0> | 62   |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP EXT ML F P<3..0> | 62   |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP EXT ML F N<3..0> | 62   |
|                           | (DP_EXT_AUX_CH) | DP 99D   | DTSDISPLAYPORT | DP EXT AUX CH C P   | 9 62 |
|                           |                 | DP 99D   | DTSDISPLAYPORT | DP EXT AUX CH C N   | 9 62 |

## Power Net Properties

| ELECTRICAL_CONSTRAINT_SET                                                           |                | NET_TYPE       |           |                        |
|-------------------------------------------------------------------------------------|----------------|----------------|-----------|------------------------|
|                                                                                     |                | PHYSICAL       | SPACING   |                        |
|  | CPU1THMSNS_D2  | THERM 1T01 55S | THERM     | DRAMTHMSNS D2 P 45     |
|  |                | THERM 1T01 55S | THERM     | DRAMTHMSNS D2 N 45     |
|  | CPU_THERMD     | THERM 1T01 55S | THERM     | CPU_THERMD P 10 45     |
|  |                | THERM 1T01 55S | THERM     | CPU_THERMD N 10 45     |
|  | MCP1THMSNS_D2  | THERM 1T01 55S | THERM     | MLBR THMDIODE P 45     |
|  |                | THERM 1T01 55S | THERM     | MLBR THMDIODE N 45     |
|  | MCP_THMDIODE   | THERM 1T01 55S | THERM     | MCP_THMDIODE P 19 45   |
|  |                | THERM 1T01 55S | THERM     | MCP_THMDIODE N 19 45   |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | ISNS 1V5 S3 P 42 53    |
|  |                | SENSE 1T01 55S | SENSE     | ISNS 1V5 S3 N 43 53    |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | ISNS AIRPORT P 34 43   |
|  |                | SENSE 1T01 55S | SENSE     | ISNS AIRPORT N 34 43   |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | ISNS CSREG P 44        |
|  |                | SENSE 1T01 55S | SENSE     | ISNS CSREG N 44        |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | ISNS HDD P 35 43       |
|  |                | SENSE 1T01 55S | SENSE     | ISNS HDD N 35 43       |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | ISNS LCDBKLT P 43 63   |
|  |                | SENSE 1T01 55S | SENSE     | ISNS LCDBKLT N 43 63   |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | CPUVTT50 CS P 56       |
|  |                | SENSE 1T01 55S | SENSE     | CPUVTT50 CS N 56       |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | IMVP6 CS P 54          |
|  |                | SENSE 1T01 55S | SENSE     | IMVP6 CS N 54          |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | IMVP6 CS R P 54        |
|  |                | SENSE 1T01 55S | SENSE     | IMVP6 CS R N 54        |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | CPU_VTTSENSE P 56      |
|  |                | SENSE 1T01 55S | SENSE     | CPU_VTTSENSE N 56      |
|  | SENSE_D1FFPAIR | SENSE 1T01 55S | SENSE     | MCPCORE50_VSEN P 22 55 |
|  |                | SENSE 1T01 55S | SENSE     | MCPCORE50_VSEN N 22 55 |
|  |                |                | MEM_POWER | PP1V5R1V35 S3 7 8      |
|  |                |                | SB_POWER  | PP3V3 S5 7 8 58        |
|  |                |                | SB_POWER  | PP3V3 S0 7 8 58        |
|  |                |                | SB_POWER  | PP1V5 S0 7 8 58        |
|  |                |                | GND       | GND                    |

## Audio Net Properties

|                                                                                     |             | NET_TYPE  |         |               |         |
|-------------------------------------------------------------------------------------|-------------|-----------|---------|---------------|---------|
| ELECTRICAL_CONSTRAINT_SET                                                           |             | PHYSICAL  | SPACING |               |         |
|  | SPKRAMP_INR | DIEFEPAIR | AUDIO   | SPKRAMP_INR P | 7 37 49 |
|  |             | DIEFEPAIR | AUDIO   | SPKRAMP_INR P | 7 37 49 |
|  | MAX98300_R  | DIEFEPAIR | AUDIO   | MAX98300 R P  | 49      |
|  |             | DIEFEPAIR | AUDIO   | MAX98300 R N  | 49      |



## K99 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

| BOARD LAYERS                                                              | BOARD AREAS  | BOARD UNITS<br>(MIL or MM) | ALLEGRO<br>VERSION |
|---------------------------------------------------------------------------|--------------|----------------------------|--------------------|
| TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM | NO_TYPE, BGA | MM                         | 15.5.1             |

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| DEFAULT           | *     | Y                     | 0.100 MM           | 0.076 MM           | 30 MM               | 0 MM                 | 0 MM              |
| STANDARD          | *     | Y                     | =DEFAULT           | =DEFAULT           | 12.7 MM             | =DEFAULT             | =DEFAULT          |

| PHYSICAL_RULE_SET | LAYER       | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 27P4_OHM_SE       | *           | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 27P4_OHM_SE       | ISL3, ISL10 | Y                     | 0.250 MM           | 0.250 MM           |                     |                      |                   |
| 27P4_OHM_SE       | ISL4, ISL9  | Y                     | 0.250 MM           | 0.250 MM           |                     |                      |                   |

| PHYSICAL_RULE_SET                 | LAYER      | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-----------------------------------|------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 40_OHM_SE                         | *          | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 40_OHM_SE                         | TOP,BOTTOM | Y                     | 0.170 MM           | 0.170 MM           |                     |                      |                   |
| 40_OHM_SE ISL3, ISL4, ISL9, ISL10 |            | Y                     | 0.140 MM           | 0.140 MM           |                     |                      |                   |

| PHYSICAL_RULE_SET                 | LAYER      | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-----------------------------------|------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 50_OHM_SE                         | *          | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 50_OHM_SE                         | TOP,BOTTOM | Y                     | 0.110 MM           | 0.110 MM           |                     |                      |                   |
| 50_OHM_SE ISL3, ISL4, ISL9, ISL10 |            | Y                     | 0.090 MM           | 0.090 MM           |                     |                      |                   |

| PHYSICAL_RULE_SET | LAYER                   | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 55_OHM_SE         | *                       | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 55_OHM_SE         | TOP,BOTTOM              | Y                     | 0.090 MM           | 0.090 MM           |                     |                      |                   |
| 55_OHM_SE         | ISL3, ISL4, ISL9, ISL10 | Y                     | 0.076 MM           | 0.076 MM           |                     |                      |                   |

| PHYSICAL_RULE_SET | LAYER       | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 70_OHM_DIFF       | *           | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 70_OHM_DIFF       | TOP,BOTTOM  | Y                     | 0.175 MM           | 0.175 MM           |                     | 0.130 MM             | 0.130 MM          |
| 70_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.135 MM           | 0.135 MM           |                     | 0.130 MM             | 0.130 MM          |
| 70_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.155 MM           | 0.155 MM           |                     | 0.130 MM             | 0.130 MM          |

| PHYSICAL_RULE_SET | LAYER       | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 80_OHM_DIFF       | *           | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 80_OHM_DIFF       | TOP, BOTTOM | Y                     | 0.140 MM           | 0.140 MM           |                     | 0.160 MM             | 0.160 MM          |
| 80_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.109 MM           | 0.109 MM           |                     | 0.160 MM             | 0.160 MM          |
| 80_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.125 MM           | 0.125 MM           |                     | 0.160 MM             | 0.160 MM          |

| PHYSICAL_RULE_SET | LAYER       | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 75_OHM_DIFF       | *           | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 75_OHM_DIFF       | TOP, BOTTOM | Y                     | 0.160 MM           | 0.160 MM           |                     | 0.160 MM             | 0.160 MM          |
| 75_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.120 MM           | 0.120 MM           |                     | 0.140 MM             | 0.140 MM          |
| 75_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.140 MM           | 0.140 MM           |                     | 0.140 MM             | 0.140 MM          |

| PHYSICAL_RULE_SET | LAYER       | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 90_OHM_DIFF       | *           | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 90_OHM_DIFF       | TOP,BOTTOM  | Y                     | 0.115 MM           | 0.115 MM           |                     | 0.210 MM             | 0.210 MM          |
| 90_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.089 MM           | 0.089 MM           |                     | 0.210 MM             | 0.210 MM          |
| 90_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.105 MM           | 0.105 MM           |                     | 0.210 MM             | 0.210 MM          |

| PHYSICAL_RULE_SET | LAYER       | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 95_OHM_DIFF       | *           | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 95_OHM_DIFF       | TOP,BOTTOM  | Y                     | 0.115 MM           | 0.115 MM           |                     |                      | 0.210 MM          |
| 95_OHM_DIFF       | ISL3, ISL10 | Y                     | 0.089 MM           | 0.089 MM           |                     | 0.210 MM             | 0.210 MM          |
| 95_OHM_DIFF       | ISL4, ISL9  | Y                     | 0.105 MM           | 0.105 MM           |                     | 0.210 MM             | 0.210 MM          |

| PHYSICAL_RULE_SET | LAYER       | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 100_OHM_DIFF      | *           | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | =STANDARD            | =STANDARD         |
| 100_OHM_DIFF      | TOP, BOTTOM | Y                     | 0.091 MM           | 0.091 MM           |                     | 0.200 MM             | 0.200 MM          |
| 100_OHM_DIFF      | ISL3, ISL10 | Y                     | 0.075 MM           | 0.075 MM           |                     | 0.300 MM             | 0.300 MM          |
| 100_OHM_DIFF      | ISL4, ISL9  | Y                     | 0.085 MM           | 0.085 MM           |                     | 0.200 MM             | 0.200 MM          |

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 1:1_DIFFPAIR      | *     | Y                     | =STANDARD          | =STANDARD          | =STANDARD           | 0.1 MM               | 0.1 MM            |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| DEFAULT          | *     | 0.1 MM               | ?      |
| STANDARD         | *     | =DEFAULT             | ?      |
| BGA_P1MM         | *     | 0.1 MM               | ?      |
| BGA_P2MM         | *     | 0.2 MM               | ?      |
| BGA_P3MM         | *     | 0.3 MM               | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| 1:1_SPACING      | *     | 0.1 MM               | ?      |
| 1.5:1_SPACING    | *     | 0.15 MM              | ?      |
| 1.8:1_SPACING    | *     | 0.18 MM              | ?      |
| 2:1_SPACING      | *     | 0.2 MM               | ?      |
| 2.28:1_SPACING   | *     | 0.228 MM             | ?      |
| 2.5:1_SPACING    | *     | 0.25 MM              | ?      |
| 3:1_SPACING      | *     | 0.3 MM               | ?      |
| 4:1_SPACING      | *     | 0.4 MM               | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| GND              | *     | =STANDARD            | ?      |
| PP1V5_MEM        | *     | =STANDARD            | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| GND_P2MM         | *     | 0.2 MM               | 1000   |
| PWR_P2MM         | *     | 0.2 MM               | 1000   |

|                  |       |                      |        |
|------------------|-------|----------------------|--------|
| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
| NB_STATIC        | *     | =STANDARD            | ?      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| 2X_DIELECTRIC    | *     | 0.140 MM             | ?      |
| 3X_DIELECTRIC    | *     | 0.210 MM             | ?      |
| 4X_DIELECTRIC    | *     | 0.280 MM             | ?      |
| 1.5X_DIELECTRIC  | *     | 0.105 MM             | ?      |
| 5X_DIELECTRIC    | *     | 0.350 MM             | ?      |

|                                                                                                                                                                                                                                                                                                                                       |                |                      |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|--|
| SYNC MASTER=K99 MLB                                                                                                                                                                                                                                                                                                                   |                | SYNC DATE=04/08/2016 |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                            |                |                      |  |
| K99 RULE DEFINITIONS                                                                                                                                                                                                                                                                                                                  |                |                      |  |
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